



WHO INFLUENZA CENTRE LONDON

**Report prepared for the WHO annual
consultation on the composition of influenza
vaccine for the Northern Hemisphere 2014/15**

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Northern Hemisphere Vaccine Recommendation Meeting for 2014-15 influenza season.

WHO CC, London: Final Report for WHO meeting: 17-19 February 2014.

Influenza Activity in the WHO European Region, October 2013 - February 2014

Weekly reporting on influenza activity started in week 40/2013. The 2013-2014 season began later compared to the 2012-2013 season with levels of transmission increasing slowly following week 50/2013. The numbers of weekly influenza detections has been much lower than the numbers of detections reported in the previous year and remained so in week 05/2014 (**Figure 1**). The proportion of respiratory specimens from ILI and ARI patients testing positive has likely reached its peak in some countries, while increasing trends in ILI and ARI have been reported by some other countries.

More than 11,000 influenza detections have been reported to EuroFlu across the whole WHO European Region with type A detections (95%) predominating over type B (5%) (**Table 1**). Within type A, the A(H1N1)pdm09 subtype has been slightly more prevalent (59%) compared to A(H3N2) viruses (41%) (**Figure 1 & Table 1**). Within type B, 92% of the viruses for which lineage-determination was performed were of the Yamagata lineage.

Both the intensity of influenza activity and the degree of geographic spread of influenza within countries have started to increase in countries in southern and western parts of Europe (**Figure 2**). Countries in the northern part of Europe have, so far, experienced low intensity. Within individual countries, while influenza A virus detections have greatly outnumbered those of influenza B, levels of influenza A subtype detections (H1/H3) have shown some variation (**Figure 3**). Overall, influenza activity across the Region has been considerably lower than that seen over the same period of the 2012-2013 season, particularly for influenza B. However, influenza B detections often increase towards the end of an influenza season.

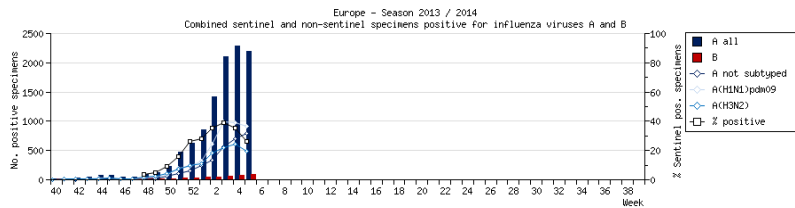
There are two systems for reporting severe disease associated with influenza infection in the WHO European Region. Severe Acute Respiratory Infections (SARI) have been reported to EuroFlu by 13 countries located in eastern European and Asiatic subregions: 191 influenza virus detections since week 40/2014 have been reported of which 113 (59%) were influenza A and 78 (41%) influenza B. Ninety-five of the influenza A viruses were subtyped: 59 (62%) were A(H1N1)pdm09 and 36 (38%), A(H3N2). EU/EEA countries have reported on hospitalization of severe laboratory-confirmed influenza cases to the European Centre for Disease Prevention and Control (ECDC). Up to week 5/2014, 1605 cases had been reported: 1588 (99%) were influenza A and 17 (1%) influenza B. Of the influenza A viruses 1078

were subtyped: 878 (81%) as A(H1N1)pdm09 and 200 (19%) as A(H3N2). The differences in proportions of type A and B viruses, and A(H1N1)pdm09 and A(H3N2) viruses, between the two reporting systems may be due, in part, to the low influenza activity in most of the countries involved in SARI surveillance up to week 05/2014.

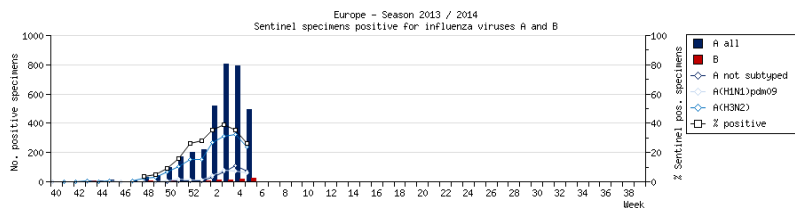
Seven countries have reported results to EuroFlu on susceptibility to antiviral drugs, oseltamivir and zanamivir - covering 220 A(H1N1)pdm09, 65 A(H3N2) and 21 type B influenza viruses. Two A(H1N1)pdm09 viruses carrying the neuraminidase H275Y amino acid substitution, causing resistance to oseltamivir, were detected in hospitalized patients in the UK. One influenza A(H3N2) virus carrying the neuraminidase E119V amino acid substitution, detected in a hospitalized patient treated with oseltamivir in the UK, showed reduced inhibition by oseltamivir but normal inhibition by zanamivir. All influenza B viruses tested showed normal susceptibility to oseltamivir and zanamivir. Overall, there has been no indication of increased resistance to the neuraminidase inhibitors oseltamivir and zanamivir during the winter of 2013/2014. All A(H1N1)pdm09 (60 of) and A(H3N2) (47 of) influenza viruses screened for susceptibility to adamantanes were found to be resistant.

Figure 1. Influenza Activity in WHO European Region: 2013-2014 Season (weeks 40-05)

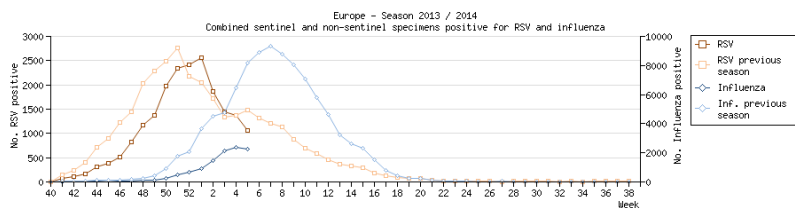
Total Influenza Detections



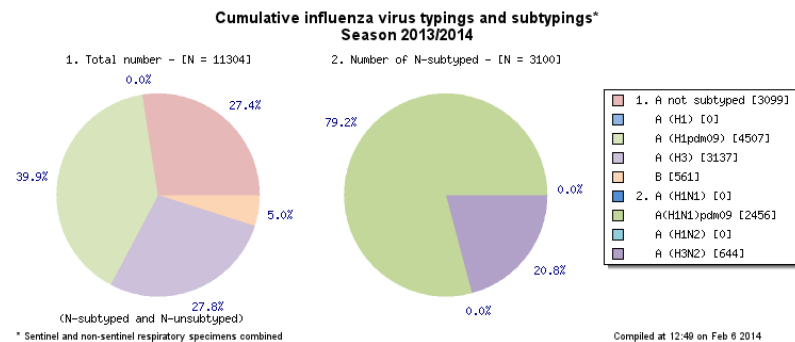
Sentinel Influenza Detections



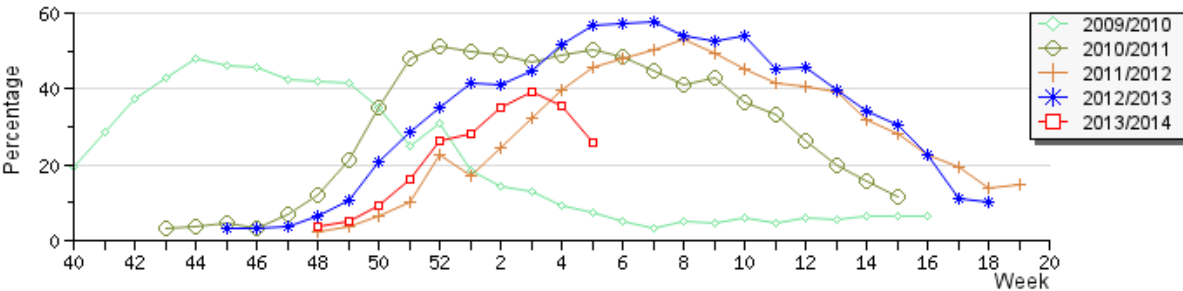
Past Season RSV and Influenza



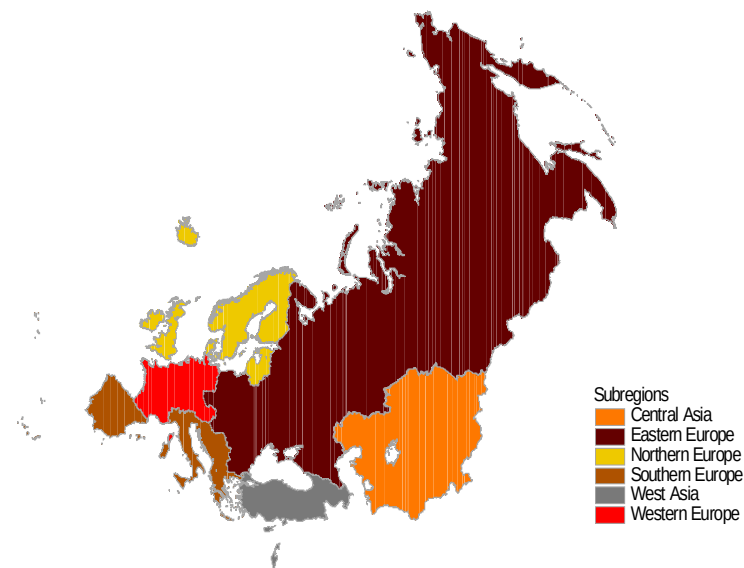
Influenza Typing and Subtyping



Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



**Table 1. Distribution of Influenza Detection in the WHO European Region:
2013-2014 Season (weeks 40-05)**



Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	4308	99%	1911	98%	906	93%	2546	91%	1023	87%	165	100%	10859	95%
Influenza A subtyped	3301		775		801		1761		950		164		7752	
<i>A (H1N1)pdm09</i>	2120	64%	451	58%	434	54%	1408	80%	59	6%	86	52%	4558	59%
<i>A (H3N2)</i>	1181	36%	324	42%	367	46%	353	20%	891	94%	78	48%	3194	41%
Influenza B	52	1%	45	2%	68	7%	244	9%	153	13%	0	0%	562	5%
B lineage determined	3		5		2		53		1		0		64	
<i>B/Yamagata lineage</i>	2	67%	4		2	100%	50	94%	1	100%	0		59	92%
<i>B/Victoria lineage</i>	1	33%	1		0	0%	3	6%	0	0%	0		5	8%
Total	4360		1956		974		2790		1176		165		11421	

*combined ILI, ARI and SARI sentinel and non-sentinel respiratory specimens testing positive for influenza

Figure 2. Development of the 2013-2014 Influenza Season in Europe

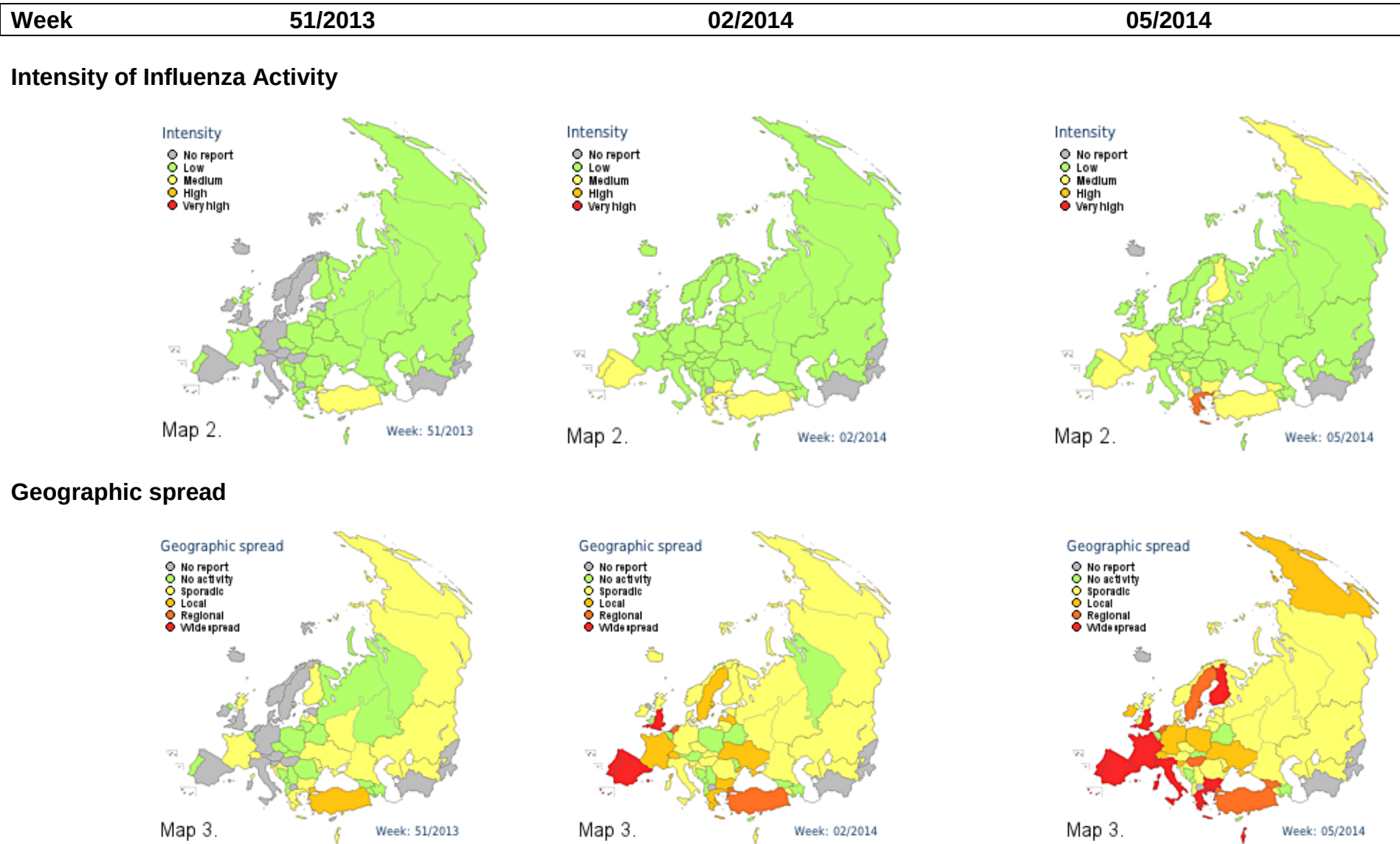
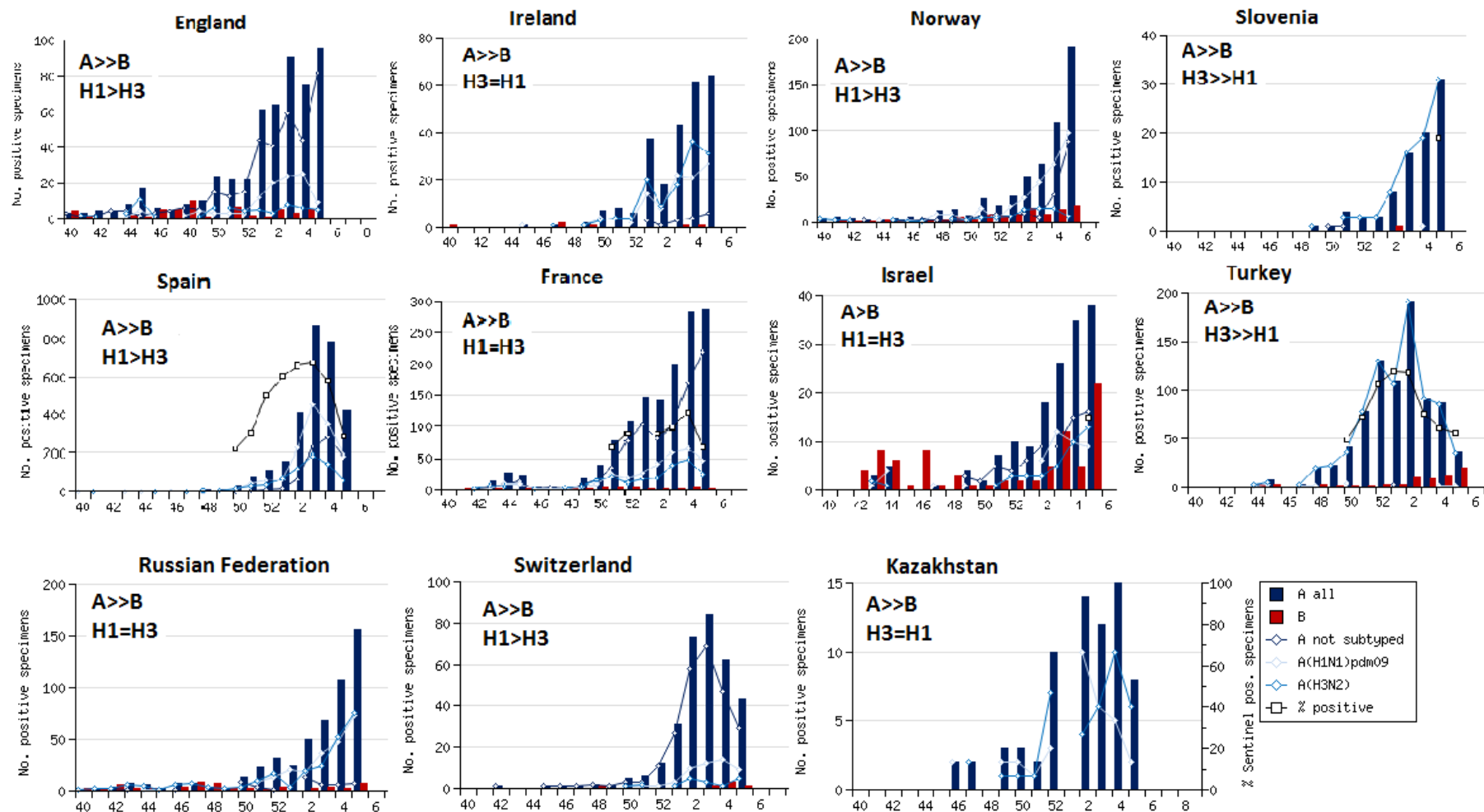


Figure 3. Influenza Type/Subtype Circulation Patterns in the WHO European Region: sentinel and non-sentinel detections



= represents similar ratio; > represents a ratio of at least 1.5:1; >> represents a ratio of ≥5

Antigenic and genetic characteristics of influenza viruses recovered from clinical specimens collected after 31 August 2013

Influenza positive specimens received at NIMR

Samples, viruses or clinical specimens, with collection dates after 2013-08-31 have been received from 38 countries in Europe, Africa, the Middle East and the Far East. The large majority (90%) were type A viruses, with A(H3N2) viruses predominating over A(H1N1)pdm09 viruses by a ratio of 1.3:1 (**Table 2**). Of the type B viruses (just under 10% of all specimens received) B/Yamagata lineage viruses predominated over those of the Victoria lineage by a ratio of 4:1. These proportions of influenza types/subtypes/lineages are similar to those reported within the WHO Euro Region (**Figures 1 & Table 1**). **Table 3** shows the isolation rates from specimens received with collection dates after 2013-08-31. Isolates from each type/subtype/lineage were recovered efficiently. **Table 4** summarises the number of viruses of the A(H1N1), A(H3N2) subtypes, and the two B/lineages that showed ≤ 2 -, 4- and ≥ 8 -fold reductions in HI titres with post-infection ferret antisera compared with the titres against their respective homologous vaccine/reference viruses. For A(H3N2) viruses a summary of the results of plaque-reduction neutralisation assays is also shown.

Influenza A(H1N1)pdm09 virus analysis.

H1N1pdm09 viruses were received from NICs in Europe, Africa/Indian Ocean (Cameroon, Cote d'Ivoire, Egypt, Ghana, Madagascar, Mauritius and Senegal), the Middle-East (Iran, Israel, Turkey), Central Asia (Kazakhstan), the Russian Far East (Vladivostok), South America (Guyane) and Hong Kong SAR. **Table 5** summarises the antigenic results of the received viruses that were propagated successfully.

The vast majority (99%) of the H1N1 viruses collected after 2013-08-31 were antigenically similar to the vaccine virus (**Table 4**); only two test viruses (1% of the total tested) showed 4-fold reductions and none showed ≥ 8 -fold reductions in HI titre compared with the titre of the vaccine virus (A/California/7/2009) with the homologous post-infection ferret antiserum. The HI results for all viruses tested since the September 2013 Vaccine Composition Meeting (VCM) are shown in **Tables 6-1 to 6-8**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-08-31. Viruses for which gene sequences are included in phylogenetic trees are highlighted and, where known, the HA genetic group is indicated. The test viruses A/Denmark/106/2013 and A/Lyon/2694/2013, an egg-propagated cultivar of A/Kazakhstan/3314/2014 and the cell-propagated cultivar of A/New Hampshire/04/2013 (received from CDC USA) showed 4-fold reduced recognition by the ferret antiserum raised against A/California/07/2009 (**Tables 6-3, 6-4, 6-8**)

Table 7 shows antigenic analysis of a set of H1N1 viruses analysed by the plaque reduction neutralisation assays using MDCK cells. All viruses were recognised well with post-infection ferret antiserum raised against the vaccine virus (A/California/7/2009) and by antisera raised against other reference viruses.

Figure 4 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. The HA genes cluster into eight designated genetic groups, with A/California/7/2009 representing group 1. Viruses collected after 2013-08-31 fell into genetic group 6. The viruses clustered into two of the three subgroups of group 6, subgroups 6B and 6C (**Figure 4**). These two subgroups carry the substitutions **D97N**, **S185T**, **S203T** and **K283E** in **HA1** and **E47K**, **S124N** and **E172K** in **HA2** compared with A/California/7/2009. The two subgroups 6B and 6C are defined by the following amino acid substitutions in **HA1** with none in **HA2**:

- Viruses in **subgroup 6B** carry substitutions **K163Q**, & **A256T** in **HA1**, e.g. A/South Africa/3626/2013
- Viruses in **subgroup 6C** carry substitutions **V234I** in **HA1**; some also carry the substitutions **V30A** and **A186T** in **HA1**, e.g. A/Paris/2496/2013 and A/Ghana/DARI-224/2013.

An amino acid sequence alignment of the HAs for a selected group of viruses is shown in **Figure 5**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted and the genetic group is shown in green.

Figure 6 illustrates the location of amino acid residues on an H1-HA structure, defining genetic subgroups 6B and 6C.

Figure 7 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 8** shows an amino acid alignment for a selected group of these viruses. The HA and NA phylogenetic trees are generally congruent with the exception of A/Mississippi/10/2013. The NA gene sequences of viruses from China and Japan resistant to oseltamivir cluster closely in both the HA and NA genes.

Antiviral analysis.

Figure 23 shows a phylogenetic tree of representative H1N1 M genes. Amino acid substitutions in M1 and M2 are highlighted. All sequences retained coding for the S31N substitution in the M2 ion-channel protein, known to confer resistance to adamantanes, with the exception of A/Puerto Rico/18/2013. An examination of sequences of the M2 gene in GISAID on 2014-02-07 revealed that, of 356 sequences derived from viruses collected after 2013-08-31, only two encoded an M2 protein expected to be inhibited by the adamantanes, A/Puerto Rico/18/2013 and A/Pennsylvania/07/2013.

Phenotypic testing of all viruses recovered from specimens with collection dates after 2013-08-31 were sensitive (normal inhibition) to the neuraminidase inhibitors zanamivir and oseltamivir (**Table 15**). Two viruses collected in January 2013, A/IIV-Moscow/34/2013 and A/Austria/71362S/2013, carried NA-H275Y substitution and showed highly reduced inhibition to oseltamivir.

Conclusion.

The vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus, A/California/7/2009.

Influenza A(H3N2) virus analysis.

Influenza A(H3N2) viruses collected since 2013-08-31 have been received from Europe, Africa/Indian Ocean (Ghana, Madagascar), Central Asia (Kazakhstan), the Middle East (Iran, Turkey) and from Hong Kong SAR. As described in previous reports, these viruses have continued to be difficult to characterise antigenically by HI assay due to variable agglutination of red blood cells from guinea pig, turkey and humans. All viruses had sufficient HA titre in assays conducted using guinea pig red blood cells in the presence of 20nM oseltamivir, added to circumvent any NA-mediated binding of H3N2 viruses to the red blood cells, to be analysed by HI assay. Virus neutralisation assays were used to complement HI tests; approximately 20% of the H3N2 viruses received were analysed in the plaque reduction neutralisation assay (PRNA).

A summary of the HI results is shown in **Table 8** and the HI results are shown in **Tables 9-1 to 9-11**. Viruses with collection dates after 2013-08-31 are shown in **Tables 9-3 to 9-11**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-08-31. Viruses for which gene sequences are included in phylogenetic trees are highlighted and, where known, the HA genetic group is indicated.

Nearly 90% of test viruses with collection dates after 2013-08-31 that were propagated in MDCK-SIAT1 cells reacted poorly in HI assays (≥ 8 -fold decrease) with post-infection ferret antiserum raised against the egg-propagated vaccine virus, A/Texas/50/2012, compared with the titre of the antiserum with the homologous virus. Similar low levels of reactivity were seen with antisera raised against three other egg-propagated reference viruses A/Serbia/NS-210/2013, A/Hong Kong/146/2013 and A/Almaty/2958/2013. Compared with the corresponding homologous titres, approximately 12%, 37% and 65% of test viruses reacted within 4-fold of the titre with the homologous egg-propagated virus for antisera raised against A/Hong Kong/146/2013, A/Almaty/2958/2013 and A/Serbia/NS-210/2013 respectively. Better reactivity was seen with antiserum raised against egg-propagated A/New York/39/2012 that had an homologous titre of 640; all test viruses were recognised by this antiserum at a titre within 4-fold of the titre with the homologous virus (**Table 9-9**).

Ferret antisera raised against reference viruses propagated in tissue culture cells A/Alabama/5/2010, A/Stockholm/18/2011, A/Berlin/93/2011, A/Athens/112/2012, A/Samara/73/2013 and A/Victoria/361/2011 recognised the test viruses more effectively. Notably, between 93% and 96% of test viruses propagated in MDCK-SIAT1 cells were recognised at titres within 4-fold of those for the antisera with their corresponding homologous viruses for antisera raised against cell-propagated A/Victoria/361/2011, A/Athens/112/2012 and A/Samara/73/2013, reference viruses with HA genes from genetic subgroups 3B and 3C.

The results of PRNA performed with MDCK-SIAT1 cells are shown in **Tables 10-1 to 10-3**. A total of 40 test viruses with collection dates after 2013-08-31 were analysed and are shown below the red horizontal lines. Also shown on the tables are the genetic groups to which the HA genes of the reference viruses and, where known, the test viruses belong. The results of PRNA were consistent with the results from HI assays done in the presence of oseltamivir. In

PRNA less than half of the test viruses reacted within 4-fold of the titre of the homologous virus with antisera raised against the egg-propagated vaccine virus, A/Texas/50/2012. Somewhat improved recognition of the test viruses was observed when antisera raised against the egg-propagated reference viruses A/Serbia/NS-210/2013 and A/Hong Kong/146/2013 were used; with these antisera ~65% and ~57%, respectively, of viruses showed titres with the antisera within 4-fold of the titres of the antisera for their homologous antigens. As observed in the HI assay, test viruses were recognised well by ferret antisera raised against reference viruses propagated in tissue culture cells: antisera raised against cell-propagated cultivars of A/Victoria/361/2011 and A/Samara/73/2013 recognised 90% and >95%, respectively, of test viruses at titres within 4-fold of the titres of the antisera for their homologous antigens. As in the HI analyses, antisera raised against both the cell-propagated and the egg-propagated cultivars of A/New York/39/2012 recognised over 90% of the test viruses at titres within 4-fold of the titres of the antisera for the homologous antigens, although note should be taken of the low homologous titre of the antiserum raised against the cell-propagated cultivar of A/New York/39/2012.

Notably, one test virus, A/Stockholm/18/2013, showed evidence of altered antigenic characteristics in both the HI and the PRNA assays. This virus belongs to the minority 3C.2 genetic subgroup. The HA gene sequence of this virus encodes substitutions at residues 144, which results in the loss of a potential glycosylation site at this position, 160, which results in the introduction of a glycosylation site at position 158, and changes at residues 225 and 311. Also, a polymorphism in the HA gene encoding residue 221 was seen. The antigenic characteristics of A/Stockholm/18/2013 require further examination.

Previously seven genetic groups based on the HA gene have been defined for H3N2 viruses, two derived from A/Perth/16/2009 and five from A/Victoria/208/2009. Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 9**. The HA genes fell within the A/Victoria/208 genetic clade and predominantly into the genetic subgroup 3C of group 3. This subgroup has three subdivisions: 3C.1, 3C.2 and 3C.3

The vaccine virus A/Texas/50/2012 belongs to genetic subgroup 3C.1. Most of the viruses received since 2013-08-31 fall into subgroups 3C.2 and 3C.3. Amino acid substitutions that define subgroups 3C.2 and 3C.3 are:

(3C.2) **N145S** and **V186G**[@] in **HA1**, and **D160N** in **HA2**, e.g. A/Hong Kong/146/2013;

(3C.3) **T128A** (resulting in the loss of a potential glycosylation site), **R142G**, **N145S** and **V186G**[@] in **HA1**, e.g. A/Samara/73/2013.

[@]Note: **G186V** substitutions in HA1 occur during adaptation of H3N2 viruses to passage in hens' eggs.

Figure 10 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny (**Figure 9**). The vaccine virus A/Texas/50/2012 is shown in red, glycosylation motifs are highlighted and the reference viruses are coloured blue. The genetic group to which each of the sequences belongs is also shown.

Figure 11 shows the location on a 2005 H3 HA structure of amino acid substitutions that define genetic groups 3C.2 and 3C.3.

Figure 12 shows a NA phylogenetic tree for representative viruses with the HA genetic groups indicated. **Figure 13** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of viruses with HA genes in genetic subgroup 3C do not cluster in the same manner as the HA sequences. The NA substitutions **Y155F** and **D251V** define two major divisions, both carrying **D93G** substitution, in recently collected viruses. Two viruses shown in **Figure 9** have HA genes in sub-group 3C.3 and NA genes in sub-group 3A and are intraclade reassortants. The majority of recently collected viruses have lost a potential glycosylation site in NA due to **N402D** substitution. **Figure 14** localises the predominant variable sites on the structure of an N2 NA that define the two clusters of NA genes that associate with 3C.3 HA genes, along with amino residues that influence the ability of viruses to agglutinate red blood cells through the NA.

Antiviral resistance

Figure 24 shows a phylogenetic tree of representative H3N2 M genes. Amino acid substitutions in M1 and M2 are highlighted. All sequences retained coding for the S31N substitution in the M2 ion-channel protein, known to confer resistance to adamantanes. An examination of sequences of the M2 gene in GISAID on 2014-02-07 revealed that all 213 sequences, derived from viruses collected after 2013-08-31, encoded the S31N substitution associated with resistance to adamantanes.

Phenotypic testing was performed on 130 viruses recovered from specimens with collection dates after 2013-08-31. All but one were sensitive (normal inhibition) to the neuraminidase inhibitors zanamivir and oseltamivir (**Table 15**). The virus A/England/651/2013, recovered from an oseltamivir treated, hospitalised patient carried an NA E119V substitution causing reduced inhibition by oseltamivir while retaining normal inhibition by zanamivir.

Conclusion.

The vast majority of recent test viruses analysed show poor reactivity with antisera raised against the egg-propagated vaccine virus A/Texas/50/2012 in both virus neutralisation and HI assays, with the latter being carried out in the presence of oseltamivir. Somewhat better reactivity was seen with antisera raised against other egg-propagated viruses. Conversely, test viruses reacted well with antisera raised against reference viruses that are genetically similar to A/Victoria/361/2011 but propagated in cells. There is no evidence of antigenic drift from the cell-propagated reference viruses but the test viruses are antigenically distinguishable from egg-propagated viruses.

Influenza B viruses.

Influenza B viruses represented just under 10% of samples received with collection dates after 2013-08-31 (**Table 2**). As described above, B/Yamagata lineage viruses predominated over those of the B/Victoria lineage at a ratio of approximately 4:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from three countries in Europe, and one each in Africa (Ghana) and the Middle East (Turkey), and Hong Kong SAR. The collection details and summary results are shown in **Table 11**.

HI results for the propagated viruses are shown in **Tables 12-1 and 12-2**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are those with collection dates after 2013-08-31. Viruses for which gene sequences are included in phylogenetic trees are highlighted and, where known, the HA genetic group is indicated. Approximately one quarter of cell-propagated test viruses showed ≥ 8 -fold reductions in HI titres compared with the titres with the homologous viruses, with post-infection ferret antisera raised against the recommended vaccine virus for quadrivalent vaccines, B/Brisbane/60/2008. Approximately two thirds of the test viruses were poorly recognised by post-infection ferret antisera raised against the egg-propagated reference viruses B/Malta/636714/2011, B/Johannesburg/3964/2012 and B/South Australia/81/2012. In contrast, all viruses showed reactivity within 4-fold of the titre of the homologous virus with antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells. As shown in **Tables 12-1 and 12-2**, these antisera were raised against B/Paris/1762/2008, B/Hong Kong/514/2009, B/Odessa/3886/2010 and B/Formosa/V2367/2012. These viruses are considered as surrogate cell-propagated antigens representing the egg-propagated prototype strain. Two viruses, collected in March and May 2013, showing increased reactivity with antiserum raised against egg-propagated viruses carried mutations in their HA gene sequences that cause loss of glycosylation at amino acid **residue 197** of HA1, a glycosylation site that is generally lost on egg passage.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 15** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 16**. The HA genes of the majority of recently collected viruses sent to NIMR fall into the B/Brisbane/60/2008 genetic clade in subgroup 1A. **Figure 17** shows a phylogenetic tree for the NA gene and **Figure 18** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa (Ghana), the Middle East (Iran and Turkey) and Hong Kong SAR. The collection details and results are summarised in **Table 13**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 14-1 to 14-5**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines are viruses with collection dates after 2013-08-31. The clade into which the HA falls is shown, where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. Post infection ferret antiserum raised against the current, egg-propagated, vaccine virus B/Massachusetts/02/2012 recognised approximately 50% of viruses collected after 2013-08-31 at titres within 4-fold of the titre for the homologous virus, but note should be made of the variation of the homologous titre between tests. A ferret antiserum raised against a cell-propagated cultivar of B/Massachusetts/02/2012 recognised 80% of viruses collected since 2013-08-31 at titres within 4-fold of its titre with the homologous virus. Approximately 60% of the test viruses were recognised by ferret antisera raised against two other viruses with HA genes belonging to the B/Massachusetts/02/2012 clade (Clade 2), B/Estonia/55669/2011 and B/Hong Kong/3577/2012, at titres within 4-fold of the titres of the antisera with their homologous viruses. Approximately 80% of test viruses were recognised by an antiserum raised against the previous vaccine virus B/Wisconsin/1/2010 at titres within 4-fold of the titre of the antiserum with the homologous virus and ~70% were similarly recognised by antiserum raised against cell-propagated B/Novosibirsk/01/2012, a virus belonging to the B/Wisconsin/1/2010 clade, Clade 3.

Many, but not all, viruses carrying Clade 2 HA genes could be differentiated antigenically from those with HA genes falling within Clade 3. However, not all antisera were able to antigenically differentiate between viruses falling in the alternate clades.

Figure 19 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 20**. **Figure 21** shows a phylogenetic tree for NA genes and **Figure 22** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. The HA and NA genes of recently collected viruses fell into the B/Massachusetts/02/2012 clade (Clade 2) and the B/Wisconsin/1/2010 clade (Clade 3) in approximately equal proportions.

Antiviral Analysis.

No evidence for reduced sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was seen for influenza B viruses collected after 2013-08-31 (**Table 15**).

Conclusion.

Viruses of the B/Yamagata lineage have predominated. For viruses of the B/Victoria lineage most were antigenically similar to viruses closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell into two antigenically distinguishable groups.

Table 2. Summary of clinical samples and isolates received, with collection dates after 2013-08-31, by country

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated ¹	Number received	Number propagated ²		Number received	Number propagated ²	Number received	Number propagated ²
2013											
SEPTEMBER											
France	2				2	2					
Ghana	9	2			7	6					
Madagascar	2						2				
Senegal	4		3	in process			1				
Sweden	2				2	2					
United Kingdom	2		1	1	1	1					
OCTOBER											
Cote d'Ivoire	4	1			3	in process					
Denmark	3		1	0	1	0				1	0
Egypt	48	33	15	in process							
Finland	1		1	1							
France	4		1	1	1	1		1	1	1	1
Ghana	5		1	0	4	1					
Iran	8				4	4				4	4
Israel	4				2	in process	2				
Madagascar	2	1					1				
Netherlands	3		1	1	2	2					
Norway	6				6	6					
Poland	1	1									
Portugal	1									1	1
Russia	1		1	1							
Senegal	3		2	in process			1				
Spain	2		2	2							
Sweden	2		2	2							
Turkey	3		1	in process	2	in process					
United Kingdom	2				1	1				1	1
NOVEMBER											
Belgium	1		1	1							
Cote d'Ivoire	17	1	4	in process	11	in process	1				
Denmark	2		1	0	1	1					
Egypt	29	21	8	in process							
Finland	1		1	1							
France	9		3	3	3	3				3	3
Germany	4		1	1	2	2		1	1		
Ghana	6		2	2	1	1		2	1	1	0
Iran	1		1	1							
Ireland	3		1	0			1			1	1
Israel	1									1	1
Italy	1				1	1					
Kazakhstan	3		1	1	2	1					
Norway	10		5	5	2	2		1	1	2	2
Poland	2	1			1	0					
Portugal	2				2	0					
Romania	1				1	in process					
Senegal	11	4	6	in process			1				
Spain	7		3	3	4	4					
Sweden	5		3	3	2	2					
Tunisia	2		1	0	1	0					
Turkey	22		3	3	17	in process	1			1	1
United Kingdom	2		2	2							
DECEMBER											
Algeria	4				3	in process				1	in process
Belgium	8		2	0	5	5				1	0
Bosnia and Herzegovina	4		1	0			2			1	1
Cote d'Ivoire	27	1	6	in process	18	in process	2				
Croatia	5				3	3	2				
Denmark	7		4	2	2	2				1	0
Egypt	30	17	13	in process							
Finland	4		1	1	3	3					
France	33		17	17	14	14				2	2
Germany	3				3	3					
Hong Kong	14		3	3	6	6		2	2	3	3
Iceland	2		1	1						1	1
Ireland	6		3	0	2	2				1	1
Israel	4		1	1	1	1	2				
Italy	15		1	1	14	14					
Kazakhstan	17		11	8	6	6					
Latvia	3		2	2	1	1					
Madagascar	3				3	in process					
Malta	1		1	in process							
Moldova	1				1	1					
Netherlands	1				1	1					
Norway	13		3	3	6	6				4	4
Portugal	9		6	5	3	2					
Romania	1		1	in process							
Russia	7		4	4	3	3					
Senegal	2		2	in process							
Slovenia	4				4	4					
Spain	28		16	13	12	10					
Sweden	6		1	1	4	4				1	1
Switzerland	2				2	in process					
Turkey	57		2	1	53	in process	1			1	1
Ukraine	7	2			5	4					
United Kingdom	5		3	3	2	2					
2014											
JANUARY											
Algeria	1				1	in process					
Belgium	3		2	2	1	1					
Bosnia and Herzegovina	3						3				
Cote d'Ivoire	4		3	in process	1	in process					
Croatia	1						1				
Germany	13		2	2	10	10		1	1		
Greece	35		32	in process	3	in process					
Iceland	4		4	4							
Ireland	3		1	1	2	2					
Israel	1		1	1							
Italy	2				1	1				1	1
Latvia	1		1	1							
Madagascar	8	3	2	in process	3	in process					
Malta	4		4	in process							
Moldova	1				1	1					
Portugal	9		6	6	3	3					
Romania	9		5	in process	4	4					
Slovenia	5				5	2					
Spain	23		19	19	4	4					
Switzerland	12		6	in process	6	in process					
Turkey	54				46	in process	4	2	2	2	in process
Ukraine	4	3	1	0							
United Kingdom	4		2	2	2	2					
38 Countries	803	91	275	139	362	170	28	10	9	37	30
			34.2%		45.1%			1.2%		4.6%	
			90.7%				9.3%				

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in presence of 20mM oseltamivir (the totalled number does not include any from batches that are in process)

Table 3. Isolation rates for samples collected after 2013-08-31

	Clinical sample				Virus isolate		
	number processed	number propagated ¹	percent recovered		number processed	number propagated ¹	percent recovered
H1	32	15	47%		146	128	88%
H3	27	22	81%		207	190	92%
B Victoria	2	1	50%		10	10	100%
B Yamagata	5	1	20%		31	31	100%

1. Propagated to sufficient titre to perform HI assay

Table 4. Summary of isolates with collection dates after 2013-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays

	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1¹	140	138 (99%)	2 (1%)	0 (0%)
H1^b	11	11 (100%)	0 (0%)	0 (0%)
H3²	211	1 (0%)	23 (11%)	187 (89%)
H3^c	40	3 (7.5%)	18 (45%)	19 (47.5)
B/Victoria³	6	5 (83%)	1 (17%)	0 (0%)
B/Yamagata⁴	30	0 (0%)	11 (37%)	19 (63%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009

b. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009 in plaque-reduction neutralisation assays

2. Compared to homologous titres for ferret antisera raised against vaccine virus A/Texas/50/2012

c. Compared to homologous titres for ferret antisera raised against vaccine virus A/Texas/50/2012 in plaque-reduction neutralisation assays

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

4. Compared to homologous titres for ferret antisera raised against vaccine virus B/Massachusetts/02/2012

Table 5. Summary of influenza A(H1N1)pdm09 samples analysed, collected after 2013-08-31

MONTH	H1N1pdm09					
	Country	Number received	Number propagated ¹	Fold reduction in HI titre		
				≤2 fold ²	4 fold ²	≥8 fold ²
2013						
SEPTEMBER						
Senegal	3	in process				
United Kingdom	1	1	1			
OCTOBER						
Denmark	1	0				
Egypt	15	in process				
Finland	1	1	1			
France	1	1		1		
Ghana	1	0				
Netherlands	1	1	1			
Russia	1	1	1			
Senegal	2	in process				
Spain	2	2	2			
Sweden	2	2	2			
Turkey	1	in process				
NOVEMBER						
Belgium	1	1	1			
Cote d'Ivoire	4	in process				
Denmark	1	0				
Egypt	8	in process	1			
Finland	1	1	1			
France	3	3	3			
Germany	1	1	1			
Ghana	2	2	2			
Iran	1	1	1			
Ireland	1	0				
Kazakhstan	1	1	1			
Norway	5	5	5			
Senegal	6	in process				
Spain	3	3	3			
Sweden	3	3	3			
Tunisia	1	0				
Turkey	3	3	3			
United Kingdom	2	2	2			
DECEMBER						
Belgium	2	0				
Bosnia and Herzegovina	1	0				
Cote d'Ivoire	6	in process				
Denmark	4	2	1	1		
Egypt	13	in process				
Finland	1	1	1			
France	17	17	17			
Hong Kong	3	3	3			
Iceland	1	1	1			
Ireland	3	0				
Israel	1	1	1			
Italy	1	1	1			
Kazakhstan	11	8	8			
Latvia	2	2	2			
Malta	1	in process				
Norway	3	3	3			
Portugal	6	5	5			
Romania	1	in process				
Russia	4	4	4			
Senegal	2	in process				
Spain	16	13	13			
Sweden	1	1	1			
Turkey	2	1	1			
United Kingdom	3	3	3			
2014						
JANUARY						
Belgium	2	2	2			
Cote d'Ivoire	3	in process				
Germany	2	2	2			
Greece	32	in process				
Iceland	4	4	4			
Ireland	1	1	1			
Israel	1	1	1			
Latvia	1	1	1			
Madagascar	2	in process				
Malta	4	in process				
Portugal	6	6	6			
Romania	5	in process				
Spain	19	19	19			
Switzerland	6	in process				
Ukraine	1	0				
United Kingdom	2	2	2			
33 Countries	275	139	138 99%	2 1%	0 0%	

1. Propagated to sufficient titre to perform HI assays

2. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Of 15 viruses sequenced with >2 fold reduction in HI titre, 9 contained HA1 substitutions in the 153-155 region known to affect receptor binding/antigenicity, as a result of cell propagation

Table 6-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses

			Haemagglutination inhibition titre ¹								
			Post-infection ferret antisera								
Viruses	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
Genetic group			4	3	5	6A	7	6A			
REFERENCE VIRUSES											
A/California/7/2009	2009-04-09	E1/E2	1280	640	640	320	320	320	160	320	320
A/Bayern/69/2009	2009-07-01	MDCK5/MDCK1	160	320	160	40	40	40	80	40	80
A/Lviv/N6/2009	2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	80	160	160	320
A/Christchurch/16/2010	4 2010-07-12	E2/E1	320	1280	1280	5120	1280	1280	1280	2560	2560
A/Hong Kong/3934/2011	3 2011-03-29	MDCK2/MDCK4	640	160	320	320	1280	640	640	1280	1280
A/Astrakhan/1/2011	5 2011-02-28	MDCK1/MDCK5	320	640	320	320	640	640	640	1280	1280
A/St. Petersburg/27/2011	6A 2011-02-14	E1/E3	640	320	640	320	640	640	640	1280	1280
A/St. Petersburg/100/2011	7 2011-03-14	E1/E3	640	640	640	320	2560	2560	1280	2560	1280
A/Hong Kong/5659/2012	6A 2012-05-21	MDCK4/MDCK2	1280	320	640	640	1280	2560	1280	1280	2560
TEST VIRUSES											
A/Moldova/157/2013	6C 2013-02-12	MDCK3	1280	640	1280	1280	2560	1280	2560	2560	2560
A/Moldova/158/2013	2013-02-12	MDCK3	1280	640	1280	640	2560	1280	1280	2560	2560
A/Guyane/91/2013	7 2013-02-15	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560
A/Moldova/179/2013	6C 2013-02-15	MDCK3	640	320	640	640	1280	1280	1280	2560	2560
A/Moldova/229/2013	6C 2013-02-18	MDCK3	1280	640	1280	640	2560	1280	2560	2560	2560
A/Centre/972/2013	2013-02-20	MDCK1/MDCK1	1280	640	1280	640	2560	1280	1280	2560	2560
A/Moldova/216/2013	2013-02-20	MDCK3	1280	640	1280	640	1280	1280	2560	5120	2560
A/Paris/1095/2013	2013-02-25	MDCK1/MDCK1	1280	640	640	640	2560	1280	1280	2560	2560
A/Lithuania/6526/2013	6C 2013-02-26	MDCK3	1280	640	1280	640	1280	1280	1280	5120	2560
A/Moldova/258/2013	6C 2013-02-27	MDCK3	640	320	640	640	1280	1280	1280	2560	1280
A/Centre/1179/2013	2013-03-01	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560
A/Moldova/294/2013	6C 2013-03-01	MDCK2	640	320	640	640	1280	1280	1280	2560	1280
A/Centre/1290/2013	6C 2013-03-12	MDCK1/MDCK1	1280	320	640	640	1280	640	1280	2560	2560
A/Mauritius/209/2013	6C 2013-03-20	MDCK2/MDCK1	1280	640	1280	1280	1280	1280	2560	2560	2560
A/Mauritius/214/2013	6C 2013-03-20	MDCK3/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Moldova/433/2013	6C 2013-03-29	MDCK3	1280	640	1280	640	2560	1280	2560	5120	5120
A/Mauritius/288/2013	6C 2013-04-09	MDCK2/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560
A/Shanghai-Putuo/1203/2013	6C 2013-04-10	E1/E1/E1	1280	640	640	640	1280	1280	1280	5120	2560
A/Guizhou-Qingzhen/SWL1251/2013	6C 2013-06-04	C1/C1/MDCK1	640	320	320	640	640	640	1280	2560	1280
A/Guangdong-Yuexiu/SWL1651/2013	2013-07-03	E1/E1/E1	640	640	640	640	1280	1280	1280	2560	2560
1. < = <40			Vaccine								

1. < = <40

Vaccine

Table 6-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses

Haemagglutination inhibition titre ¹												
Post-infection ferret antisera												
Viruses		Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10 4	A/HK 3934/11 F21/11 3	A/Astrak 1/11 F22/11 5	A/St. P 27/11 F23/11 6A	A/St. P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A
Genetic group												
REFERENCE VIRUSES												
A/California/7/2009		2009-04-09	E1/E2	640	640	640	320	320	320	320	640	320
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	320	160	160	80	160	160	160	160
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	320	160	320
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	1280	2560	5120	2560	2560	2560	5120	2560
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	1280	320	640	640	1280	1280	1280	5120	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	320	640	640	1280	1280	1280	2560	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	320	640	640	1280	1280	1280	1280	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	2560	2560	2560	2560	5120
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	160	640	640	640	1280	1280
TEST VIRUSES												
A/Cameroon/13V-1101/2013	6C	2013-02-25	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560
A/Cameroon/13V-1538/2013	6C	2013-03-13	MDCK3	2560	1280	2560	2560	5120	5120	5120	5120	5120
A/Cameroon/13V-1940/2013		2013-04-01	MDCK4	2560	1280	2560	2560	5120	2560	5120	5120	5120
A/Cameroon/13V-2186/2013	6C	2013-04-11	MDCK2	320	160	320	320	640	640	640	1280	1280
A/Cameroon/13V-2052/2013		2013-04-13	MDCK2	1280	640	1280	640	1280	640	1280	2560	1280
A/Cameroon/13V-2201/2013		2013-04-23	MDCK3	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Cameroon/13V-2465/2013		2013-05-02	MDCK3	1280	640	1280	640	2560	1280	2560	5120	2560
A/Cameroon/13V-2463/2013		2013-05-07	MDCK3	1280	640	1280	640	2560	2560	2560	5120	2560
A/Cameroon/13V-2476/2013	6C	2013-05-07	MDCK3	1280	640	1280	640	2560	1280	2560	5120	2560
A/Cameroon/13V-2479/2013		2013-05-07	MDCK2	640	320	640	640	1280	640	1280	2560	1280
A/Cameroon/13V-2635/2013		2013-05-10	MDCK3	1280	640	1280	1280	2560	1280	2560	5120	2560
A/Cameroon/13V-2644/2013	6C	2013-05-15	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560
A/Cameroon/13V-2645/2013	6C	2013-05-15	MDCK2	320	160	640	320	640	640	640	1280	1280
A/Cameroon/13V-2719/2013		2013-05-17	MDCK2	2560	1280	2560	2560	5120	5120	5120	5120	5120
A/Cameroon/13V-2715/2013		2013-05-21	MDCK2	2560	1280	2560	1280	5120	2560	2560	5120	5120
A/Cameroon/13V-3043/2013	6C	2013-05-30	MDCK2	2560	1280	2560	2560	5120	2560	5120	5120	5120
A/Cameroon/13V-3089/2013	6C	2013-06-01	MDCK4	640	320	640	640	1280	640	640	1280	1280
A/Cameroon/13V-3217/2013		2013-06-03	MDCK3	1280	640	1280	1280	2560	1280	1280	2560	2560
A/Cameroon/13V-3052/2013		2013-06-04	MDCK3	1280	320	640	640	1280	1280	1280	2560	2560
A/Cameroon/13V-3210/2013	6C	2013-06-06	MDCK4	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Cameroon/13V-3234/2013	6C	2013-06-06	MDCK2	1280	1280	1280	320	1280	640	640	1280	640
Sequence in phylogenetic trees				1. < = <40		Vaccine						

Sequence in phylogenetic trees

1. < = <40

Vaccine

Table 6-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2013-12-17, 2014-01-15)

Haemagglutination inhibition titre ¹												
Post infection ferret antisera												
Viruses		Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
Genetic group							4	3	5	6	7	6A
REFERENCE VIRUSES												
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	320	320	320	320	320
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	320	160	160	80	160	160	160	160
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	320	160	160	320	160	320
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	1280	2560	5120	2560	2560	2560	5120	2560
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	1280	320	640	640	1280	1280	1280	5120	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	320	640	640	1280	1280	1280	2560	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	320	640	640	1280	1280	1280	1280	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	2560	2560	2560	2560	5120
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	160	640	640	640	1280	1280
TEST VIRUSES												
A/La Reunion/2514/2013		2013-07-12	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280
A/England/621/2013	6C	2013-09-30	SIAT1/MDCK1	1280	640	1280	640	1280	1280	1280	2560	2560
A/Asturias/1161/2013	6B	2013-10-01	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560
A/Stockholm/19/2013	6B	2013-10-03	MDCK0/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560
A/Netherlands/2248/2013	6B	2013-10-14	MDCK4/MDCK1	1280	320	1280	1280	1280	1280	1280	2560	2560
A/Stockholm/23/2013	6B	2013-10-15	MDCK2/MDCK1	640	160	640	320	1280	640	640	1280	1280
A/Moscow/18/2013	6B	2013-10-21	C2/MDCK1	640	160	320	640	640	640	640	1280	1280
A/Valladolid/142/2013	6B	2013-10-22	SIAT1/MDCK1	1280	320	1280	640	1280	1280	1280	2560	2560
A/Lyon/2694/2013	6B	2013-10-29	MDCK2/MDCK1	320	160	320	320	640	640	640	1280	1280
A/Bayern/138/2013	6B	2013-11-01	C3/MDCK1	1280	320	1280	1280	2560	1280	1280	2560	2560
A/Stockholm/21/2013	6B	2013-11-01	MDCK2/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560
A/England/624/2013		2013-11-04	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	1280	2560	2560
A/Aragon/1368/2013	6B	2013-11-04	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560
A/Sweden/138/2013	6B	2013-11-08	MDCK0/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560
A/Lyon-CHU/46.08/2013	6B	2013-11-10	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280
A/Galicia/1485/2013	6B	2013-11-11	SIAT1/MDCK1	1280	320	1280	640	2560	1280	1280	2560	2560
A/Norway/3073/2013	6B	2013-11-12	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Galicia/1484/2013	6B	2013-11-14	SIAT1/MDCK1	640	320	640	640	1280	1280	640	1280	1280
A/Ghana/FS-1090/2013	6C	2013-11-15	SIAT1/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560
A/Ghana/DARI-224/2013	6C	2013-11-18	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120
A/England/640/2013	6B	2013-11-24	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Lyon/2868/2013		2013-11-25	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Stockholm/33/2013	6B	2013-11-28	MDCK3/MDCK1	1280	640	640	320	640	640	320	640	640
A/Lyon/2899/2013	6B	2013-12-02	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280
A/Stockholm/35/2013	6B	2013-12-14	MDCK1/MDCK1	1280	320	640	640	1280	1280	1280	2560	2560
Sequences in phylogenetic trees				Vaccine								

G155E

G155E>G, D222G

N156N=D

Table 6-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-01-22)

				Haemagglutination inhibition titre ¹								
				Post infection ferret antisera								
Viruses		Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
Genetic group							4	3	5	6	7	6A
REFERENCE VIRUSES												
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	320	320	320	320	320
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	320	160	40	40	80	40	40	40
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	160	160	320
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	1280	1280	2560	5120	2560
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK5	640	160	640	320	1280	640	640	1280	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK7	1280	640	1280	640	1280	1280	1280	2560	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	640	1280	1280	1280	5120	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	640	1280	1280	1280	5120	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	80	160	160	320	320	320	640	1280
TEST VIRUSES												
A/New Hampshire/04/2013	6B	2013-09-01	C2/MDCK1	320	80	160	160	320	160	160	640	640
A/New Hampshire/04/2013	6B	2013-09-01	Ex/E1	1280	640	1280	1280	1280	1280	2560	5120	2560
A/Mississippi/10/2013	6B	2013-10-21	Ex/E1	1280	1280	1280	1280	1280	1280	1280	2560	2560
A/Paris/2401/2013	6C	2013-12-10	MDCK1/MDCK1	1280	640	1280	640	2560	2560	1280	5120	2560
A/Hong Kong/6015/2013	6B	2013-12-11	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120
A/Hong Kong/6019/2013	6B	2013-12-12	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120
A/Hong Kong/6030/2013	6B	2013-12-13	MDCK1/MDCK1	2560	1280	1280	2560	5120	2560	5120	5120	5120
A/Latvia/12027346/2013	6B	2013-12-13	MDCK2/MDCK1	640	160	320	320	640	640	640	1280	1280
A/Paris/2481/2013		2013-12-17	MDCK2/MDCK1	1280	320	640	640	1280	1280	1280	2560	2560
A/Paris/2496/2013	6C	2013-12-17	MDCK1/MDCK1	1280	1280	1280	640	2560	2560	2560	5120	2560
A/Denmark/106/2013	6B	2013-12-18	MDCK2/MDCK1	320	320	320	160	640	160	640	640	640
A/Paris/2506/2013		2013-12-20	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120
A/Centre/2508/2013	6B	2013-12-23	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120
A/Nord Pas de Calais/2519/2013	6B	2013-12-23	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120
A/Latvia/12046984/2013	6C	2013-12-23	MDCK1/MDCK1	1280	320	640	640	1280	1280	1280	5120	2560
A/Paris/2516/2013	6B	2013-12-24	MDCK1/MDCK1	1280	640	1280	1280	1280	1280	2560	5120	2560
A/Latvia/01009567/2014	6B	2014-01-05	MDCK1/MDCK1	640	320	320	640	1280	1280	1280	2560	2560

G155E
G155E>G, D222G

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 6-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-01-24)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹								
				Post infection ferret antisera								
				A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
				4				3	5	6	7	6A
REFERENCE VIRUSES												
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	640	320	320	320	320	320
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	640	160	40	40	<	<	40	80
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	160	80	<	320	160	320
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	5120	1280	1280	1280	2560	2560
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	320	640	320	1280	1280	1280	2560	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	1280	1280	1280	2560	2560	5120	5120
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	1280	1280	2560	1280	5120	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	1280	640	1280	1280	1280	2560	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	160	640	640	640	1280	1280
TEST VIRUSES												
A/Kazakhstan/3271/2013	6B	2013-11-08	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560
A/Belguim/13G1251/2013	6B	2013-11-16	MDCK2	1280	1280	1280	1280	2560	2560	2560	5120	5120
A/Kazakhstan/3314/2013		2013-12-10	MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Kazakhstan/3313/2013		2013-12-10	MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Kazakhstan/3315/2013	6B	2013-12-15	MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Kazakhstan/3317/2013	6B	2013-12-19	MDCK2	1280	640	1280	1280	1280	1280	1280	5120	2560
A/Kazakhstan/3320/2013	6B	2013-12-20	MDCK2	640	320	320	320	640	640	640	1280	1280
A/Kazakhstan/3348/2013	6B	2013-12-27	MDCK2	1280	1280	1280	2560	2560	2560	2560	5120	5120
A/Denmark/101/2013	6B	2013-12-29	MDCK2	1280	1280	1280	1280	2560	1280	2560	5120	5120
A/Kazakhstan/3350/2013	6B	2013-12-30	MDCK2	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Belguim/14G0003/2014	6B	2014-01-02	MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120
A/Belguim/14G0004/2014	6B	2014-01-03	MDCK2	1280	640	1280	1280	2560	1280	1280	5120	2560

G155E
G155E>G, D222G

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 6-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-01-28)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre ¹								
				Post infection ferret antisera								
				A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12
				4			4	3	5	6	7	6A
REFERENCE VIRUSES												
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	320	320	320	320	320
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	320	160	80	40	80	80	80	80
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	320
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	2560	2560	5120	5120
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	160	640	320	1280	640	640	1280	1280
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK7	1280	640	1280	1280	2560	1280	2560	2560	2560
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	2560	5120	2560
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	320	640	640	1280	640	1280	2560	1280
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	160	640	640	640	1280	1280
TEST VIRUSES												
A/Finland/380/2013	6B	2013-10-14	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Toulouse/2730/2013	6B	2013-11-02	MDCK3/MDCK1	1280	640	1280	1280	1280	1280	1280	2560	2560
A/Norway/3159/2013	6B	2013-11-22	MDCK2/MDCK1	640	640	640	1280	1280	1280	1280	2560	5120
A/Norway/3214/2013		2013-11-25	LLCMK2MDCK1/MDCK1	640	320	1280	1280	1280	1280	1280	2560	2560
A/Norway/3175/2013		2013-11-27	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560
A/Tehran/15381/2013	6B	2013-11-27	E2/E1	1280	1280	1280	1280	1280	1280	1280	2560	2560
A/Norway/3230/2013	6B	2013-11-28	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	5120
A/Finland/383/2013	6B	2013-11-30	MDCK1/MDCK1	2560	1280	2560	1280	5120	2560	2560	5120	5120
A/Toulon/2949/13	6B	2013-12-02	MDCK2/MDCK1	640	320	640	640	1280	1280	640	2560	1280
A/Norway/3234/2013	6B	2013-12-04	MDCK1/MDCK1	640	320	640	640	1280	1280	2560	2560	2560
A/Paris/2444/2013	6C	2013-12-06	MDCK2/MDCK2	1280	320	1280	640	1280	1280	2560	2560	2560
A/Lyon/2954/13		2013-12-09	MDCK2/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560
A/Lyon/2963/13		2013-12-09	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	2560
A/Finland/384/2013	6B	2013-12-11	MDCK1/MDCK1	2560	1280	1280	1280	2560	2560	2560	5120	5120
A/Lyon-CHU/ 51.200/13		2013-12-16	MDCK2/MDCK1	640	320	640	640	1280	1280	1280	2560	1280
A/Lyon/3020/13		2013-12-17	MDCK2/MDCK1	640	160	640	640	640	640	640	2560	1280
A/Lisboa/148/2013		2013-12-17	MDCK1/MDCK1	640	320	640	640	2560	1280	1280	2560	2560
A/Lyon/3058/13	6B	2013-12-18	MDCK3/MDCK1	640	320	640	640	1280	1280	1280	2560	2560
A/Pavia/155/2013	6B	2013-12-20	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120
A/England/666/2013		2013-12-21	SIAT2/MDCK1	1280	320	640	1280	2560	1280	1280	2560	2560
A/Lisboa/156/2013		2013-12-23	MDCK2/MDCK1	640	320	1280	640	2560	1280	1280	2560	2560
A/Iceland/67/2013	6B	2013-12-24	MDCK1/MDCK1	1280	320	640	640	1280	1280	1280	2560	2560
A/Lyon/3101/13		2013-12-26	MDCK4/MDCK1	640	320	640	640	1280	1280	1280	2560	2560
A/Kazakhstan/3346/2013		2013-12-27	MDCK3	640	320	640	640	1280	1280	1280	2560	1280
A/Desgnettes/3083/13		2013-12-27	MDCK2/MDCK1	640	320	640	1280	1280	1280	1280	2560	2560
A/Lisboa/152/2013	6B	2013-12-27	MDCK2/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	5120
A/England/672/2013		2013-12-29	SIAT2/MDCK1	640	320	640	640	640	1280	640	2560	1280
A/England/677/2013	6B	2013-12-30	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120
A/Lisboa/157/2013		2013-12-30	SIAT1/MDCK1	1280	1280	2560	1280	5120	2560	2560	5120	5120
A/Lisboa/153/2013		2013-12-31	SIAT1/MDCK1	1280	1280	1280	1280	5120	2560	2560	5120	5120
A/Ireland/14M02257/2014	6B	2014-01-01	MDCK3	640	320	640	640	1280	1280	1280	2560	2560
A/Lisboa/154/2013	6B	2014-01-01	MDCK1/MDCK1	1280	640	1280	1280	5120	2560	2560	5120	5120
A/Lisboa/4/2014	6B	2014-01-04	SIAT1/MDCK1	640	320	640	640	2560	1280	1280	2560	2560
A/Lisboa/6/2014		2014-01-04	SIAT1/MDCK1	1280	320	640	640	2560	1280	1280	2560	2560
A/Lisboa/11/2014		2014-01-05	SIAT1/MDCK1	1280	640	640	640	5120	1280	2560	5120	5120
A/England/5/2014	6B	2014-01-06	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560
A/Lisboa/10/2014	6B	2014-01-06	MDCK1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	2560
A/Lisboa/12/2014		2014-01-06	SIAT1/MDCK1	2560	1280	2560	1280	5120	2560	2560	5120	5120
A/England/3/2014	6B	2014-01-08	SIAT1/MDCK1	1280	640	640	1280	2560	1280	1280	5120	2560
A/Iceland/1/2014		2014-01-10	MDCK1/MDCK1	1280	320	640	640	1280	1280	1280	2560	2560
A/Iceland/2/2014	6B	2014-01-10	MDCK1/MDCK1	1280	640	640	1280	1280	1280	1280	2560	2560
A/Iceland/3/2014	6B	2014-01-15	MDCK1/MDCK1	640	320	640	640	2560	1280	1280	2560	1280
A/Iceland/4/2014	6B	2014-01-16	MDCK1/MDCK1	1280	640	1280	1280	5120	2560	2560	5120	2560

1. < = <40

Vaccine

Sequences in phylogenetic trees

G155E
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Table 6-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-02-05)

Haemagglutination inhibition titre ¹													
Viruses	Collection date	Passage History	Post infection ferret antisera										
			A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/S. Africa 3626/13 F3/14	
Genetic group													
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E2	1280	640	640	320	320	320	320	320	320	160
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	320	160	80	40	80	80	80	80	40
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	320	160
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	2560	2560	2560	5120	5120	2560
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK4	640	160	640	640	1280	1280	1280	1280	2560	640
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	640	320	1280	640	1280	2560	2560	640
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	1280	2560	2560	2560	5120	5120	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	640	1280	1280	1280	2560	2560	640
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	320	640	640	640	1280	1280	320
A/South Africa/3626/2013	6B	2013-06-06	MDCK2/MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280
TEST VIRUSES													
A/Izmir/1149/2013	6B	2013-11-01	SIAT1/MDCK2	640	320	1280	640	1280	1280	1280	2560	2560	1280
A/Agri/1176/2013	6B	2013-11-04	SIAT1/MDCK2	640	320	640	640	1280	1280	1280	2560	2560	640
A/Egypt/68/2013	6B	2013-11-12	E2/E1	1280	1280	1280	1280	2560	2560	1280	2560	2560	1280
A/Belgium/13G1251/2013	6B	2013-11-16	E2	1280	640	1280	640	1280	1280	1280	2560	2560	1280
A/Denizli/1258/2013	6B	2013-11-19	SIAT1/MDCK2	1280	640	1280	1280	2560	1280	1280	2560	2560	1280
A/Ankara/3935/2013	6B	2013-12-09	SIAT1/MDCK2	1280	640	2560	1280	2560	2560	2560	5120	5120	2560
A/Andalucia/1585/2013		2013-12-14	SIAT1/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	1280
A/Norway/3322/2013	6B	2013-12-16	MDCK1/MDCK2	1280	320	640	640	1280	1280	1280	2560	2560	1280
A/Norway/3293/2013	6B	2013-12-17	MDCK1/MDCK2	640	320	640	640	1280	1280	1280	2560	2560	640
A/Castilla La Mancha/131557/2013		2013-12-17	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560	1280
A/Andalucia/14175/2013	6B	2013-12-17	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/Israel/31/2013	6B	2013-12-18	C0/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	1280
A/Kazakhstan/3271/2013	6B	2013-12-18	E2	1280	320	640	320	1280	1280	1280	2560	1280	1280
A/Madrid/SO11395/2013	6B	2013-12-21	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/Aragon/14040/2013	6B	2013-12-26	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/Aragon/14041/2013		2013-12-26	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/Aragon/14042/2013		2013-12-26	SIAT1/MDCK1	640	160	640	640	640	640	1280	2560	1280	640
A/Aragon/14276/2013	6B	2013-12-26	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/Kazakhstan/3348/2013	6B	2013-12-27	E2	1280	640	1280	1280	1280	1280	1280	2560	2560	2560
A/Andalucia/14173/2013		2013-12-29	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/IV-Moscow/139/2013	6B	2013-12-30	MDCK2/MDCK1	640	160	320	320	640	640	640	1280	1280	640
A/Vladivostok/91/2013		2013-12-30	MDCK1/MDCK1	640	160	640	320	640	1280	640	2560	2560	640
A/Vladivostok/92/2013		2013-12-30	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280
A/Vladivostok/93/2013	6B	2013-12-30	MDCK1/MDCK1	640	320	1280	640	1280	2560	1280	2560	2560	1280
A/Madrid/31/2014		2014-01-03	SIAT1/MDCK1	1280	320	1280	1280	2560	2560	1280	5120	2560	1280
A/Navarra/220/2014		2014-01-03	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280
A/Belguim/14G0004/2014	6B	2014-01-03	E2	1280	640	640	640	1280	1280	1280	2560	2560	1280
A/Madrid/SO11443/2014		2014-01-04	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	2560	1280
A/Israel/33/2013	6B	2014-01-05	Cx/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560	1280
A/Castilla La Mancha/147/2014		2014-01-05	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	1280	2560	2560	1280
A/Castilla La Mancha/148/2014	6B	2014-01-07	SIAT1/MDCK1	1280	320	1280	1280	1280	1280	1280	2560	2560	1280
A/Castilla La Mancha/145/2014		2014-01-08	SIAT1/MDCK1	640	320	640	640	1280	640	1280	2560	2560	1280
A/Castilla La Mancha/106/2014		2014-01-09	SIAT1/MDCK1	640	640	1280	640	1280	1280	1280	2560	2560	1280
A/Castilla La Mancha/104/2014		2014-01-09	SIAT1/MDCK1	1280	320	1280	1280	1280	1280	2560	2560	2560	1280
A/Aragon/267/2014	6B	2014-01-09	SIAT1/MDCK1	2560	1280	2560	2560	5120	2560	5120	5120	5120	2560
A/Aragon/268/2014		2014-01-09	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	2560	5120	2560	1280
A/Aragon/269/2014		2014-01-10	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	2560	1280
A/Brandenburg/1/2014	6B	2014-01-10	C2/MDCK1	1280	1280	2560	1280	2560	2560	2560	5120	2560	2560
A/Madrid/107/2014		2014-01-13	SIAT1/MDCK1	1280	320	1280	1280	2560	1280	1280	5120	2560	1280
A/Castilla La Mancha/119/2014	6B	2014-01-13	SIAT1/MDCK1	1280	320	1280	1280	2560	2560	1280	5120	2560	2560
A/Madrid/125/2014		2014-01-13	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	1280	2560	2560	1280
A/Madrid/127/2014		2014-01-13	SIAT1/MDCK1	640	160	1280	640	1280	1280	1280	2560	2560	640
A/Aragon/270/2014	6B	2014-01-13	SIAT1/MDCK1	1280	640	1280	1280	2560	1280	1280	5120	2560	1280
A/Madrid/120/2014	6B	2014-01-14	SIAT1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280
A/Andalucia/174/2014		2014-01-14	SIAT1/MDCK1	1280	640	2560	1280	2560	2560	2560	5120	5120	2560
A/Schleswig-Holstein/1/2014	6B	2014-01-15	C2/MDCK1	640	320	1280	640	1280	1280	1280	2560	2560	1280
A/Andalucia/171/2014	6B	2014-01-16	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560

1. <= <40

Vaccine

G155E
G155E>G, D222G

Table 6-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-02-07)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹									
			Post infection ferret antisera									
			A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/09/34	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/S. Africa 3626/13 F3/14
Genetic group						4	3	5	6	7	6A	
REFERENCE VIRUSES												
A/California/7/2009	2009-04-09	E1/E2	1280	640	640	320	320	320	320	320	320	160
A/Bayern/69/2009	2009-07-01	MDCK5/MDCK1	320	320	160	80	40	80	80	80	80	40
A/Lviv/N6/2009	2009-10-27	MDCK4/S1/MDCK3	640	1280	640	160	80	160	320	160	320	160
A/Christchurch/16/2010	2010-07-12	E1/E3	2560	2560	2560	5120	2560	2560	2560	5120	5120	2560
A/Hong Kong/3934/2011	2011-03-29	MDCK2/MDCK4	640	160	640	640	1280	1280	1280	1280	2560	640
A/Astrakhan/1/2011	2011-02-28	MDCK1/MDCK5	640	320	640	320	1280	640	1280	2560	2560	640
A/St. Petersburg/27/2011	2011-02-14	E1/E3	1280	1280	1280	1280	2560	2560	2560	5120	5120	1280
A/St. Petersburg/100/2011	2011-03-14	E1/E3	640	640	640	640	1280	1280	1280	2560	2560	640
A/Hong Kong/5659/2012	2012-05-21	MDCK4/MDCK2	320	160	320	320	640	640	640	1280	1280	320
A/South Africa/3626/2013	2013-06-06	MDCK2/MDCK2	1280	640	1280	1280	2560	2560	2560	5120	5120	1280
TEST VIRUSES												
A/Kazakhstan/3313/2013	2013-12-10	E3	1280	640	1280	640	1280	1280	1280	2560	2560	1280
A/Kazakhstan/3314/2013	2013-12-10	E3	320	160	320	320	320	320	320	640	640	640
A/Andalucia/1584/2013	2013-12-11	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560
A/Hong Kong/6015/2013	2013-12-11	E3	1280	640	1280	1280	1280	1280	1280	2560	2560	1280
A/Navarra/1564/2013	2013-12-12	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560
A/Andalucia/1586/2013	2013-12-12	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560
A/Hong Kong/6019/2013	2013-12-12	E3	2560	1280	2560	2560	2560	2560	2560	5120	5120	2560
A/Pais Vasco/1569/2013	2013-12-13	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560
A/Kazakhstan/3315/2013	2013-12-15	E3	1280	1280	1280	1280	2560	1280	1280	2560	2560	2560
A/Kazakhstan/3317/2013	2013-12-19	E3	640	640	640	640	1280	1280	1280	2560	2560	1280
A/Kazakhstan/3320/2013	2013-12-20	E3	1280	320	640	640	1280	1280	1280	2560	2560	1280
A/Kazakhstan/3346/2013	2013-12-27	E3	1280	640	1280	1280	1280	1280	1280	2560	2560	2560
A/Kazakhstan/3350/2013	2013-12-30	E3	1280	1280	1280	1280	2560	2560	1280	5120	5120	2560
A/Ireland/14M02257/2014	2014-01-01	E3	1280	640	1280	640	1280	1280	1280	2560	2560	1280
A/Belgium/14G0003/2014	2014-01-02	E3	1280	640	1280	640	1280	1280	1280	2560	2560	1280

G155E
G155E>G, D222G

1. < = <40

Vaccine

Table 7. Antigenic analyses of influenza A(H1N1)pdm09 viruses - Plaque Reduction Neutralisation (MDCK-Parent) (2014-02-05)

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre ¹				
				Post infection ferret sera				
				A/Cal 07/09 F05/10	A/Bayern 69/09 F11/11	A/HK 3934/11 F21/11	A/St. P 27/11 F23/11	A/S. Africa 3626/13 F3/14
						3	6	6B
REFERENCE VIRUSES								
A/California/7/2009		2009-04-09	E1/E2	1280	1280	2560	640	ND
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	640	320	80	ND
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK5	2560	5120	2560	1280	ND
A/St.Petersburg/27/2011	6	2011-02-14	E1/E3	640	1280	640-1280	320	ND
A/South Africa/3626/2013	6B	2013-06-06	MDCK2/MDCK2	640-1280	640	640-1280	640-1280	1280
TEST VIRUSES								
A/New Hampshire/04/2013	6B	2013-09-01	Ex/E1	1280	1280	1280	1280	1280
A/Mississippi/10/2013	6B	2013-10-01	Ex/E1	1280	640	1280	640	640
A/Kazakhstan/3313/2013		2013-12-10	MDCK2	2560	1280	2560	5120	2560
A/Hong Kong/6015/2013	6B	2013-12-11	MDCK1/MDCK1	2560	1280	1280	2560	2560
A/Hong Kong/6019/2013	6B	2013-12-12	MDCK1/MDCK1	5120	2560	2560	2560	2560
A/Denmark/106/2013	6B	2013-12-18	MDCK2/MDCK1	1280	640	1280	1280	1280
A/Nord Pas de Calais/2519/2013	6B	2013-12-23	MDCK1/MDCK1	1280	1280	1280	2560	2560
A/Latvia/12046984/2013	6C	2013-12-23	MDCK1/MDCK1	1280	640	1280	2560	1280
A/Kazakhstan/3348/2013	6B	2013-12-27	MDCK2	5120	2560	5120	5120	5120
A/Belguim/14G0003/2014	6B	2014-01-02	MDCK2	1280	1280	1280	2560	2560
A/Latvia/01009567/2014	6B	2014-01-05	MDCK1/MDCK1	1280	1280	2560	5120	1280

1. 80% reduction

Vaccine

Figure 5. HA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100	
A/California/7/2009	DTLCIGYHANNST	DTVDTVLEKNVT	VTHTSVNLLLEDKHNGK	LCKLRGVAPLHLGKCNIAGWILGN	PECESLSTASSWSYIVETPSSD	NGTCYPGDFIDYEE					
A/Bayern/69/2009								S			
A/Ghana/DILI-0007/2013			I	D				S	E		Out
A/Lviv/N6/2009								S			3
A/Hong Kong/3934/2011								S			3
A/Astrakhan/1/2011								S		N	5
A/Christchurch/16/2010								S		N	7
A/St Petersburg/100/2011								S			7
A/Hong Kong/5659/2012								S		N	6A
A/St Petersburg/27/2011								S		N	6A
A/Latvia/12046984/2013								S		N	6C
A/Ghana/FS-1090/2013								S		N	6C
A/Paris/2496/2013			A					S		N	6C
A/Gothenburg/5/2013 se								S		N	6B
A/Belgium/1460004/2014								S		N	6B
A/England/5/2014								S		N	6B
A/Kazakhstan/3320/2013								S		N	6B
A/South Africa/3626/2013								S		N	6B
A/Lisboa/10/2014								S		N	6B
A/England/3/2014								S		N	6B
A/Iceland/4/2014								S		N	6B
A/Denmark/106/2013								S		N	6B
A/Hong Kong/6019/2013								S		N	6B
A/Ireland/14M02257/2014								S		N	6B
A/Bayern/138/2013								S		N	6B
A/Sapporo/9/2014 niid		S						S		N	6B
A/Yamagata/158/2013 niid		S						S		N	6B
A/Athens GR/18/2014 grc								S		N	6B
A/Lyon/3058/2013								S		N	6B
A/Tehran/15381/2013								S		N	6B

	110	120	130	140	150	160	170	180	190	200		
A/California/7/2009	LREQLVSSVSSFERFE	IFPKTSSWP	NHDSNKGVT	AACPHAGAKS	FFYKNLIWL	VKKGNSY	PKLSKSYIND	KGEVLVL	WGIHHP	STSA	QQSLYQ	NADAYVF
A/Bayern/69/2009						E						
A/Ghana/DILI-0007/2013			T						M			
A/Lviv/N6/2009												
A/Hong Kong/3934/2011		X		T	S				P			
A/Astrakhan/1/2011												
A/Christchurch/16/2010			D				T					
A/St Petersburg/100/2011					G				T		T	
A/Hong Kong/5659/2012		N			R							
A/St Petersburg/27/2011									T			
A/Latvia/12046984/2013				T					T			
A/Ghana/FS-1090/2013												
A/Paris/2496/2013									TT			
A/Gothenburg/5/2013 se							Q		T			
A/Belgium/1460004/2014			E				Q		T			
A/England/5/2014							Q		T			
A/Kazakhstan/3320/2013							Q		T			
A/South Africa/3626/2013							Q		T			
A/Lisboa/10/2014							Q		T			
A/England/3/2014							Q		T			
A/Iceland/4/2014							Q		T			
A/Denmark/106/2013		S					Q		T		N	
A/Hong Kong/6019/2013							Q		T			
A/Ireland/14M02257/2014							Q		T			
A/Bayern/138/2013							Q		T			
A/Sapporo/9/2014 niid							Q		T			
A/Yamagata/158/2013 niid							Q		T			
A/Athens GR/18/2014 grc							NQ		T			
A/Lyon/3058/2013							Q		T			
A/Tehran/15381/2013							Q		T			

	210	220	230	240	250	260	270	280	290	300	
A/California/7/2009	VGSSRYSKFKPEIA	IRPKVRDQ	EGRMNYWT	LVEPGDKITFEAT	GNLVVP	RYAFAMERNAGSGII	ISDTPV	HCNTT	TCQTPKGA	INTSL	LPFQNIHPITI
A/Bayern/69/2009											
A/Ghana/DILI-0007/2013						K	A				
A/Lviv/N6/2009	T		G								
A/Hong Kong/3934/2011	T		N							V	
A/Astrakhan/1/2011	T	K	XV			L					
A/Christchurch/16/2010	T		N		A						
A/St Petersburg/100/2011	T		R								
A/Hong Kong/5659/2012	T				L						
A/St Petersburg/27/2011	T		R								
A/Latvia/12046984/2013	T			I					E		
A/Ghana/FS-1090/2013	T	T		I					E		
A/Paris/2496/2013	T	K	G		I				E		
A/Gothenburg/5/2013 se	T					T			E		
A/Belgium/1460004/2014	T					T			E		
A/England/5/2014	T					T			E		
A/Kazakhstan/3320/2013	T					T			E		
A/South Africa/3626/2013	T					T			E		
A/Lisboa/10/2014	T					T			E		
A/England/3/2014	T					T			E		
A/Iceland/4/2014	T					T			E		
A/Denmark/106/2013	T	K				T	F		E		
A/Hong Kong/6019/2013	T					T		E			
A/Ireland/14M02257/2014	T					T			E		
A/Bayern/138/2013	T					T			E		
A/Sapporo/9/2014 niid	T					T			D		
A/Yamagata/158/2013 niid	T					T			D		
A/Athens GR/18/2014 grc	T					T			E		
A/Lyon/3058/2013	T					T			E		
A/Tehran/15381/2013	T		G			T			E		

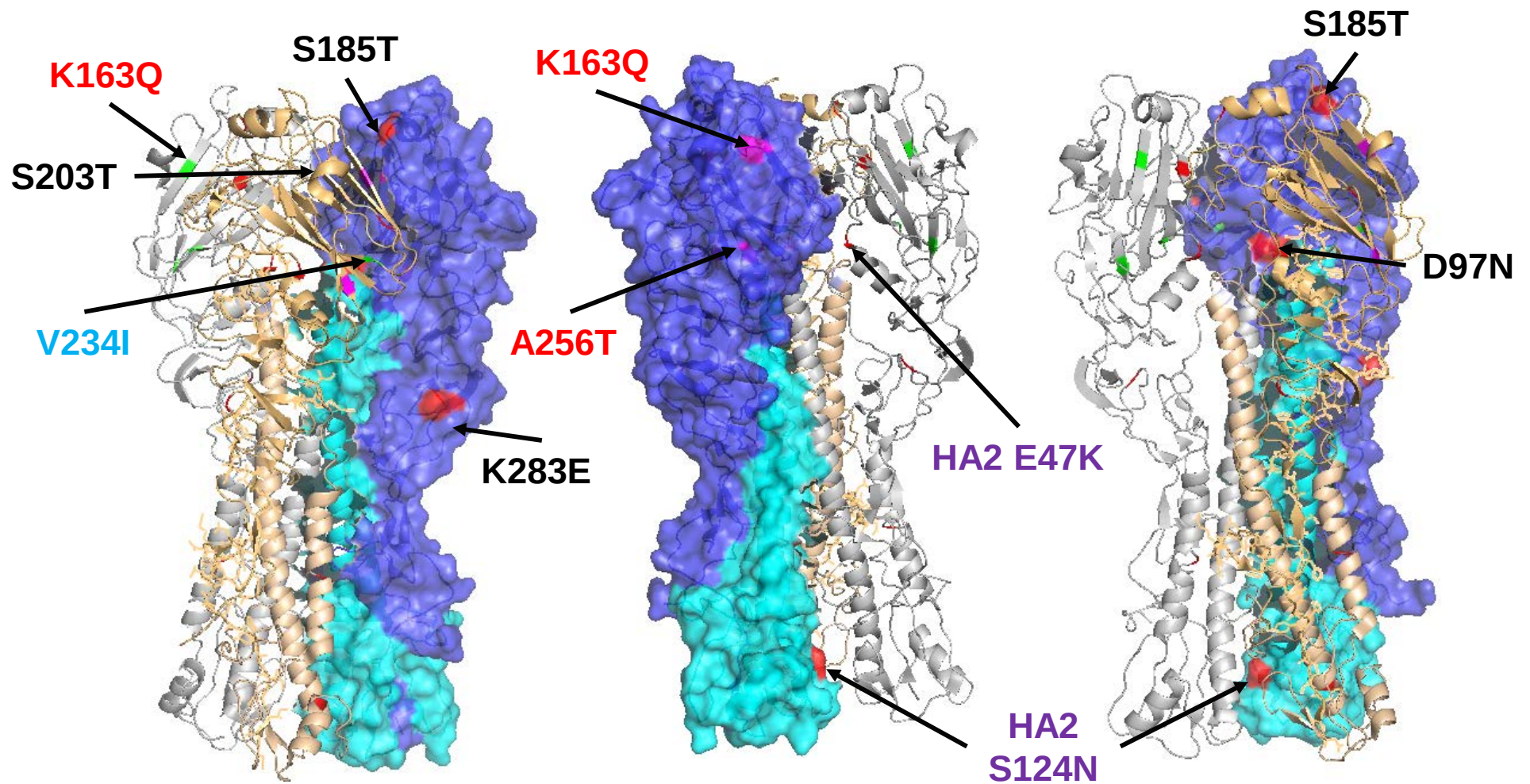
	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQSGSYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL									
A/Bayern/69/2009	V									
A/Ghana/DILI-0007/2013	V							K		
A/Lviv/N6/2009	V									
A/Hong Kong/3934/2011	V									
A/Astrakhan/1/2011	V							K		
A/Christchurch/16/2010	V							K		
A/St Petersburg/100/2011	V							K		
A/Hong Kong/5659/2012	V							K		
A/St Petersburg/27/2011	V							K		
A/Latvia/12046984/2013	V							K		
A/Ghana/FS-1090/2013	V							K		
A/Paris/2496/2013	V							K		
A/Gothenburg/5/2013 se	V						Q	K		
A/Belgium/1460004/2014	V							K		
A/England/5/2014	V							K		
A/Kazakhstan/3320/2013	V							K		
A/South Africa/3626/2013	V							K		
A/Lisboa/10/2014	V							K		
A/England/3/2014	V							K		
A/Iceland/4/2014	V							K		
A/Denmark/106/2013	V							K		
A/Hong Kong/6019/2013	V							K		
A/Ireland/14M02257/2014	V							K		
A/Bayern/138/2013	V							K		
A/Sapporo/9/2014 niid	V							K		
A/Yamagata/158/2013 niid	V							K	X	
A/Athens GR/18/2014 grc	V							K		
A/Lyon/3058/2013	V							K		
A/Tehran/15381/2013	V							K		

	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	EKRINLNKKVDDGFLDIWTYNAELLVLLLENERTLDYHDSNVKNLYEKVRSQKNNAKEIGNGCFEFYHKCDNTCMESVKN							NGTYDYPKYSEEAKLNREEI		
A/Bayern/69/2009										
A/Ghana/DILI-0007/2013								V		
A/Lviv/N6/2009										
A/Hong Kong/3934/2011									I	
A/Astrakhan/1/2011										
A/Christchurch/16/2010										
A/St Petersburg/100/2011					N					
A/Hong Kong/5659/2012					N					
A/St Petersburg/27/2011					N					
A/Latvia/12046984/2013					N					K
A/Ghana/FS-1090/2013					N					K
A/Paris/2496/2013					N			A	E	K
A/Gothenburg/5/2013 se					N					K
A/Belgium/1460004/2014					N				V	K
A/England/5/2014					N					K
A/Kazakhstan/3320/2013					N					K
A/South Africa/3626/2013					N					K
A/Lisboa/10/2014					N			M		K
A/England/3/2014					N			M		K
A/Iceland/4/2014					I	N		M		K
A/Denmark/106/2013					N					K
A/Hong Kong/6019/2013					N					K
A/Ireland/14M02257/2014					N					K
A/Bayern/138/2013					N					K
A/Sapporo/9/2014 niid					N					K
A/Yamagata/158/2013 niid					N					K
A/Athens GR/18/2014 grc					N					K
A/Lyon/3058/2013					N					K
A/Tehran/15381/2013					N					K

	510	520	530	540
A/California/7/2009	DGVKLESTRIYQILAIYSTVASSLLVLVSLGAISFWMCNS		NGSLQCRICI	
A/Bayern/69/2009				
A/Ghana/DILI-0007/2013		A		
A/Lviv/N6/2009				
A/Hong Kong/3934/2011				
A/Astrakhan/1/2011				
A/Christchurch/16/2010				
A/St Petersburg/100/2011				
A/Hong Kong/5659/2012				
A/St Petersburg/27/2011				
A/Latvia/12046984/2013		P		
A/Ghana/FS-1090/2013		V		
A/Paris/2496/2013				
A/Gothenburg/5/2013 se				
A/Belgium/1460004/2014				
A/England/5/2014				
A/Kazakhstan/3320/2013				
A/South Africa/3626/2013			V	
A/Lisboa/10/2014				
A/England/3/2014				
A/Iceland/4/2014				
A/Denmark/106/2013				
A/Hong Kong/6019/2013				
A/Ireland/14M02257/2014				
A/Bayern/138/2013				
A/Sapporo/9/2014 niid				
A/Yamagata/158/2013 niid				
A/Athens GR/18/2014 grc				
A/Lyon/3058/2013				
A/Tehran/15381/2013				

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 6. H1-HA locations of genetic group defining amino acid substitutions



HA1 substitutions D97N, S185T, S203T & K283E and HA2 substitutions E47K, S124N & E172K (not in structure) differentiate subgroup 6B and 6C viruses from A/California/7/2009. HA1 substitutions K163Q and A256T define subgroup 6B and V234I defines subgroup 6C.

0.002

	310	320	330	340	350	360	370	380	390	400						
A/California/7/2009	RPWVSFNQ	LEYGIGYICS	GF6DNP	RPNDKTG	SCGPVSS	NGANGV	KGFSKY	GNGVW	IGRTKSI	SSRN	GFEM	IWDPN	GWGTG	TDNNFS	IKQDIV	GINEWS
A/Bayern/69/2009														TK		
A/Ghana/DILI-0007/2013																
A/Lviv/N6/2009																
A/Hong Kong/3934/2011		R													I	
A/Astrakhan/1/2011									K							
A/Christchurch/16/2010										I						
A/St Petersburg/100/2011							I		K							
A/Hong Kong/5659/2012									K					S		
A/St Petersburg/27/2011									K							
A/Latvia/12046984/2013									K							
A/Ghana/FS-1090/2013									K							
A/Paris/2496/2013			H						K							
A/Gothenburg/5/2013 swe			V						K							K
A/Belgium/14G0004/2014			V						K							K
A/England/5/2014			V				I		K							K
A/Kazakhstan/3320/2013			V						K							K
A/South Africa/3626/2013			V						K							
A/Lisboa/10/2014			V						K						K	
A/England/3/2014			V						K						K	
A/Iceland/4/2014			V						K						K	
A/Denmark/106/2013			V						K							
A/Hong Kong/6019/2013			V						K							
A/Ireland/14M02257/2014			V						K							
A/Bayern/138/2013			V						K							
A/Sapporo/9/2014 niid			V						K						K	
A/Yamagata/158/2013 niid			V						K						K	
A/Athens GR/18/2014 grc			V						K						K	
A/Lyon/3058/2013			V				S		K						K	
A/Tehran/15381/2013			V						K						K	

	410	420	430	440	450	460
A/California/7/2009	GYSGSFVQHPELT	GLDCIRPCFW	VELIRGRPK	ENTIWTSG	SSISFCGVNS	DTVGSWPDGAELPFTIDK
A/Bayern/69/2009						
A/Ghana/DILI-0007/2013						
A/Lviv/N6/2009						
A/Hong Kong/3934/2011						V
A/Astrakhan/1/2011						
A/Christchurch/16/2010						
A/St Petersburg/100/2011						
A/Hong Kong/5659/2012					G	
A/St Petersburg/27/2011						
A/Latvia/12046984/2013						
A/Ghana/FS-1090/2013						
A/Paris/2496/2013						
A/Gothenburg/5/2013 swe			E			
A/Belgium/14G0004/2014			E		F	
A/England/5/2014			E	K		
A/Kazakhstan/3320/2013			E			
A/South Africa/3626/2013			E			
A/Lisboa/10/2014			E		K	
A/England/3/2014			E		K	
A/Iceland/4/2014			E		K	
A/Denmark/106/2013			E		G	
A/Hong Kong/6019/2013			E			
A/Ireland/14M02257/2014			E			
A/Bayern/138/2013			E			
A/Sapporo/9/2014 niid			E			
A/Yamagata/158/2013 niid			E			
A/Athens GR/18/2014 grc			E			
A/Lyon/3058/2013		R			E	
A/Tehran/15381/2013			E			

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 8. Summary of influenza A(H3N2) samples analysed, collected after 2013-08-31

MONTH	H3N2				
Country	Number received	Number propagated ¹	≤2 fold ³	4 fold ³	≥8 fold ³
2013					
SEPTEMBER					
France	2	2			2
Ghana	7	6			6
Sweden	2	2			2
United Kingdom	1	1			1
OCTOBER					
Cote d'Ivoire	3	in process			
Denmark	1	0			
France	1	1			1
Ghana	4	1			1
Iran	4	4			4
Israel	2	in process			
Netherlands	2	2			2
Norway	6	6			6
Turkey	2	in process			
United Kingdom	1	1			1
NOVEMBER					
Cote d'Ivoire	11	in process			
Denmark	1	1			1
France	3	3			3
Germany	2	2			2
Ghana	1	1			1
Italy	2	2		1	1
Kazakhstan	2	1			1
Norway	2	2			2
Poland	1	0			
Portugal	2	0			
Romania	1	in process			
Spain	4	4			4
Sweden	2	2			2
Tunisia	1	0			
Turkey	17	in process			
DECEMBER					
Algeria	3	in process			
Belgium	5	5			5
Cote d'Ivoire	18	in process			
Croatia	3	3			3
Denmark	2	2			2
Finland	3	3			3
France	14	14			14
Germany	3	3			3
Hong Kong	6	6			6
Ireland	2	2			2
Israel	1	1			1
Italy	14	14		5	9
Kazakhstan	6	6			6
Latvia	1	1			1
Madagascar	3	in process			
Moldova	1	1			1
Netherlands	1	1			1
Norway	6	6			6
Portugal	3	2			2
Russia	3	3		1	2
Slovenia	4	4			4
Spain	12	10		4	6
Sweden	4	4	1	1	2
Switzerland	2	in process			
Turkey	53	in process		2	14
Ukraine	5	4			4
United Kingdom	2	2			2
2014					
JANUARY					
Algeria	1	in process			
Belgium	1	1			1
Cote d'Ivoire	1	in process			
Germany	10	10		1	9
Greece	3	in process			
Ireland	2	2			2
Italy	1	1			1
Madagascar	3	in process			
Moldova	1	1			1
Portugal	3	3			3
Romania	4	4			4
Slovenia	5	2		1	1
Spain	4	4		1	3
Switzerland	6	in process			
Turkey	46	in process		6	18
United Kingdom	2	2			2
	363	171	1	23	187
33 Countries			0%	11%	89%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against egg grown A/Texas/50/2012 vaccine virus

Table 9-1 . Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2013-09-18, 2013-09-20, 2013-09-25, 2013-10-15)

Viruses		Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post-infection ferret antisera									
				A/Perth 16/09 F35/11	A/Ala 5/10 F27/10	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 Egg F35/12	A/Berlin 93/11 T/C F17/12	A/Vic 361/11 F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F36/12	A/Samara 73/13 F24/13
Genetic group				5	3A	6	3C	3C.1	3C.1	3B	3C.1	3C.3	
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	640	160	160	320	160	640	320	640	320	320
A/Alabama/5/2010	5	2010-07-13	MK1/C1/SIAT2	40	80	160	80	80	320	160	320	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	160	160	640	320	320	1280	640	2560	1280	1280
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	1280	1280	2560	1280	2560	2560	2560	5120	1280
A/Victoria/361/2011	3C	2011-10-24	E3/E2	320	640	320	1280	2560	1280	640	320	5120	640
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	320	160	320	1280	640	1280	640	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	80	160	160	160	160	640	320	640	640	320
A/Athens/112/2012	3B	2012-02-01	SIAT8	80	160	320	160	160	640	320	640	640	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	640	1280	1280	1280	2560	1280	1280	2560	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	640	320	640	1280	640	1280	1280	1280
TEST VIRUSES													
A/Cameroon/13V-228/2013		2012-12-14	SIAT3	<	80	320	80	80	320	160	320	640	320
A/Cameroon/13V-229/2013		2012-12-19	SIAT2	40	160	320	160	160	640	320	640	640	640
A/Cameroon/13V-231/2013	3A	2012-12-20	SIAT2	<	80	320	160	80	320	160	320	640	320
A/Cameroon/13V-232/2013	3A	2012-12-24	SIAT2	40	160	640	320	160	640	160	1280	1280	640
A/Kazakstan/2969/2013		2013-01-30	SIAT3	<	40	160	40	80	320	160	320	320	320
A/Bretagne/995/2013		2013-02-18	MDCK1/SIAT1	40	160	160	160	160	640	320	640	640	640
A/Moldova/262/2013	3C.3	2013-02-27	SIAT2	<	80	160	80	160	640	320	320	640	640
A/Picardie/1274/2013		2013-03-09	MDCK1/SIAT1	80	160	160	160	160	640	320	640	640	640
A/Moldova/368/2013	3C.3	2013-03-14	SIAT2	<	40	160	160	160	640	320	320	640	640
A/Lorraine/1348/2013		2013-03-18	MDCK1/SIAT1	40	160	160	160	160	640	320	640	640	640
A/Lithuania/9696/2013		2013-03-28	SIAT3	40	80	320	160	160	640	320	640	640	640
A/Paris/1516/2013	3C.3	2013-03-30	MDCK1/SIAT1	<	80	160	80	160	320	320	320	640	640
A/Lithuania/9520/2013	3C.3	2013-04-02	SIAT3	<	40	160	80	80	640	160	320	640	1280
A/Moldova/440/2013	3C.3	2013-04-02	SIAT2	<	40	40	40	80	320	320	320	640	640
A/Picardie/1572/2013	3C.3	2013-04-11	MDCK1/SIAT1	40	80	160	80	160	640	320	640	640	640
A/Jiangxi-Yushui/1220/2013	3C.3	2013-04-17	C2/1/SIAT2	80	160	320	160	160	640	320	640	640	640
A/Hunan-Lusong/1532/2013	3C.3	2013-06-10	C1/1/SIAT2	40	160	160	160	160	640	320	640	640	320
A/Kazakstan/2999/2013		unknown	SIAT3	40	160	320	160	160	640	320	640	1280	1280
A/Kazakstan/3118/2013		unknown	SIAT2	80	160	320	160	320	1280	640	1280	1280	1280
A/Kazakstan/3121/2013	3C.3	unknown	SIAT2	40	160	320	160	160	640	320	640	640	640
A/Kazakstan/3197/2013		unknown	SIAT2	<	40	80	40	80	160	80	160	160	160
A/Kazakstan/3263/2013	3C.3	unknown	SIAT2	40	80	320	80	160	640	320	640	640	640

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 9-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2013-10-22)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹												
				Post-infection ferret antisera												
				A/Perth 16/09 F35/11	A/Ala 5/10 F27/10	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 Egg F35/12	A/Berlin 93/11 T/C F17/12	A/Vic 361/11 T/C F34/12	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F36/12	A/Samara 73/13 F24/13	A/Almaty 2958/13 F38/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13
				5	3A	6	3C	3C.1	3C.1	3B	3C.1	3C.3	3C.3	3C.3	3C.2	
REFERENCE VIRUSES																
A/Perth/16/2009		2009-07-04	E3/E3	1280	320	160	320	320	640	320	640	640	320	160	160	320
A/Alabama/5/2010	5	2010-07-13	MK1/C1/SIAT2	<	80	80	160	80	320	160	160	320	160	<	40	80
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	160	160	640	160	160	640	640	640	640	640	320	320	640
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	1280	2560	640	1280	1280	1280	5120	1280	1280	1280	640
A/Victoria/361/2011	3C	2011-10-24	E3/E2	1280	1280	320	1280	5120	2560	640	640	5120	640	2560	5120	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT6	160	160	320	320	320	1280	640	640	1280	1280	160	320	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	160	320	320	320	160	1280	640	1280	1280	640	ND	ND	ND
A/Athens/112/2012	3B	2012-02-01	SIAT6	80	160	320	320	160	640	320	640	640	320	<	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	640	640	640	640	2560	640	1280	5120	1280	1280	1280	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	640	320	320	1280	640	640	1280	1280	320	640	640
A/Almaty/2958/2013	3C.3	2013-01-27	E5	320	320	1280	640	640	1280	640	1280	5120	1280	2560	1280	1280
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5	320	640	640	640	640	1280	640	1280	5120	1280	2560	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5	320	640	1280	1280	640	1280	640	1280	5120	1280	1280	640	1280
TEST VIRUSES																
A/Mauritius/167/2013	3C.3	2013-03-11	MDCK2/SIAT1	80	160	320	160	160	640	640	1280	1280	1280	160	640	320
A/Mauritius/571/2013	3C.3	2013-07-03	MDCK2/SIAT1	80	160	320	160	160	640	640	1280	1280	1280	160	640	320
A/Mauritius/610/2013	3C.3	2013-07-15	MDCK2/SIAT1	40	160	320	160	320	640	640	1280	1280	1280	<	640	640
A/Mauritius/703/2013	3C.3	2013-08-19	MDCK2/SIAT1	40	160	320	160	160	640	640	640	1280	1280	<	320	320

1. < = <40, ND = Not Determined

Vaccine

Table 9-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2013-12-10)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret antisera									
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Berlin 93/11 T/C F17/12	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Almaty 2958/13 F38/13	A/HK 146/13 F40/13
					3A	6	3C.1	3C.1	3B	3C.1	3C.3	3C.3	3C.2
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	320	320	640	320	640	320	320	160	320
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	160	640	320	640	640	640	320	1280	320	640
A/Iowa/19/2010	6	2010-12-30	E3/E2	640	1280	2560	2560	2560	2560	1280	1280	1280	1280
A/Berlin/93/2011	3C.1	2011-12-07	NVD3/SIAT4	160	320	320	1280	640	640	320	1280	320	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	160	320	320	1280	640	1280	320	1280	160	640
A/Athens/112/2012	3B	2012-02-01	SIAT4	160	320	320	640	640	1280	320	640	320	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	2560	1280	2560	2560	1280	2560	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	320	640	320	1280	640	1280	640	1280	640	1280
A/Almaty/2958/2013	3C.3	2013-01-27	E5	320	1280	640	1280	640	1280	1280	1280	2560	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5	320	1280	1280	1280	640	1280	640	1280	1280	2560
TEST VIRUSES													
A/Ghana/DARI-111/2013	3C.3	2013-07-19	SIAT2/SIAT1	80	320	160	640	320	1280	320	640	160	320
A/Ghana/DILI-619/2013	3C.3	2013-08-14	SIAT1/SIAT1	80	320	160	640	320	640	320	640	160	320
A/Ghana/DILI-620/2013	3C.3	2013-08-14	SIAT1/SIAT1	80	320	160	640	320	1280	320	640	160	320
A/Ghana/DILI-694/2013	3C.3	2013-09-18	SAIT2/SIAT4	<	80	80	320	160	320	80	320	80	160
A/Ghana/DILI-701/2013	3C.3/NA-3A	2013-09-20	SIAT2/SIAT1	40	160	80	640	160	640	160	640	160	320
A/Ghana/FS-951/2013	3C.3	2013-09-20	SIAT4	<	80	80	640	160	320	80	320	80	160
A/Ghana/DILI-702/2013	3C.3	2013-09-23	SIAT1/SIAT3	40	160	160	640	320	640	320	640	160	320
A/Ghana/DILI-704/2013	3C.3/NA-3A	2013-09-24	SIAT1/SIAT2	<	160	160	640	320	640	160	640	160	320
A/Ghana/DILI-726/2013	3C.3	2013-09-24	SIAT1/SIAT3	<	160	160	640	320	640	320	640	160	320
A/Ghana/DILI-775/2013	3C.3	2013-10-08	SIAT2/SIAT1	<	160	160	640	320	640	160	640	160	320
A/Ghana/FS-1065/2013	3C.3/NA-3A	2013-11-05	SIAT1/SIAT1	<	640	320	640	640	640	320	1280	320	640

1. < = <40

Vaccine

Sequences in phylogenetic trees

Table 9-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-01-10)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret antisera									
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
				3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3	
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	1280	320	320	320	640	320	320	320	320	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	160	640	320	640	640	320	1280	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	2560	1280	1280	1280	1280	640	640	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	80	320	320	320	640	320	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	320	320	1280	320	1280	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	2560	2560	1280	2560	2560	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	640	640	640	1280	640	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	1280	640	1280	1280	1280	1280	640	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	1280	1280	1280	1280	640	1280	640	2560	1280
NIB-85 (A/Almaty/2958/2013)	3C.3		E5/E1	640	1280	1280	1280	1280	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Bordeaux/2456/2013	3C.2	2013-08-27	MDCK2/SIAT1	<	80	80	160	320	80	320	80	160	160
A/Toulouse/2460/2013		2013-08-29	MDCK2/SIAT1	160	320	320	640	640	320	1280	640	640	320
A/Lyon/37.477/2013	3C.3	2013-09-13	MDCK2/SIAT1	40	320	160	320	640	320	640	320	320	320
A/Norway/2916/2013	3C.3	2013-10-01	MDCK2/SIAT1	40	80	160	160	640	160	640	320	320	320
A/Norway/2920/2013	3C.2	2013-10-01	MDCK1/SIAT1	40	160	160	160	640	160	320	160	160	320
A/Norway/2940/2013	3C.3	2013-10-03	MDCK2/SIAT1	40	160	160	320	640	320	640	640	320	320
A/Norway/2938/2013	3C.3	2013-10-04	MDCK2/SIAT1	80	320	160	320	640	320	1280	640	320	320
A/Norway/2965/2013	3C.3	2013-10-08	MDCK2/SIAT1	80	320	160	320	640	160	640	320	320	320
A/Norway/2967/2013	3C.3	2013-10-10	MDCK2/SIAT1	40	160	160	320	640	160	640	320	320	320
A/England/620/2013	3C.3	2013-10-16	SIAT1/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Nimes/2658/2013	3C.3	2013-10-24	MDCK2/SIAT1	<	160	80	160	320	160	320	160	160	160
A/Clermont Ferrand/2731/2013		2013-11-02	MDCK2/SIAT1	40	160	160	320	320	160	640	320	320	320
A/Norway/3060/2013	3C.3	2013-11-06	MDCK1/SIAT1	40	160	80	320	640	320	640	320	320	320
A/Pais Vasco/1403/2013	3C.3	2013-11-08	SIAT1/SIAT1	80	320	160	320	1280	320	640	320	640	320
A/Baleares/1441/2013	3C.3	2013-11-19	SIAT1/SIAT1	80	320	160	320	640	320	640	320	320	320
A/Aix-en-Provence/2846/2013	3C.2	2013-11-22	MDCK2/SIAT1	40	320	160	320	640	320	640	640	320	320
A/Thuringen/123/2013	3C.3	2013-11-25	C2/SIAT1	<	160	80	160	320	160	320	80	160	80
A/Bayern/139/2013	3C.3	2013-11-26	C2/SIAT1	80	320	320	320	640	320	640	320	320	320
A/Navarra/1503/2013	3C.3	2013-11-27	SIAT1/SIAT1	80	320	160	640	640	320	640	320	320	320
A/Toulouse/2892/2013	3C.3	2013-11-28	MDCK2/SIAT1	<	80	80	160	320	80	320	80	160	80
A/Baleares/1498/2013	3C.3	2013-11-29	SIAT1/SIAT1	80	640	320	640	1280	320	1280	640	640	640
A/Soria/143/2013	3C.3	2013-12-03	SIAT1/SIAT1	<	320	320	320	640	320	320	320	320	320
A/Lyon/2909/2013		2013-12-05	MDCK2/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Lyon/2923/2013		2013-12-05	MDCK2/SIAT1	80	160	320	320	640	320	640	320	320	320
A/Baleares/1521/2013	3C.3	2013-12-07	SIAT1/SIAT1	80	320	160	320	640	160	640	320	320	320
A/Baleares/1520/2013	3C.3	2013-12-08	SIAT1/SIAT1	40	320	160	320	640	160	640	320	320	320
A/Navarra/1528/2013	3C.3	2013-12-10	SIAT1/SIAT1	80	320	160	640	1280	320	640	320	320	320

< = <40

Vaccine

Sequences in phylogenetic trees

Table 9-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-01-14, 2014-01-17)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret antisera									
				A/Perth 16/09	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Athens 112/12	A/Texas 50/12	A/Samara 73/13	A/Serbia NS-210/13	A/HK 146/13	NIB-85
				F35/11	F28/11	F15/11	T/C F11/13	F16/12	Egg F42/13	F24/13	F39/13	F40/13	F45/13
					3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	160	160	160	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT5	40	320	160	320	640	160	640	320	160	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	640	1280	1280	1280	1280	1280	640	640	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	80	320	160	640	640	320	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	160	640	1280	320	640	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	640	640	1280	1280	1280	1280	1280	640	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	640	320	640	1280	640	1280	640	640	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	640	640	1280	1280	1280	1280	640	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	1280	1280	1280	1280	640	1280	640	1280	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	1280	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Stockholm/17/2013	3C.2	2013-09-03	MDCK2/SIAT1	<	80	40	80	320	160	320	160	160	160
A/Stockholm/18/2013	3C.2	2013-09-04	MDCK2/SIAT1	<	<	<	<	80	40	80	40	40	40
A/England/617/2013	3C.3	2013-09-15	SIAT2/SIAT2	<	80	40	160	320	80	320	160	160	160
A/Netherlands/2244/2013	3C.3	2013-10-19	MDCK3/SIAT1	<	80	40	80	320	80	320	160	80	160
A/Netherlands/2247/2013	3C.3	2013-10-23	MDCK3/SIAT1	40	160	80	160	320	160	320	160	160	160
A/Stockholm/22/2013	3C.3	2013-11-09	MDCK1/SIAT1	<	80	80	160	320	160	320	160	160	160
A/Stockholm/30/2013	3C.2	2013-11-26	MDCK1/SIAT1	<	80	40	80	320	80	320	160	160	160
A/Bursa/177/2013	3C.2	2013-12-02	MDCK1/SIAT1	80	160	160	320	320	160	640	320	320	320
A/Stockholm/34/2013	3C.3	2013-12-04	MDCK3/SIAT1	<	160	80	160	640	160	320	320	160	320
A/Netherlands/2249/13	3C.3	2013-12-05	MDCK2/SIAT1	<	160	80	160	320	80	320	160	160	160
A/Stockholm/31/2013	3C.2	2013-12-08	MDCK1/SIAT1	40	160	80	160	640	160	640	320	320	320
A/Berlin/215/2013	3C.3	2013-12-10	C2/SIAT1	<	160	80	160	640	160	320	160	160	320
A/Stockholm/29/2013	3C.3	2013-12-11	MDCK1/SIAT1	320	640	320	640	2560	640	1280	640	640	640
A/Stockholm/36/2013	3C.3	2013-12-15	MDCK1/SIAT1	80	320	160	320	1280	320	640	640	320	640
A/Bursa/247/2013	3C.3	2013-12-16	MDCK1/SIAT1	<	160	80	160	320	160	320	160	160	160
A/Antalya/251/2013	3C.3	2013-12-16	MDCK1/SIAT1	40	160	80	320	320	160	320	160	160	160

1. < = <40

Vaccine

Sequences in phylogenetic trees

* HA substitutions N144S (-CHO), K160T (+CHO), P221L>P, N225D, Q311H

Table 9-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-01-21)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret antisera									
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
					3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	80	160	80	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	640	320	320	640	160	1280	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	1280	1280	2560	640	1280	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	80	320	160	640	640	160	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	160	640	640	160	640	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	640	1280	2560	1280	1280	1280	1280	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	640	320	1280	1280	320	1280	640	1280	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	640	1280	1280	1280	1280	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	2560	1280	640	1280	640	1280	640	2560	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	1280	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Paris/2031/2013	3C.3	2013-09-30	MDCK2/SIAT1	80	320	320	640	640	160	1280	320	320	320
A/Denmark/96/2013	3C.3	2013-11-03	SIAT4/SIAT1	40	160	40	320	640	80	640	160	320	320
A/Bursa/176/2013	3C.3	2013-12-02	MDCK1/SIAT2	40	160	160	320	640	160	640	320	320	320
A/Latvia/12019591/2013	3C.3	2013-12-10	MDCK2/SIAT1	40	160	160	320	640	160	1280	320	320	320
A/Bursa/246/2013	3C.2	2013-12-16	MDCK1/SIAT2	40	160	160	320	640	160	640	320	320	320
A/Ukraine/710/2013	3C.3	2013-12-16	SIAT2/SIAT1	40	160	80	320	640	160	640	320	320	160
A/Centre/2484/2013	3C.3	2013-12-17	MDCK2/SIAT1	80	320	320	640	1280	320	1280	640	640	320
A/Lorraine/2485/2013	3C.2	2013-12-17	MDCK1/SIAT1	40	160	160	320	640	160	640	320	320	160
A/Provence-Alpes-Côte d'Azur/2501/2013	3C.2	2013-12-18	MDCK2/SIAT1	80	320	160	320	640	160	640	320	320	320
A/Denmark/104/2013	3C.2	2013-12-23	SIAT2/SIAT1	80	320	320	320	320	160	640	320	320	320
A/Ukraine/6004/2013	3C.3	2013-12-24	C1/SIAT1	40	160	80	160	320	80	320	160	160	160
A/Ukraine/728/2013	3C.3	2013-12-25	SIAT1/SIAT1	80	160	160	640	640	160	1280	320	320	320
A/Hong Kong/23/2014	3C.3	2013-12-26	MDCK1/SIAT1	<	160	80	320	320	80	640	160	160	160
A/Hong Kong/4/2014	3C.2	2013-12-26	MDCK1/SIAT1	40	160	160	320	320	160	640	320	320	320
A/Hong Kong/12/2014	3C.2	2013-12-28	MDCK1/SIAT1	40	160	80	160	320	80	640	160	160	160
A/Hong Kong/32/2014	3C.3	2013-12-29	MDCK1/SIAT1	40	160	80	160	320	80	640	160	160	160
A/Hong Kong/19/2014	3C.3	2013-12-29	MDCK1/SIAT1	40	160	160	320	640	160	640	320	320	160
A/Hong Kong/33/2014	3C.3	2013-12-30	MDCK1/SIAT1	40	160	80	320	640	160	640	320	320	320
A/Denmark/103/2013	3C.3	2013-12-30	SIAT2/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Lorraine/004/2014	3C.2	2013-12-30	MDCK1/SIAT1	40	320	160	320	640	160	640	320	320	320
A/Bretagne/014/2014	3C.3	2013-12-30	MDCK1/SIAT1	80	320	160	320	640	160	640	320	320	160

1. < = <40

Vaccine

Table 9-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-01-24)

Viruses		Collection Date	Passage History	Haemagglutination inhibition titre ¹										
				Post infection ferret antisera										
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Texas 50/12 Egg F36/12	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
Genetic group				3A	6	3C.1	3B	3C.1	3C.1	3C.3	3C.3	3C.2	3C.3	
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	160	320	160	80	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	640	320	320	640	320	2560	1280	320	320	320
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	1280	1280	2560	1280	5120	1280	640	1280	640
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	80	320	160	640	640	320	1280	640	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	160	640	640	320	1280	640	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	1280	5120	1280	1280	1280	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	640	320	1280	1280	320	5120	1280	640	1280	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	640	1280	1280	1280	5120	1280	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	2560	1280	640	1280	640	5120	1280	640	2560	640
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	1280	1280	5120	1280	1280	1280	1280
TEST VIRUSES														
A/Kazakhstan/3372/2013	3C.3	unknown	SIAT2	<	160	80	160	640	80	640	640	160	160	320
A/Kazakhstan/3291/2013	3C.3	2013-11-12	SIAT2	<	160	80	320	640	80	640	640	160	320	160
A/Belguim/13G1319/2013	3C.3	2013-12-12	SIAT2	<	160	160	320	640	160	640	640	320	320	320
A/Kazakhstan/3371/2013	3C.3	2013-12-15	SIAT2	<	160	80	320	640	160	640	640	320	320	320
A/Belguim/13G1334/2013	3C.3	2013-12-16	SIAT2	<	80	40	320	640	80	640	640	320	160	320
A/Moldova/696/2013	3C.3	2013-12-17	MDCK2/SIAT1	40	160	160	320	640	160	640	640	320	320	320
A/Kazakhstan/3281/2013	3C.3	2013-12-18	SIAT2	<	80	80	320	640	80	640	640	160	160	160
A/Belguim/13G1326/2013	3C.2	2013-12-19	SIAT2	40	160	80	320	320	80	640	640	160	320	160
A/Kazakhstan/3319/2013	3C.3	2013-12-23	SIAT2	<	160	80	320	640	160	640	640	160	320	160
A/Kazakhstan/3318/2013	3C.3	2013-12-23	SIAT2	<	160	80	320	640	80	640	640	160	160	320
A/Belguim/14G0001/2014	3C.2	2014-01-03	SIAT2	40	160	160	320	640	80	640	640	160	320	320

1. < = <40

Vaccine

Table 9-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-01-29)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
				Post infection ferret antisera										
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13	
REFERENCE VIRUSES					3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3	
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	160	160	80	160	160	
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	640	160	320	640	160	1280	320	320	320	
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	1280	1280	1280	640	1280	640	640	640	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	160	320	320	640	1280	320	1280	320	320	320	
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	160	640	1280	160	640	320	320	320	
A/Texas/50/2012	3C.1	2012-04-15	ES/E2	640	1280	1280	1280	2560	1280	1280	1280	1280	1280	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	320	160	640	640	320	1280	320	640	640	
A/Serbia/NS-210/2013	3C.3	2013-01-18	ES/E1	320	1280	1280	1280	2560	1280	1280	1280	1280	1280	
A/Hong Kong/146/2013	3C.2	2013-01-11	ES/E1	320	1280	1280	1280	1280	640	2560	640	2560	1280	
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	ES/E1	640	2560	2560	2560	2560	2560	2560	2560	2560	2560	
TEST VIRUSES														
A/Tehran/58191/2013	3C.3	2013-10-23	MDCK3/SIAT1	40	160	80	320	640	160	640	320	320	320	
A/Tehran/63139/2013	3C.3	2013-10-27	MDCK3/SIAT1	<	160	80	320	320	160	640	320	320	320	
A/Tehran/63138/2013		2013-10-27	MDCK3/SIAT1	<	160	80	320	320	160	640	160	160	320	
A/Tehran/63151/2013	3C.3	2013-10-29	MDCK3/SIAT1	<	80	40	160	320	80	320	160	160	160	
A/Norway/3105/2013	3C.2	2013-11-11	MDCK2/SIAT1	<	80	40	80	320	40	160	40	80	80	
A/Norway/3265/2013		2013-12-04	MDCK2/SIAT1	40	160	80	320	640	160	640	160	320	320	
A/Norway/3266/2013	3C.3	2013-12-05	MDCK1/SIAT1	<	80	40	160	320	160	320	160	160	320	
A/Lyon/2959/2013	3C.2	2013-12-07	MDCK2/SIAT1	<	80	40	160	320	80	320	80	160	160	
A/England/650/2013		2013-12-09	MDCK2/SIAT1	40	160	80	320	640	160	640	320	320	320	
A/Finland/385/2013		2013-12-11	SIAT1/SIAT1	<	160	80	160	320	160	640	160	160	160	
A/Ireland/14M00263/2013		2013-12-13	SIAT3	<	80	80	160	320	80	320	160	160	160	
A/Slovenia/2527/2013		2013-12-13	SIATx/SIAT1	<	160	80	320	640	160	640	320	160	160	
A/Slovenia/2528/2013	3C.3	2013-12-14	MDCKx/SIAT1	<	80	40	160	320	80	640	160	160	160	
A/Croatia/2561/2013	3C.3	2013-12-15	MDCK1/SIAT1	40	160	80	320	640	160	1280	320	320	320	
A/Lisboa/147/2013	3C.3	2013-12-15	SIAT3/SIAT1	<	160	80	160	320	160	320	160	160	160	
A/Finland/386/2013		2013-12-16	MDCK2/SIAT1	<	160	80	320	640	160	640	320	160	320	
A/Finland/386/2013		2013-12-16	MDCK1/SIAT1/SIAT1	<	160	80	320	640	160	640	320	320	320	
A/Croatia/2570/2013		2013-12-16	MDCK1/SIAT1	<	160	80	320	640	160	640	160	320	320	
A/Toulon/3038/2013	3C.2	2013-12-16	MDCK3/SIAT1	<	160	80	320	640	160	640	160	320	320	
A/Lyon/2990/2013		2013-12-16	MDCK2/SIAT1	<	80	80	160	320	80	320	160	160	160	
A/Lyon/3014/2013		2013-12-16	MDCK2/SIAT1	<	160	160	160	640	80	320	160	160	160	
A/Milano/59/2013	3C.3	2013-12-17	SIAT1/SIAT1	<	160	80	160	640	160	640	160	160	160	
A/Milano/60/2013		2013-12-17	SIAT1/SIAT1	<	80	40	160	320	80	320	160	160	160	
A/Lyon/3018/2013		2013-12-17	MDCK2/SIAT1	40	160	160	320	640	160	1280	320	320	320	
A/Norway/3290/2013	3C.2	2013-12-17	MDCK1/SIAT1	<	80	40	160	160	80	160	40	80	80	
A/Norway/3303/2013	3C.2	2013-12-17	MDCK1/SIAT1	<	80	40	80	160	80	160	80	80	80	
A/Norway/3325/2013		2013-12-17	MDCK1/SIAT1	<	80	40	160	320	80	320	160	160	160	
A/Pavia/154/2013	3C.3	2013-12-18	MDCK1/SIAT1	<	160	80	320	640	160	640	160	160	160	
A/England/651/2013	3C.3	2013-12-18	SIAT1/SIAT1	<	40	<	80	80	40	160	40	80	40	*
A/Lyon-chu/ 51.726/2013		2013-12-20	MDCK2/SIAT1	<	160	160	320	640	160	640	160	320	320	
A/Kazakhstan/3370/2013	3C.3	2013-12-24	SIAT3	<	80	40	160	320	80	320	80	80	160	
A/Norway/3314/2013	3C.3	2013-12-24	MDCK1/SIAT1	<	80	40	160	320	80	320	160	160	160	
A/Slovenia/2576/2013	3C.2	2013-12-25	SIATx/SIAT1	40	160	80	320	640	160	640	160	320	160	
A/Belgium/13G1346/2013	3C.3	2013-12-27	SIAT3	<	160	80	320	640	80	640	160	160	160	
A/Kazakhstan/3344/2013		2013-12-27	SIAT3	<	80	40	160	320	80	320	80	160	160	
A/Croatia/2624/2013	3C.3	2013-12-29	MDCK1/SIAT1	40	320	160	640	640	160	1280	320	320	320	
A/Parma/1/2014		2013-12-30	MDCK2/SIAT1	40	320	160	640	1280	320	1280	320	320	320	
A/Parma/2/2014		2013-12-30	MDCK2/SIAT1	<	80	40	160	320	80	320	160	80	160	
A/Limoges/3105/2013	3C.3	2013-12-30	Cx/MDCK1	<	80	80	160	320	80	640	160	160	160	
A/Lisboa/149/2013		2013-12-30	SIAT1/SIAT1	<	160	80	320	640	160	320	320	320	320	
A/Ireland/14M02879/2014	3C.3	2014-01-02	SIAT3	<	80	80	160	320	80	160	80	80	80	
A/England/2/2014	3C.3	2014-01-03	SIAT1/SIAT1	<	80	40	160	320	80	320	160	160	160	
A/England/1/2014	3C.3	2014-01-03	SIAT1/SIAT1	<	160	80	160	320	80	640	160	160	160	
A/Parma/3/2014	3C.3	2014-01-04	MDCK2/SIAT1	<	160	80	320	640	160	640	320	160	320	
A/Lisboa/1/2014	3C.3	2014-01-04	SIAT1/SIAT1	<	160	80	320	640	160	320	160	160	320	
A/Ireland/14M03520/2014	3C.3	2014-01-06	SIAT3	40	160	80	320	640	160	320	160	160	160	
A/Lisboa/2/2014	3C.3	2014-01-06	SIAT1/SIAT1	<	80	80	160	320	80	320	160	160	160	
A/Lisboa/3/2014		2014-01-06	SIAT1/SIAT1	<	160	80	320	320	80	160	160	160	160	
A/Romania/157360/2014	3C.2	2014-01-07	MDCK2/SIAT1	<	80	40	160	160	40	160	80	160	160	
A/Mures/157328/2014	3C.3	2014-01-07	MDCK2/SIAT1	<	160	80	160	320	160	640	320	160	320	
A/Maramures/157552/2014	3C.3	2014-01-07	MDCK1/SIAT1	<	80	40	160	320	80	320	80	80	80	
A/Iasi/157785/2014	3C.3	2014-01-13	MDCK1/SIAT1	<	160	80	320	320	80	640	160	160	160	
A/Slovenia/108/2014	3C.2	2014-01-14	SIATx/SIAT1	40	160	80	320	640	160	640	160	320	160	
A/Moldova/7/2014	3C.3	2014-01-15	SIAT2	<	80	80	160	320	80	320	80	80	80	

1. <= <40

Sequences in phylogenetic trees

* HA substitutions A138S, N144K (-CHO), S145N, F193Y, R307I, S312N, L98F

Vaccine

Table 9-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-01-31)

Haemagglutination inhibition titre ¹																	
Viruses	Genetic group	Collection Date	Passage History	Post infection ferret antisera													
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11 3A	A/Iowa 19/10 F15/11 6	A/Vic 361/11 T/C F11/13 3C.1	A/Athens 112/12 F16/12 3B	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/Serbia NS-210/13 F39/13 3C.3	A/HK 146/13 F40/13 3C.2	NIB-85 F45/13 3C.3	A/NY 39/12 Egg F1/14 3C.3	A/NY 39/12 T/C F2/14 3C.3		
REFERENCE VIRUSES																	
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	160	160	80	160	160	80	80		
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	1280	160	320	640	160	1280	320	320	320	320	320		
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	2560	1280	1280	640	1280	640	640	640	1280	640		
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	160	320	320	640	1280	320	1280	320	320	640	320	160		
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	160	640	640	160	640	320	320	320	320	160		
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	2560	1280	1280	1280	640	1280	1280	640		
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	80	320	160	640	640	320	1280	320	640	640	320	320		
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	1280	1280	1280	2560	1280	2560	1280	1280	1280	1280	640		
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	320	1280	1280	1280	1280	640	2560	640	2560	1280	1280	640		
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	2560	2560	2560	2560	2560	2560	2560	2560	2560	1280	640		
A/New York/39/2012	3C.3	2012-10-20	E4/E1	320	320	640	640	1280	320	1280	320	640	320	640	320		
A/New York/39/2012	3C.3	2012-10-20	C2/SIAT3	40	160	80	320	320	160	640	320	320	320	320	160		
TEST VIRUSES																	
A/Trieste/37/2013	3C.3	2013-11-26	MDCK3/SIAT1	80	320	160	640	1280	320	1280	640	640	320	320	320		
A/Pavia/145/2013	3C.3	2013-11-28	MDCK2/SIAT1	40	320	160	320	640	160	640	320	320	320	320	160		
A/Pavia/146/2013		2013-12-02	MDCK2/SIAT1	40	320	160	640	1280	320	1280	640	320	320	320	160		
A/Pavia/147/2013		2013-12-03	Cx/SIAT1	80	320	320	640	1280	320	1280	640	640	320	320	320		
A/Pavia/148/2013		2013-12-05	MDCK1/SIAT1	80	320	320	640	1280	320	1280	640	640	640	320	320		
A/Pavia/149/2013		2013-12-06	MDCK1/SIAT1	80	320	320	320	1280	320	1280	640	640	320	320	320		
A/Trieste/38/2013		2013-12-06	Cx/SIAT1	<	160	160	320	640	160	640	320	320	320	160	160		
A/Trieste/39/2013		2013-12-09	Cx/SIAT1	40	160	160	320	1280	160	640	320	320	320	320	160		
A/Ireland/14M00149/2013	3C.3	2013-12-10	SIAT4	<	160	80	160	320	160	640	160	160	320	160	80		
A/Trieste/40/2013		2013-12-10	Cx/SIAT1	40	320	320	640	1280	320	1280	320	320	320	320	160		
A/Navarra/1565/2013	3C.3	2013-12-10	SIAT1/SIAT1	<	160	160	320	640	160	640	320	320	320	160	160		
A/Belgium/13G1308/2013	3C.2	2013-12-10	SIAT4	80	320	160	320	640	160	640	320	320	320	320	160		
A/Aragon/1548/2013		2013-12-11	SIAT1/SIAT1	40	320	320	640	1280	320	1280	640	640	640	320	320		
A/Baleares/1582/2013		2013-12-11	SIAT1/SIAT1	<	320	160	320	640	160	640	320	320	320	160	160		
A/Navarra/1552/2013		2013-12-12	SIAT1/SIAT1	40	320	160	640	1280	320	1280	640	640	640	320	320		
A/Milano/58/2013		2013-12-13	SIAT1/SIAT1	<	160	160	320	1280	320	640	320	320	320	320	160		
A/Ukraine/6002/2013	3C.3	2013-12-14	SIAT3	<	160	80	160	640	160	640	320	320	320	160	80		
A/Slovenia/2529/2013	3C.3	2013-12-15	MDCKx/SIAT2	80	320	160	320	640	160	640	320	320	640	320	160		
A/Trieste/41/2013	3C.3	2013-12-16	Cx/SIAT1	<	160	80	160	640	160	640	160	160	160	160	80		
A/Moscow/137/2013	3C.3	2013-12-24	MDCK2/SIAT1	<	320	80	320	640	160	640	320	320	320	320	160		
A/Israel/32/2013	3C.3	2013-12-27	C3/SIAT1	<	320	80	320	640	160	640	320	320	320	160	160		
A/Moscow/138/2013	3C.2	2013-12-27	MDCK2/SIAT1	160	320	320	640	1280	320	1280	640	640	640	320	160		
A/Baleares/14038/2013		2013-12-27	SIAT1/SIAT1	80	320	320	640	1280	320	1280	640	640	320	320	160		
A/Baleares/14037/2013	3C.3	2013-12-29	SIAT1/SIAT1	40	320	320	640	1280	320	1280	320	320	320	320	160		
A/Moscow/140/2013	3C.3	2013-12-30	MDCK2/SIAT1	40	320	160	320	640	160	640	320	320	320	160	160		
A/Slovenia/17/2014	3C.2	2014-01-03	MDCKx/SIAT2	80	320	160	320	640	320	640	640	640	640	320	320		
A/Erzurum/99/2014	3C.3	2014-01-08	SIAT1/SIAT1	<	320	160	320	640	160	640	320	320	320	320	160		
A/Adana/111/2014	3C.2	2014-01-08	SIAT1/SIAT1	80	320	320	320	640	320	640	320	320	640	320	160		
A/Adana/116/2014	3C.3	2014-01-08	SIAT1/SIAT1	<	160	80	320	640	160	640	320	320	320	160	80		
A/Adana/117/2014	3C.3	2014-01-08	SIAT1/SIAT1	<	160	80	320	640	160	640	320	320	320	320	160		
A/Samsun/125/2014	3C.3	2014-01-08	SIAT1/SIAT1	40	320	160	320	640	160	640	320	320	320	320	160		
A/Aragon/271/2014		2014-01-08	SIAT1/SIAT1	40	320	160	640	1280	320	1280	640	640	640	320	320		
A/Trabzon/163/2014	3C.2	2014-01-09	SIAT1/SIAT1	80	320	320	320	640	320	640	320	320	320	320	320		
A/Samsun/177/2014	3C.3	2014-01-09	SIAT1/SIAT1	<	160	160	320	640	160	640	320	320	320	160	160		
A/Navarra/218/2014	3C.3	2014-01-09	SIAT1/SIAT1	<	160	80	320	640	160	640	160	160	320	160	80		
A/Aragon/266/2014	3C.3	2014-01-09	SIAT1/SIAT1	<	320	160	320	640	160	640	320	320	320	320	160		
A/Adana/199/2014	3C.2	2014-01-10	SIAT1/SIAT1	40	320	160	320	640	160	640	320	320	320	320	160		
A/Diyarbakır/211/2014	3C.2	2014-01-10	SIAT1/SIAT1	40	160	160	320	640	160	640	320	320	320	320	160		
A/Ankara/221/2014	3C.3	2014-01-10	SIAT1/SIAT1	<	160	160	320	640	160	640	320	320	320	160	160		
A/Adana/223/2014	3C.2	2014-01-10	SIAT1/SIAT1	40	160	80	320	640	160	640	320	320	320	160	160		
A/Castilla La Mancha/155/2014	3C.2	2014-01-10	SIAT1/SIAT1	80	320	160	320	640	160	640	320	640	320	320	320		

1. < = <40

Vaccine

Table 9-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-02-04

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret antisera									
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13 F45/13
				3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	160	160	80	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	320	160	320	640	160	640	160	320	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	2560	1280	2560	1280	1280	640	640	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	160	320	320	640	1280	320	1280	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	320	640	1280	320	640	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	2560	1280	1280	1280	1280	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	640	1280	320	1280	640	1280	640
A/Serbia/NS210/2013	3C.3	2013-01-18	E5/E1	320	640	1280	1280	1280	1280	1280	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	2560	2560	1280	2560	320	2560	640	2560	1280
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	2560	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Niedersachsen/113/2013		2013-12-19	C4/SIAT1	40	160	160	320	640	160	640	320	320	320
A/Berlin/216/2013		2013-12-28	C2/SIAT1	<	80	80	160	320	80	320	160	160	160
A/Ankara/4247/2013	3C.3	2013-12-31	SIAT1/SIAT1	40	320	160	640	640	320	640	320	320	320
A/Erzurum/6/2014	3C.3	2014-01-02	SIAT1/SIAT1	40	320	160	320	640	320	640	320	320	320
A/Adana/9/2014		2014-01-02	SIAT1/SIAT1	40	320	160	320	640	160	640	160	320	320
A/Adana/11/2014		2014-01-02	SIAT1/SIAT1	<	160	80	320	320	160	320	160	320	160
A/Trabzon/28/2014	3C.3	2014-01-03	SIAT1/SIAT1	<	160	80	160	640	160	320	160	160	160
A/Trabzon/31/2014		2014-01-03	SIAT1/SIAT1	<	160	80	320	320	80	320	80	160	160
A/Diyarbakir/65/2014	3C.3	2014-01-04	SIAT1/SIAT1	<	160	160	320	640	160	320	160	160	320
A/Ankara/5/2014		2014-01-06	SIAT1	40	320	320	320	640	320	640	320	320	320
A/Adana/66/2014	3C.3	2014-01-06	SIAT1/SIAT1	<	80	80	160	640	160	320	160	160	160
A/Samsun/87/2014	3C.3	2014-01-07	SIAT1/SIAT1	40	160	80	160	320	160	320	160	320	320
A/Malatya/92/2014	3C.3	2014-01-07	SIAT1/SIAT1	<	160	160	320	640	160	320	320	320	320
A/Rheinland-Pfalz/1/2014	3C.3	2014-01-07	C3/SIAT1	<	160	80	320	640	160	320	160	160	160
A/Gumushane/150/2014	3C.3	2014-01-08	SIAT1/SIAT1	<	160	80	160	320	160	320	160	320	160
A/Gumushane/154/2014		2014-01-08	SIAT1/SIAT1	40	320	160	320	640	320	640	320	320	320
A/Ankara/23/2014		2014-01-08	SIAT1	80	320	320	640	640	320	640	320	320	320
A/Berlin/2/2014		2014-01-09	C2/SIAT1	40	160	160	640	320	80	320	320	320	160
A/Niedersachsen/1/2014	3C.3	2014-01-10	C2/SIAT1	<	160	80	320	640	160	320	320	160	320
A/Berlin/1/2014		2014-01-13	C2/SIAT1	<	160	80	160	320	160	320	160	160	160
A/Mecklenburg-Vorpommern/1/2014	3C.3	2014-01-13	C2/SIAT1	<	160	80	320	640	160	640	160	160	320
A/Baden-Württemberg/1/2014	3C.3	2014-01-13	C2/SIAT1	40	160	160	640	640	320	320	320	320	320
A/Rheinland-Pfalz/2/2014		2014-01-14	C2/SIAT1	<	160	80	320	640	160	640	320	320	320
A/Berlin/3/2014		2014-01-16	C2/SIAT1	<	80	80	320	320	160	320	160	160	160
A/Berlin/4/2014	3C.3	2014-01-20	C2/SIAT1	<	160	160	320	640	160	320	320	160	160
A/Bayern/1/2014	3C.3	2014-01-20	C2/SIAT1	<	160	80	320	640	160	320	160	160	160

1. < = <40

Vaccine

Table 9-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) (2014-02-07)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret antisera									
				A/Perth 16/09 F35/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 T/C F11/13	A/Athens 112/12 F16/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/Serbia NS-210/13 F39/13	A/HK 146/13 F40/13	NIB-85 F45/13
					3A	6	3C.1	3B	3C.1	3C.3	3C.3	3C.2	3C.3
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	160	160	80	160	160
A/Stockholm/18/2011	3A	2011-03-28	SIAT4	80	320	160	320	640	160	640	160	320	160
A/Iowa/19/2010	6	2010-12-30	E3/E2	320	1280	2560	1280	2560	1280	1280	640	640	1280
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	160	320	320	640	1280	320	1280	320	320	320
A/Athens/112/2012	3B	2012-02-01	SIAT4	80	320	320	640	1280	320	640	320	320	320
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	1280	1280	1280	1280	1280	640	1280	640	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	320	320	640	1280	320	1280	640	1280	640
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1	320	640	1280	1280	1280	1280	1280	1280	1280	1280
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1	640	2560	2560	1280	2560	320	2560	640	2560	1280
NIB-85 (A/Almaty/2958/2013)	3C.3	2013-01-27	E5/E1	640	1280	1280	1280	2560	1280	1280	1280	1280	1280
TEST VIRUSES													
A/Ankara/3937/2013		2013-12-10	SIAT1/MDCK2	<	40	40	80	160	40	160	40	40	40
A/Erzurum/4165/2013		2013-12-26	SIAT1/SIAT1	<	80	80	160	160	80	160	80	160	160
A/Ankara/4187/2013		2013-12-26	SIAT1/SIAT1	<	80	40	160	320	80	160	80	80	80
A/Samsun/4192/2013		2013-12-27	SIAT1/SIAT1	40	320	320	640	1280	320	640	320	320	320
A/Malatya/4202/2013		2013-12-27	SIAT1/SIAT1	<	80	80	320	320	80	160	80	160	160
A/Diyarbakir/4214/2013		2013-12-27	SIAT1/SIAT1	<	160	80	320	320	80	320	160	160	160
A/Erzurum/4221/2013		2013-12-27	SIAT1/SIAT1	40	320	160	640	1280	160	640	320	320	320
A/Ankara/4227/2013		2013-12-27	SIAT1/SIAT1	40	160	160	640	320	80	320	160	320	320
A/Cankiri/4224/2013		2013-12-28	SIAT1/SIAT1	80	320	320	640	640	160	640	320	320	320
A/Ankara/4236/2013		2013-12-30	SIAT1/SIAT1	<	80	80	160	320	80	160	80	80	80

1. < = <40

Vaccine

Table 10-1. Antigenic analyses of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) (2014-01-20)

Viruses	Genetic group	Collection Date	Passage History	plaques	Neutralisation titre ¹				
					Post infection ferret sera				
					A/Vic 361/11 T/C F11/13 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Serbia NS-210/13 Egg F39/13 3C.3	A/Samara 73/13 T/C F24/13 3C.3	A/HK 146/13 Egg F40/13 3C.2
REFERENCE VIRUSES									
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	single	320	160	160	160-320	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	single	320	640	320-640	320	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1		320	320-640	640	320	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2		160	80	80	160-320	160
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1		160	160	160	160-320	640
TEST VIRUSES									
A/Ghana/DARI-13-111/2013	3C.3	2013-07-19	SIAT2/SIAT1	single	160	160	160-320	320	160-320
A/Ghana/DILI-13-619/2013	3C.3	2013-08-14	SIAT1/SIAT1		160	160	160-320	320	160-320
A/Ghana/DILI-13-620/2013	3C.3	2013-08-14	SIAT1/SIAT1		80	80	160	160-320	80-160
A/Ghana/DILI-13-726/2013	3C.3	2013-09-24	SIAT1/SIAT3	single	80-160	80	160	160-320	80-160
A/Norway/2920/2013	3C.2	2013-10-01	MDCK1/SIAT1	single	160	80-160	160	320	160
A/Norway/2940/2013	3C.3	2013-10-03	MDCK2/SIAT1		160	80-160	160	160-320	80-160
A/Norway/2938/2013	3C.3	2013-10-04	MDCK2/SIAT1		160	160	160	320	160
A/Norway/2965/2013	3C.3	2013-10-08	MDCK2/SIAT1		160	160	160	160-320	160
A/England/620/2013	3C.3	2013-10-16	SIAT1/SIAT1		160	80	80-160	160-320	160
A/Norway/3060/2013	3C.3	2013-11-06	MDCK1/SIAT1		160	160	160-320	320	160
A/Baleares/1441/2013	3C.3	2013-11-19	SIAT1/SIAT1		160	80	80-160	160	80-160
A/Thuringen/123/2013	3C.3	2013-11-25	C2/SIAT1		80	80	80	160	80
A/Bayern/139/2013	3C.3	2013-11-26	C2/SIAT1		160	80-160	160	320	160
A/Baleares/1498/2013	3C.3	2013-11-29	SIAT1/SIAT1		80-160	80	160	160-320	80-160
A/Baleares/1521/2013	3C.3	2013-12-07	SIAT1/SIAT1	320	160	160-320	640	320	

1. 80% reduction

Vaccine

Sequences in phylogenetic trees

Table 10-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) (2014-01-30)

Viruses					Neutralisation titre ¹						
					Post infection ferret sera						
					A/Vic 361/11 T/C F11/13	A/Texas 50/12 Egg F42/13	A/Serbia NS-210/13 Egg F39/13	A/Samara 73/13 T/C F24/13	A/HK 146/13 Egg F42/13	A/NY 39/12 Egg F01/14	A/NY 39/12 T/C F02/14
Genetic group					3C.1	3C.1	3C.3	3C.3	3C.2	3C.3	3C.3
REFERENCE VIRUSES											
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	single	320	160	160	320	160	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2		320	640	320	320	320	320	320
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1		320	640	320-640	320	320	320	160
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	single	80	40	80	160-320	160	80	40
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1		160	160	160	320	640	160-320	80-160
A/New York/39/2012	3C.3	2012-10-20	E4/E1		160	160	320	320	320	160-320	160
A/New York/39/2012	3C.3	2012-10-20	C2/S3	single	40	<	40	80	<	80	40
TEST VIRUSES											
A/Lyon/37.477/2013	3C.3	2013-09-13	MDCK2/SIAT1		160-320	160	160-320	320	160	80	40
A/Norway/2916/2013	3C.3	2013-10-01	MDCK2/SIAT1	single	40	40	40	80	40	160	80-160
A/Norway/2967/2013	3C.3	2013-10-10	MDCK2/SIAT1	single	80	40	80	160	80	80	40-80
A/Nimes/2658/2013	3C.3	2013-10-24	MDCK2/SIAT1		80-160	80	80	160	80-160	40-80	40-80
A/Clermont Ferrand/2731/2013		2013-11-02	MDCK2/SIAT1		80-160	80	160	160-320	80-160	160-320	80-160
A/Pais Vasco/1403/2013	3C.3	2013-11-08	SIAT1/SIAT1		80-160	80	80	160-320	80-160	160	80
A/Aix-en-Provence/2846/2013	3C.2	2013-11-22	MDCK2/SIAT1		160-320	160	160	320	320	80-160	80
A/Navarra/1503/2013	3C.3	2013-11-27	SIAT1/SIAT1		80	40	40-80	80	40	40-80	40
A/Toulouse/2892/2013	3C.3	2013-11-28	MDCK2/SIAT1	single	40	40	40	80	40	40	40
A/Lyon/2909/2013		2013-12-05	MDCK2/SIAT1		80-160	80	80	160	80	160-320	80-160
A/Lyon/2923/2013		2013-12-05	MDCK2/SIAT1	single	40	40	40	80	40	80-160	80
A/Baleares/1520/2013	3C.3	2013-12-08	SIAT1/SIAT1		160-320	160	160	320	160	160	80
A/Navarra/1528/2013	3C.3	2013-12-10	SIAT1/SIAT1		160	80	160	160-320	160	160	80-160

1. 80% reduction; < = <40

Vaccine

Sequences in phylogenetic trees

Table 10-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) (2014-02-03)

					Neutralisation titre ¹						
					Post infection ferret sera						
Viruses		Collection Date	Passage History	plaques	A/Vic 361/11 T/C F11/13	A/Texas 50/12 Egg F42/13	A/Serbia NS-210/13 Egg F39/13	A/Samara 73/13 T/C F24/13	A/HK 146/13 Egg F42/13	A/NY 39/12 Egg F01/14	A/NY 39/12 T/C F02/14
	Genetic group				3C.1	3C.1	3C.3	3C.3	3C.2	3C.3	3C.3
REFERENCE VIRUSES											
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT6	single	320	160	160	160-320	160	ND	ND
A/Texas/50/2012	3C.1	2012-04-15	E5/E2		320	640	320-640	320	320	ND	ND
A/Serbia/NS-210/2013	3C.3	2013-01-18	E5/E1		320	320-640	640	320	320	ND	ND
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	single	160	80	80	160-320	160	ND	ND
A/Hong Kong/146/2013	3C.2	2013-01-11	E5/E1		160	160	160	160-320	640	ND	ND
A/New York/39/2012	3C.3	2012-10-20	E4/E1		160	160	320	320	320	160-320	160
A/New York/39/2012	3C.3	2012-10-20	C2/SIAT3	single	40	<	40	80	<	80	40
TEST VIRUSES											
A/Toulouse/2460/2013		2013-08-29	MDCK2/SIAT1		160	160	160	320	160-320	160	80-160
A/Stockholm/18/2013	3C.2	2013-09-04	MDCK2/SIAT1	single	<	<	<	<	<	<	<
A/England/617/2013	3C.3	2013-09-15	SIAT2/SIAT2		160	80	80	160	80-160	80	160
A/Paris/2031/2013	3C.3	2013-09-30	MDCK2/SIAT1		160	160	160	320	160	160	80-160
A/Netherlands/2247/2013	3C.3	2013-10-23	MDCK3/SIAT1	single	160	80	160	320	160	80-160	80
A/Soria/143/2013	3C.3	2013-12-03	SIAT1/SIAT1		160-320	160	160	320	160-320	160	80-160
A/Berlin/215/2013	3C.3	2013-12-10	C2/SIAT1		320	160	160	320-640	160-320	160	160
A/Latvia/12019591/2013	3C.3	2013-12-10	MDCK2/SIAT1		80-160	80	80	160	80	80	40
A/Stockholm/29/2013	3C.3	2013-12-11	MDCK1/SIAT1		320	160-320	320	640	640	160	160-320
A/Bursa/246/2013	3C.2	2013-12-16	MDCK1/SIAT2		320	160-320	320	640	640	320	160-320
A/Centre/2484/2013	3C.3	2013-12-17	MDCK2/SIAT1		160-320	160	160-320	320	160-320	160-320	160
A/Lorraine/2485/2013	3C.2	2013-12-17	MDCK1/SIAT1		320	160-320	160-320	320	320	160-320	160
A/Ukraine/728/2013	3C.3	2013-12-25	SIAT1/SIAT1		320	160	320	320	320	160	160
A/Hong Kong/32/2014	3C.3	2013-12-29	MDCK1/SIAT1		320	160	160	320	320	320	160
A/Lorraine/004/2014	3C.2	2013-12-30	MDCK1/SIAT1		160	160	160	160-320	160	160	160
A/Bretagne/014/2014	3C.3	2013-12-30	MDCK1/SIAT1		160-320	160	160	320	160	160	160

1. 80% reduction; < = <40

Vaccine

Sequences in phylogenetic trees

* HA substitutions N144S (-CHO), K160T (+CHO), P221L>P, N225D, Q311H

Figure 9. Phylogenetic comparison of influenza A(H3N2) HA genes

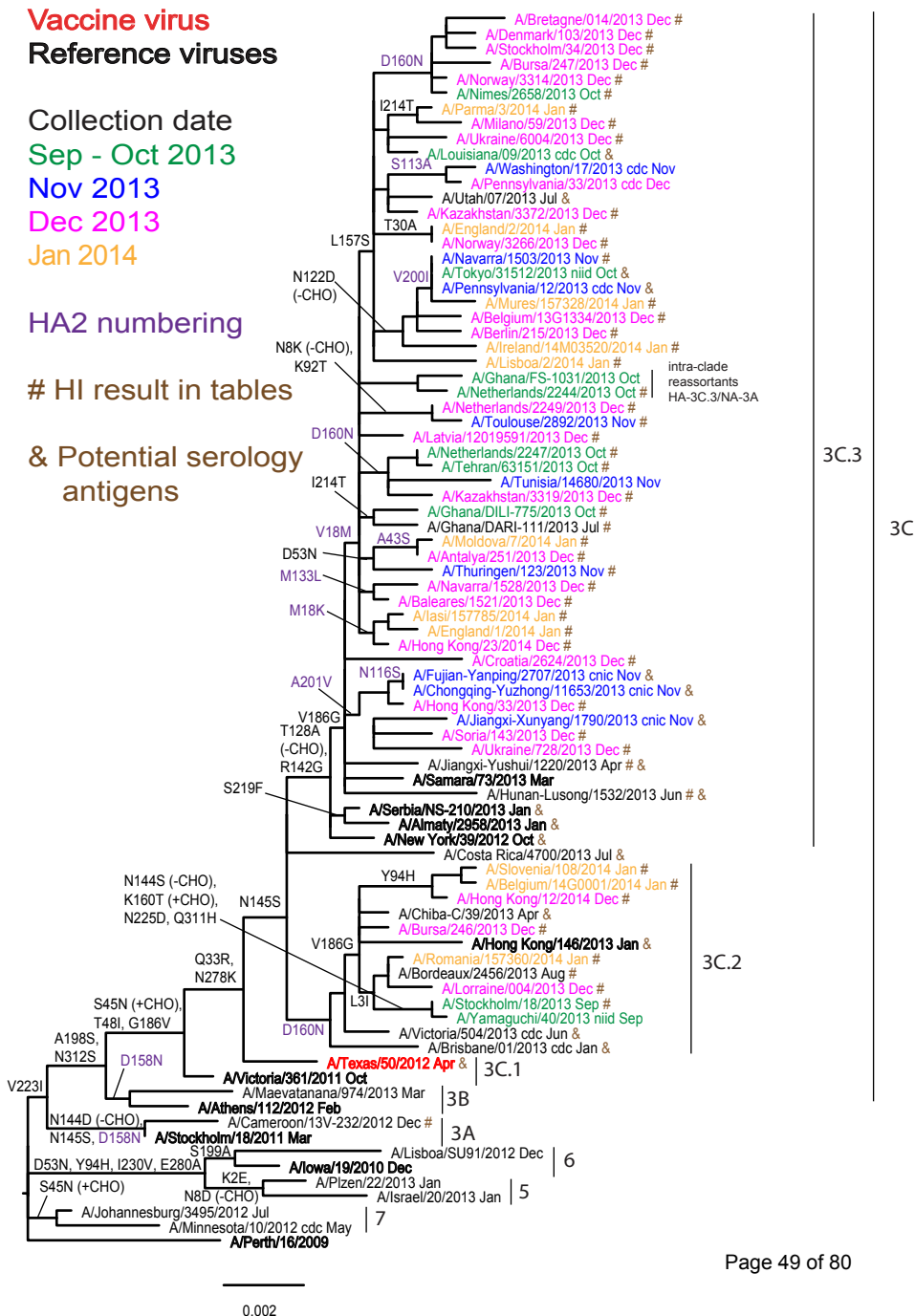


Figure 10. HA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100										
A/Perth/16/2009	QKLPGNDNST	ATLCLGHHA	VPPNGT	IVKTI	TNDQIE	VNTAT	ELVQSS	STGEIC	DSPHQ	ILDGKN	NCTL	IDALL	LGDPQ	CDGFQ	NKKK	WDLF	VERSK	AYSNC	YPY	
A/Iowa/19/2010							N				E							H		6
A/Stockholm/18/2011											E									3A
A/Athens/112/2012									R		E									3B
A/Victoria/361/2011							N	I			E									3C.1
A/Texas/50/2012					R		N	I			E									3C.1
A/Stockholm/18/2013							N	I			E									3C.2
A/Romania/157360/2014					R		N	I			E									3C.2
A/Hong Kong/146/2013							N	I			E									3C.2
A/Slovenia/108/2014					R		N	I			E						H			3C.2
A/Almaty/2958/2013							R	N	I		E									3C.3
A/Serbia/NS210/2013							R	N	I		E									3C.3
A/Samara/73/2013							R	N	I		E									3C.3
A/Soria/143/2013							R	N	I		E									3C.3
A/Hong Kong/33/2013							R	N	I		E									3C.3
A/England/1/2014							R	N	I		E									3C.3
A/Baleares/1521/2013							R	N	I		E									3C.3
A/Moldova/7/2014							R	N	I	N	E									3C.3
A/Kazakhstan/3319/2013							R	N	I		E									3C.3
A/Tunisia/14680/2013							R	N	I		E									3C.3
A/Toulouse/2892/2013		K			T		R	N	I	G	E							T		3C.3
A/Netherlands/2244/2013							R	N	I		E									3C.3
A/Ireland/14M03520/2014							R	N	I		E						R			3C.3
A/Mures/157328/2014							R	N	I		E						K			3C.3
A/Navarra/1503/2013							R	N	I		E									3C.3
A/Norway/3266/2013					A	R		N	I		E									3C.3
A/Milano/59/2013							R	N	I		E									3C.3
A/Parma/3/2014							R	N	I		E									3C.3
A/Norway/3314/2013							R	N	I		E									3C.3
A/Denmark/103/2013							R	N	I		E									3C.3

	110	120	130	140	150	160	170	180	190	200												
A/Perth/16/2009	DVPDYASLRSLV	ASSGTL	EFN	ESF	NW	TGVTQ	NGTSS	ACIR	RKSN	SNF	FSRLN	WTLH	NFKYP	ALNVT	MPN	NEQ	FDKLY	IWGL	HPGT	DKDQ	IFLYA	QASG
A/Iowa/19/2010							N												H	X	T	A
A/Stockholm/18/2011							DS												H			S
A/Athens/112/2012							NS												H			S
A/Victoria/361/2011							N				R								H	V		S
A/Texas/50/2012					N		N												H	V		P
A/Stockholm/18/2013							SS												H			S
A/Romania/157360/2014							NS												H			S
A/Hong Kong/146/2013							K	NS											H	W		S
A/Slovenia/108/2014							NS												H			S
A/Almaty/2958/2013					A		K	G	NS										H	V		S
A/Serbia/NS210/2013					A		G	NS											H	V		S
A/Samara/73/2013					A		K	G	NS										H	V		S
A/Soria/143/2013					A		G	NS											H			S
A/Hong Kong/33/2013					A		G	NS											H			S
A/England/1/2014					A		G	NS											H			S
A/Baleares/1521/2013					A		G	NS											H			S
A/Moldova/7/2014					A		G	NS											H			S
A/Kazakhstan/3319/2013					A		G	NS											H			S
A/Tunisia/14680/2013					K		A	G	NS										H			S
A/Toulouse/2892/2013							A	G	NS										H			S
A/Netherlands/2244/2013							A	G	NS										H			S
A/Ireland/14M03520/2014					D		A	G	NS				S						H			S
A/Mures/157328/2014					D		A	G	NS				S						H			S
A/Navarra/1503/2013					D		A	G	NS				S						H			S
A/Norway/3266/2013							A	G	NS				S						H			S
A/Milano/59/2013							A	G	NS				S						H			S
A/Parma/3/2014							A	G	NS				S						H			S
A/Norway/3314/2013							A	G	NS				S						H			S
A/Denmark/103/2013							A	G	NS				S						H			S

	210	220	230	240	250	260	270	280	290	300													
A/Perth/16/2009	RTTVSTKR	SQQT	VP	SNIGSR	PRVRN	IPSRIS	YTWI	VKPGD	ILL	INST	GNLI	APRGY	FKIR	SGKSS	IMRS	DAPIG	KCN	SECIT	PN	GSIP	NDK	PFQ	NVNR
A/Iowa/19/2010		A	I																				
A/Stockholm/18/2011					I																		
A/Athens/112/2012					I																		
A/Victoria/361/2011					I																		
A/Texas/50/2012					F	I																	
A/Stockholm/18/2013					X	I	D																
A/Romania/157360/2014					I																		
A/Hong Kong/146/2013					I																		
A/Slovenia/108/2014		K	A	I		I																	
A/Almaty/2958/2013					F	I																	
A/Serbia/NS210/2013					F	I																	
A/Samara/73/2013					I																		
A/Soria/143/2013					I																		
A/Hong Kong/33/2013					I																		
A/England/1/2014					I																		
A/Baleares/1521/2013					I								N	R									
A/Moldova/7/2014					I																		
A/Kazakhstan/3319/2013					I																		
A/Tunisia/14680/2013					I																		
A/Toulouse/2892/2013					I	D																	
A/Netherlands/2244/2013					I	D																	
A/Ireland/14M03520/2014					I																		
A/Mures/157328/2014					I									Q									
A/Navarra/1503/2013					I																		
A/Norway/3266/2013					I																		
A/Milano/59/2013					I																		
A/Parma/3/2014					I																		
A/Norway/3314/2013					I																		
A/Denmark/103/2013					I																		

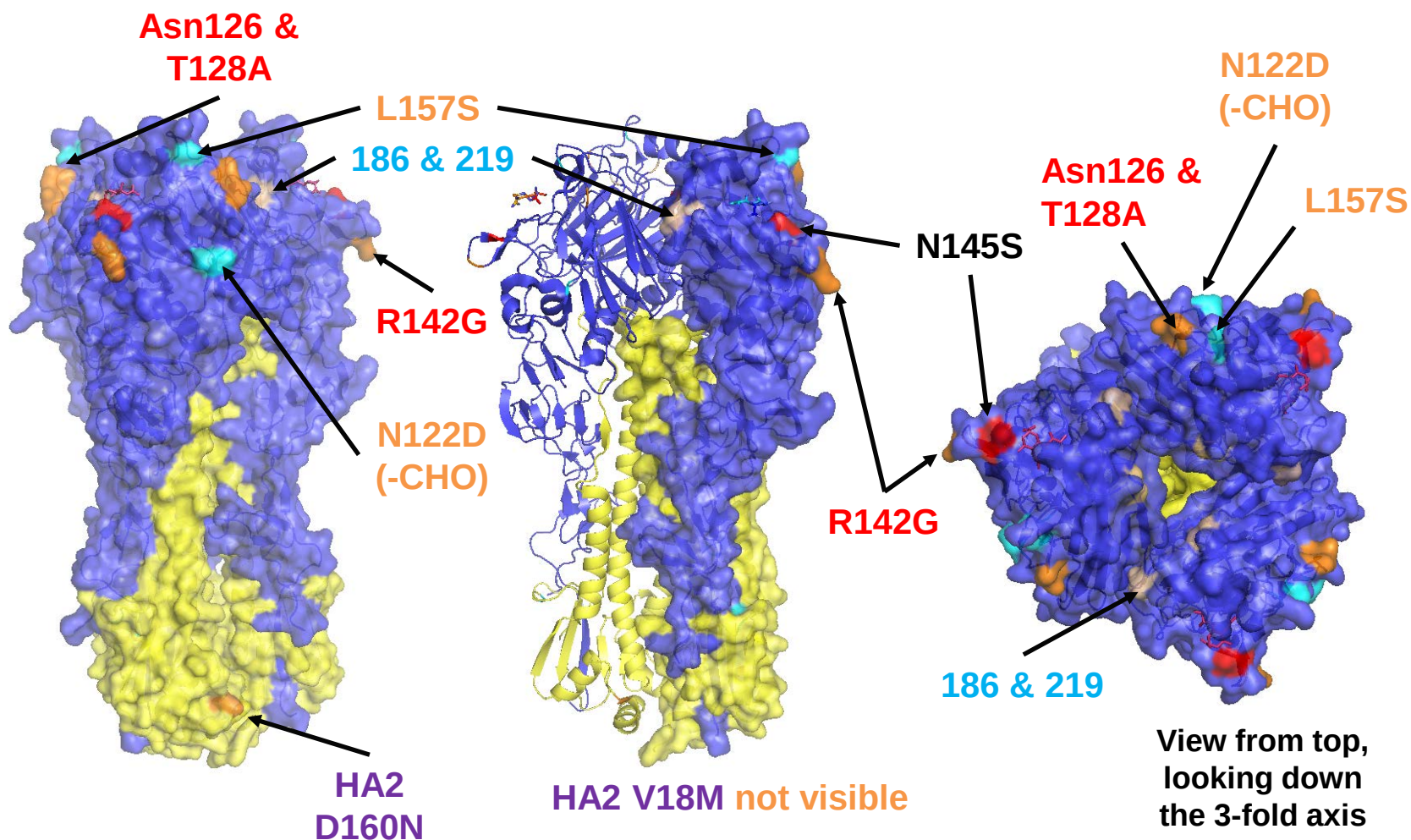
	310	320	330	340	350	360	370	380	390	400
A/Perth/16/2009	TYGACPRYVKQNTLKLATGMRNVPEKQTR	GI	FGA	IAGFIENGWEGMVDGWYGRHQN	SEGRGQAADLKSTQA	AI	DQINGKLNRLIG	TNEKFHQI	KEFS	
A/Iowa/19/2010										
A/Stockholm/18/2011										
A/Athens/112/2012			S							
A/Victoria/361/2011			S							
A/Texas/50/2012			S							
A/Stockholm/18/2013			HS							
A/Romania/157360/2014			S							
A/Hong Kong/146/2013										
A/Slovenia/108/2014			S							
A/Almaty/2958/2013			S			F				
A/Serbia/NS210/2013			S							
A/Samara/73/2013			S							
A/Soria/143/2013			S							
A/Hong Kong/33/2013			S							
A/England/1/2014			S			K				
A/Baleares/1521/2013			S			M				
A/Moldova/7/2014			S			M		S		
A/Kazakhstan/3319/2013			S			M				
A/Tunisia/14680/2013		F	S			M				
A/Toulouse/2892/2013			S			M				
A/Netherlands/2244/2013			S			M				
A/Ireland/14M03520/2014			S			K				
A/Mures/157328/2014			S			M		R		
A/Navarra/1503/2013			S			M				
A/Norway/3266/2013			S			M				
A/Milano/59/2013			S			M				
A/Parma/3/2014			S			M				
A/Norway/3314/2013			S			M				
A/Denmark/103/2013			S			M				

	410	420	430	440	450	460	470	480	490	500
A/Perth/16/2009	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLT	DSEMNKLF	EKKQLRENAEDMGNGCFKIYHKCDNACIGSIR	NGT	YDHDVYRDEALNNRF					
A/Iowa/19/2010										
A/Stockholm/18/2011									N	
A/Athens/112/2012									N	
A/Victoria/361/2011										
A/Texas/50/2012										
A/Stockholm/18/2013									N	
A/Romania/157360/2014									N	
A/Hong Kong/146/2013									N	
A/Slovenia/108/2014									N	
A/Almaty/2958/2013										
A/Serbia/NS210/2013										
A/Samara/73/2013										
A/Soria/143/2013										
A/Hong Kong/33/2013										
A/England/1/2014						L				
A/Baleares/1521/2013										
A/Moldova/7/2014										
A/Kazakhstan/3319/2013									N	
A/Tunisia/14680/2013									N	
A/Toulouse/2892/2013										
A/Netherlands/2244/2013										
A/Ireland/14M03520/2014										
A/Mures/157328/2014										
A/Navarra/1503/2013										
A/Norway/3266/2013										
A/Milano/59/2013	A									
A/Parma/3/2014										
A/Norway/3314/2013									N	
A/Denmark/103/2013									N	

	510	520	530	540	550
A/Perth/16/2009	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFMWACQGNIRNCICI				
A/Iowa/19/2010					
A/Stockholm/18/2011					
A/Athens/112/2012					
A/Victoria/361/2011					
A/Texas/50/2012					
A/Stockholm/18/2013					
A/Romania/157360/2014					
A/Hong Kong/146/2013					
A/Slovenia/108/2014					
A/Almaty/2958/2013					
A/Serbia/NS210/2013					
A/Samara/73/2013					
A/Soria/143/2013					V
A/Hong Kong/33/2013					V
A/England/1/2014					
A/Baleares/1521/2013					
A/Moldova/7/2014					
A/Kazakhstan/3319/2013					
A/Tunisia/14680/2013					
A/Toulouse/2892/2013					
A/Netherlands/2244/2013					
A/Ireland/14M03520/2014					I
A/Mures/157328/2014					I
A/Navarra/1503/2013					
A/Norway/3266/2013					
A/Milano/59/2013					
A/Parma/3/2014					
A/Norway/3314/2013					
A/Denmark/103/2013					

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 11. H3-HA locations of genetic group defining amino acid substitutions



The HA1 N145S substitution defines subgroups 3C.2 and 3C.3. The subgroups are defined by **T128A & R142G** substitutions (3C.3) and **HA2 D160N** (3C.2). Within subgroup 3C.3 clusters of viruses are defined by individual substitutions: **HA2 V18M** or **N122D (-CHO)** or **L157S**.

Figure 12. Phylogenetic comparison of influenza A(H3N2) NA genes

Vaccine virus

Reference viruses

Collection date

Sep - Oct 2013

Nov 2013

Dec 2013

Jan 2014

HI result in tables

& Potential serology
antigens

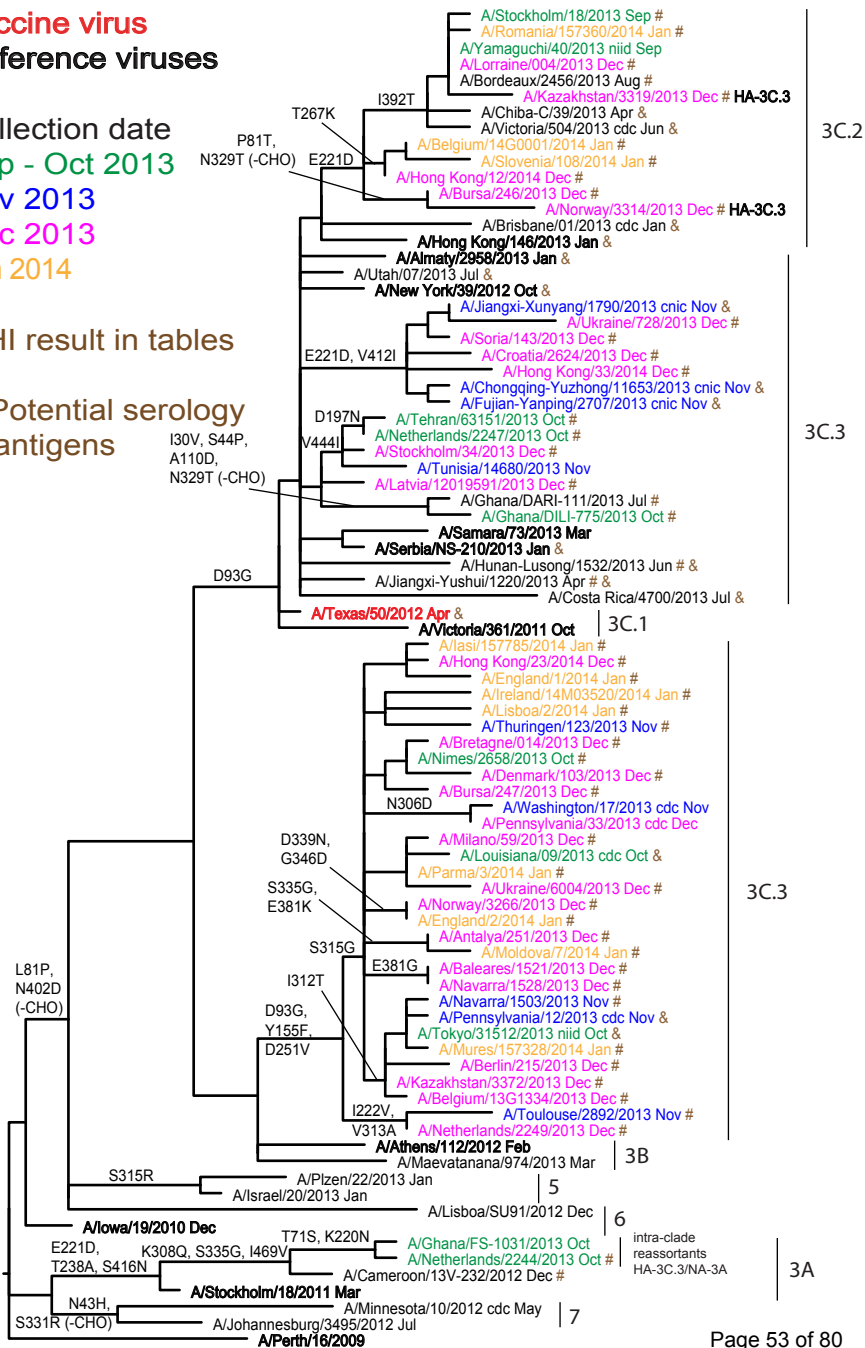
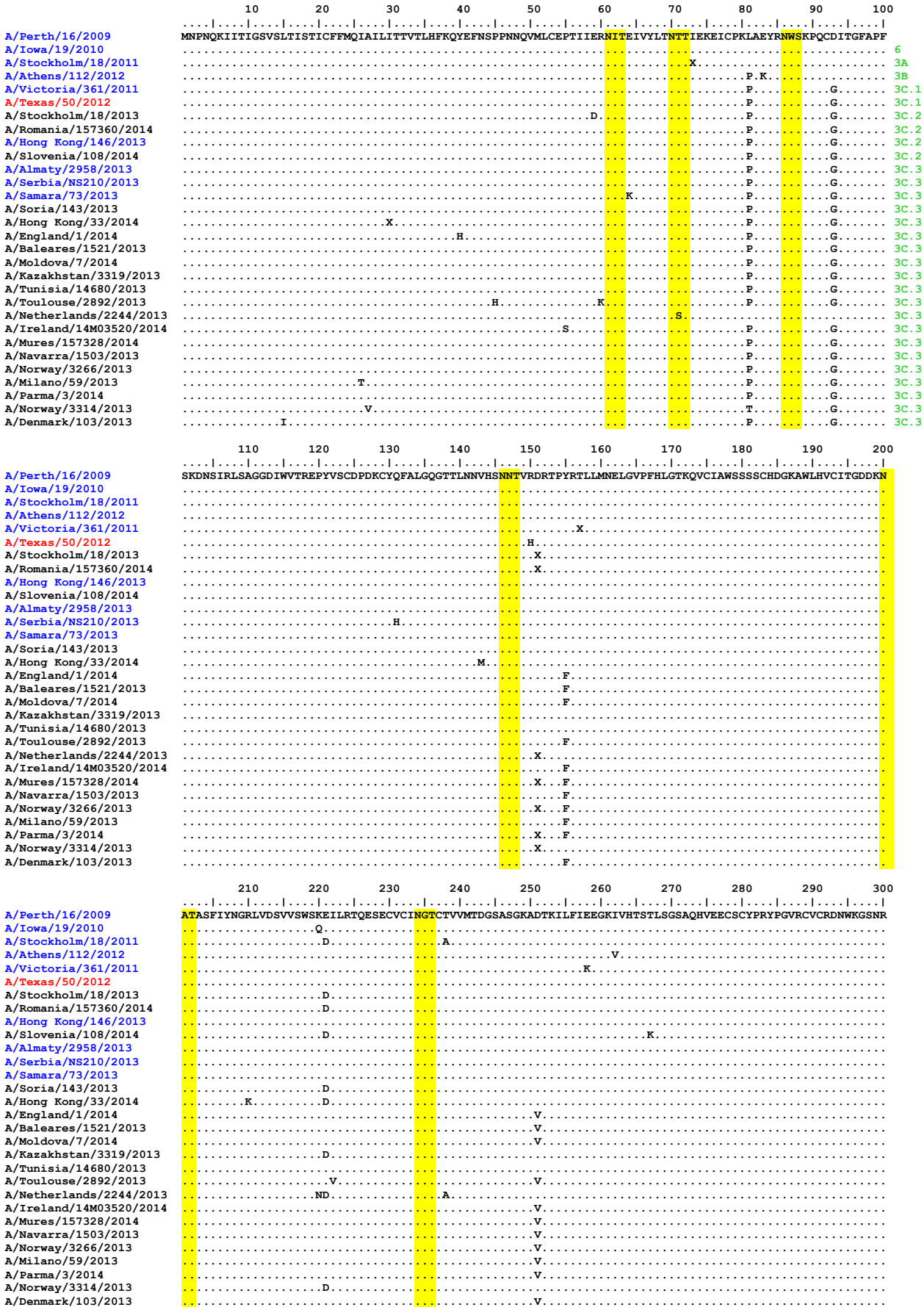


Figure 13. NA protein alignment for A(H3N2) viruses.

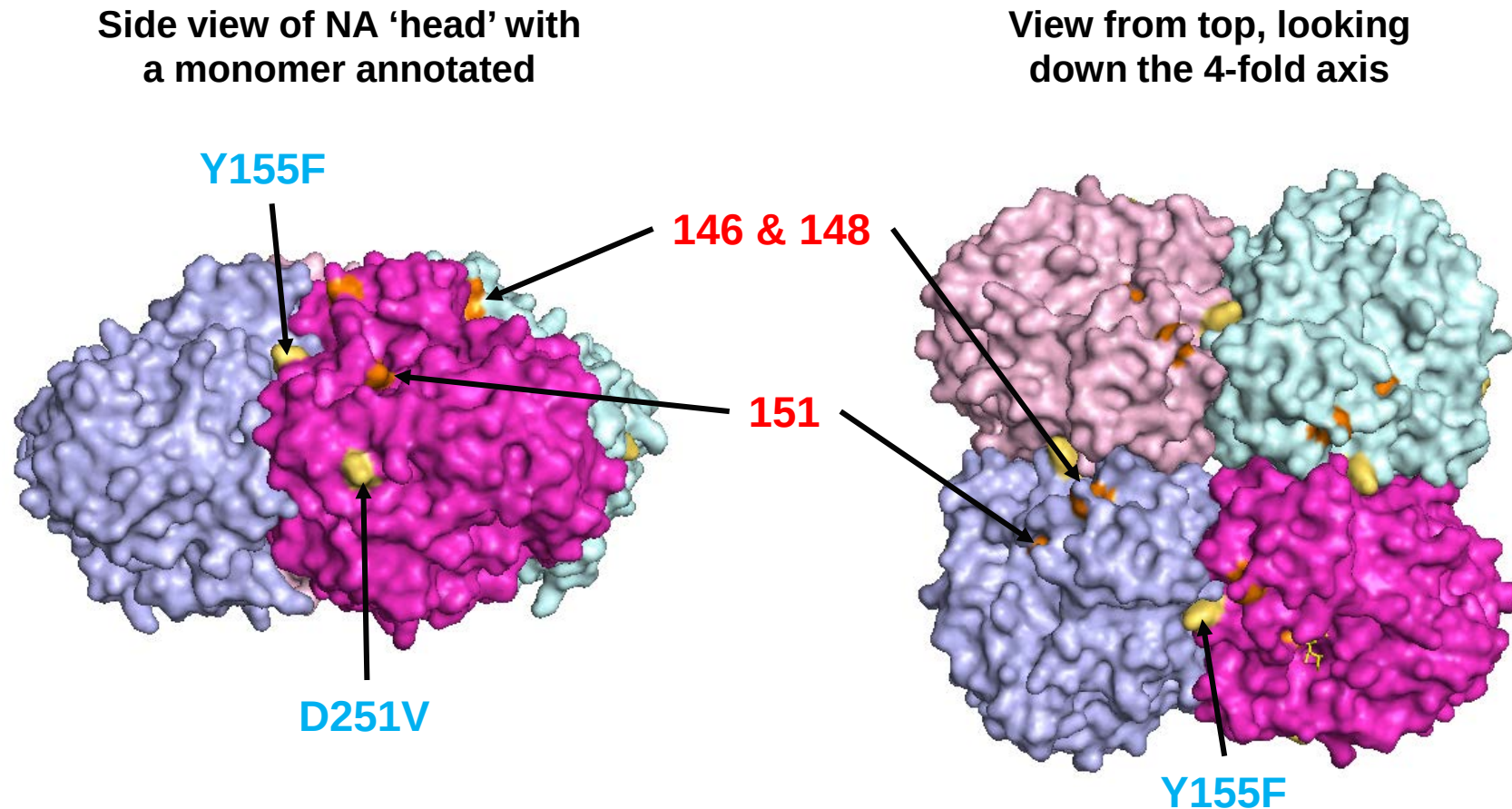
Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400
A/Perth/16/2009	PIVDINIKDHSIVSSYVCSGLVGDTPRKND	SSSSSHCLDPNNEEGGGHGVKGWAFDDGNDVMMGR	TISEKSR	LG	YETFKV	IEGWSNPKSKLQINRQVIVDR				
A/Iowa/19/2010							N.T		M.	
A/Stockholm/18/2011							N.T			
A/Athens/112/2012							N.T			
A/Victoria/361/2011			T.				N.T			
A/Texas/50/2012							N.T			
A/Stockholm/18/2013							N.T		T.	
A/Romania/157360/2014							N.T		T.	K
A/Hong Kong/146/2013							N.T			
A/Slovenia/108/2014							N.T			
A/Almaty/2958/2013							N.T			
A/Serbia/NS210/2013							N.T			
A/Samara/73/2013							N.T			
A/Soria/143/2013			F.				N.T			
A/Hong Kong/33/2014							N.T			
A/England/1/2014		G.					N.T			
A/Baleares/1521/2013		G.					N.T	G.		
A/Moldova/7/2014		G.		G.			N.T	K.		
A/Kazakhstan/3319/2013			Q.				N.T		T.	
A/Tunisia/14680/2013							N.T			
A/Toulouse/2892/2013	A.						N.T			
A/Netherlands/2244/2013	Q.			G.			N.T			
A/Ireland/14M03520/2014		V.G.				D.	N.T			
A/Mures/157328/2014		T.G.					N.T			
A/Navarra/1503/2013		T.G.					N.T			
A/Norway/3266/2013		G.		N.	D.		N.T			
A/Milano/59/2013		G.					N.T			
A/Parma/3/2014		G.					N.T			
A/Norway/3314/2013			T.				N.T			
A/Denmark/103/2013		G.					N.T			
	410	420	430	440	450	460				
A/Perth/16/2009	GNRS	GYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTS	SGTYGTG	SWPDGADINLMPI						
A/Iowa/19/2010							L.			
A/Stockholm/18/2011		N.					L.			
A/Athens/112/2012	D.						LK.			
A/Victoria/361/2011	D.						L.			
A/Texas/50/2012	D.						L.			
A/Stockholm/18/2013	D.						L.			
A/Romania/157360/2014	D.						L.			
A/Hong Kong/146/2013	D.						L.			
A/Slovenia/108/2014	D.						L.			
A/Almaty/2958/2013	D.						L.			
A/Serbia/NS210/2013	D.						L.			
A/Samara/73/2013	D.		T.				L.			
A/Soria/143/2013	D.	I.					L.			
A/Hong Kong/33/2014	D.	I.					L.			
A/England/1/2014	D.						L.			
A/Baleares/1521/2013	D.						L.			
A/Moldova/7/2014	D.						L.			
A/Kazakhstan/3319/2013	D.						L.			
A/Tunisia/14680/2013	D.				I.		L.F			
A/Toulouse/2892/2013	D.			D.			L.			
A/Netherlands/2244/2013		N.					L.V			
A/Ireland/14M03520/2014	D.						L.			
A/Mures/157328/2014	D.						L.			
A/Navarra/1503/2013	D.						L.			
A/Norway/3266/2013	D.						L.			
A/Milano/59/2013	D.						L.			
A/Parma/3/2014	D.						L.			
A/Norway/3314/2013	D.				I.		L.			
A/Denmark/103/2013	D.						L.			

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 14. N2-NA locations of amino acids influencing NA-mediated agglutination and defining genetic subgroups



Positions where amino acid substitutions influence NA-mediated agglutination are shown in **red** type. Positions shown in **cyan** type define NA subgroups within viruses carrying 3C.3 HA genes.

Table 11. Summary of influenza B (Victoria-lineage) samples analysed, collected after 2013-08-31

MONTH Country	B Victoria lineage				
	Number received	Number propagated ¹	≤2 fold ²	4 fold ²	≥8 fold ²
2013					
SEPTEMBER					
OCTOBER					
France	1	1	1		
NOVEMBER					
Germany	1	1	1		
Ghana	2	1	1		
Norway	1	1	1		
DECEMBER					
Hong Kong	2	2	1	1	
2014					
JANUARY					
Germany	1	in process			
Turkey	2	in process			
	10	6	5	1	0
6 Countries			83.3%	16.7%	0.0%

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 12-1. Antigenic analyses of influenza B viruses (Victoria lineage) (2013-09-17)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
				Post-infection ferret sera ¹									
				B/Bris ² 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Eng 393/08 F31/08	B/Bris 60/08 F26/13	B/Paris 1762/08 F7/11	B/HK 514/09 F9/13	B/Odessa 3886/10 F33/11	B/Malta 636714/11 F29/13	B/Jhb 3964/12 F1/13	B/For V2367/12 F4/13
				1A		1A	1A	1A	1B	1B	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	1280	640	80	80	<	20	<	80	160	80
B/England/393/2008	1A	2008-08-29	E1/E2	2560	80	320	640	40	80	80	320	640	320
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	160	320	640	80	160	160	320	640	640
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	1280	<	<	80	40	40	80	<	40	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	1280	<	<	80	40	80	160	<	40	40
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	20	80	320	40	80	80	80	160	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	640	40	80	80	320	320	320
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	640	1280	80	320	80	640	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	80	640	40	40	40	80	160	320
TEST VIRUSES													
B/Centre/890/2013	1A	2013-02-14	MDCK1/MDCK1	1280	<	<	80	40	40	40	<	40	80
B/Paris/958/2013	1A	2013-02-18	MDCK1/MDCK1	1280	<	<	80	40	40	80	<	40	80
B/Paris/1087/2013	1A	2013-02-25	MDCK1/MDCK1	2560	<	<	160	80	80	80	20	160	160
B/Bretagne/1138/2013		2013-02-25	MDCK1/MDCK1	2560	<	<	160	80	80	160	<	80	80
B/Paris/1116/2013		2013-02-26	MDCK1/MDCK1	2560	<	<	160	80	80	80	20	80	80
B/Pays de Loire/1222/2013	1A	2013-03-05	MDCK1/MDCK1	2560	<	<	80	80	40	80	<	40	80
B/Picardie/1254/2013	1A	2013-03-06	MDCK1/MDCK1	2560	<	<	160	40	40	80	<	40	80
B/Jiangsu-Qingpu/1256/2013	1A	2013-03-06	E3/E3/E1	2560	160	160	640	80	160	160	320	640	320
B/Haute Normandie/1452/2013	1A	2013-03-23	MDCK1/MDCK1	2560	<	<	80	80	40	80	<	40	80
B/Centre/1496/2013	1A	2013-03-28	MDCK1/MDCK1	2560	<	<	160	80	80	80	20	80	160
B/Bretagne/1591/2013	1A	2013-04-17	MDCK1/MDCK1	2560	<	<	80	40	40	80	<	40	80
B/Guangdong-Duanzhou/1160/2013	1A	2013-05-05	C1/C1/MDCK1	1280	<	<	80	80	40	80	<	80	80
B/Jiangsu-Chongchuan/11052/2013	1A	2013-06-20	E3/E2/E1	320	40	80	320	40	80	80	80	160	160
1. < = <10; 2. hyperimmune sheep serum							Vaccine*						

1. < = <10; 2. hyperimmune sheep serum

Vaccine*

Sequences in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 12-2. Antigenic analyses of influenza B viruses (Victoria lineage) (2013-12-17, 2014-01-15, 2014-01-21, 2014-01-29)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre									
				Post infection ferret sera ¹									
				B/Bris ² 60/08 Sh 522	B/Mal 2506/05 F37/11	B/Bris 60/08 F26/13	B/Paris 1762/08 F7/11	B/HK 514/09 F9/13	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F29/13	B/Jhb 3964/12 F1/13	B/For V2367/12 F4/13	B/Sth Aus 81/12 F41/13
				1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	2560	640	20	<	<	<	80	160	80	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	40	80	80	320	320	320	1280
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	10	40	40	40	80	<	40	80	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK4	2560	10	40	80	80	160	<	40	80	160
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	2560	40	80	40	80	160	80	160	160	640
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	40	80	80	320	320	320	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	80	160	160	640	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK2	1280	20	160	40	40	80	80	80	160	640
B/South Australia/81/2012	1A	2012-11-28	E4/E1	5120	80	1280	40	80	80	320	320	320	1280
TEST VIRUSES													
B/La Reunion/2500/2013		2013-03-13	MDCK2/MDCK1	5120	20	160	80	80	160	40	<	80	160
B/Toulon/2950/2013	1A	2013-10-23	MDCK2/MDCK1	5120	10	160	40	80	80	40	40	80	160
B/Wisconsin/05/2013	1A	2013-11-01	C1/MDCK1	1280	20	80	40	40	80	20	<	80	80
B/Florida/07/2013	1A	2013-11-01	C1/MDCK1	2560	20	160	40	40	80	40	<	160	160
B/Ghana/FS-1118/2013	1A	2013-11-21	MDCK2	2560	10	80	40	40	80	40	<	40	40
B/Sachsen Anhalt/13/2013	1A	2013-11-29	MDCK1	2560	40	80	40	40	80	20	<	40	80
B/Norway/3235/2013	1A	2013-11-30	MDCK1	2560	10	80	40	40	80	40	20	40	80
B/Hong Kong/439/2013	1A	2013-12-13	MDCK1/MDCK1	2560	20	160	20	40	80	10	<	80	80
B/Hong Kong/445/2013	1A	2013-12-12	MDCK1/MDCK1	1280	20	40	40	40	40	20	<	80	80

1. < = <10; 2. hyperimmune sheep serum

Vaccine*

Sequence in phylogenetic trees

* B/Victoria-lineage virus recommended for use in quadrivalent vaccines

Figure 15. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus
Reference viruses

Collection date
Oct 2013
Nov 2013
Dec 2013

HA2 numbering

HI result in tables

& Potential serology
antigens

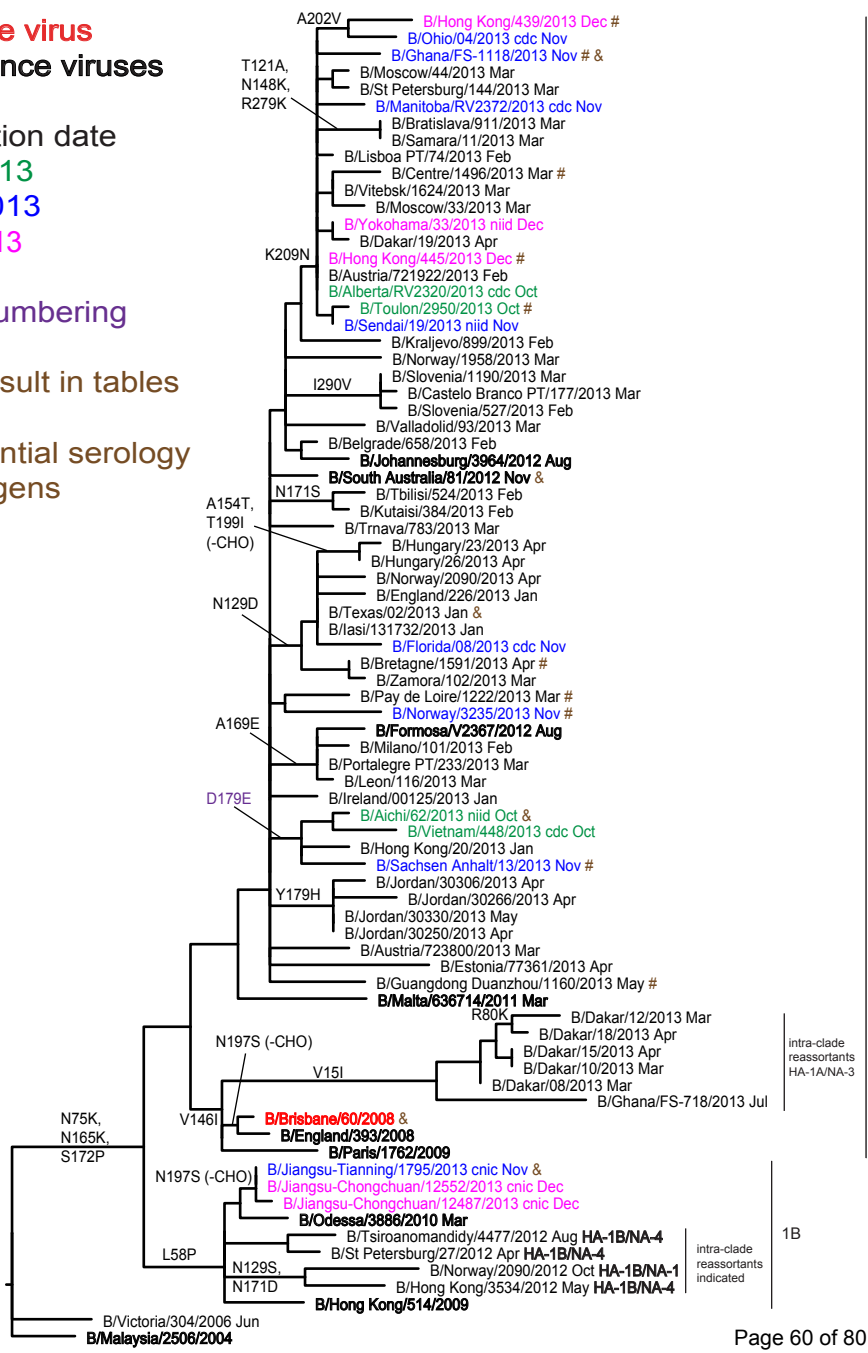
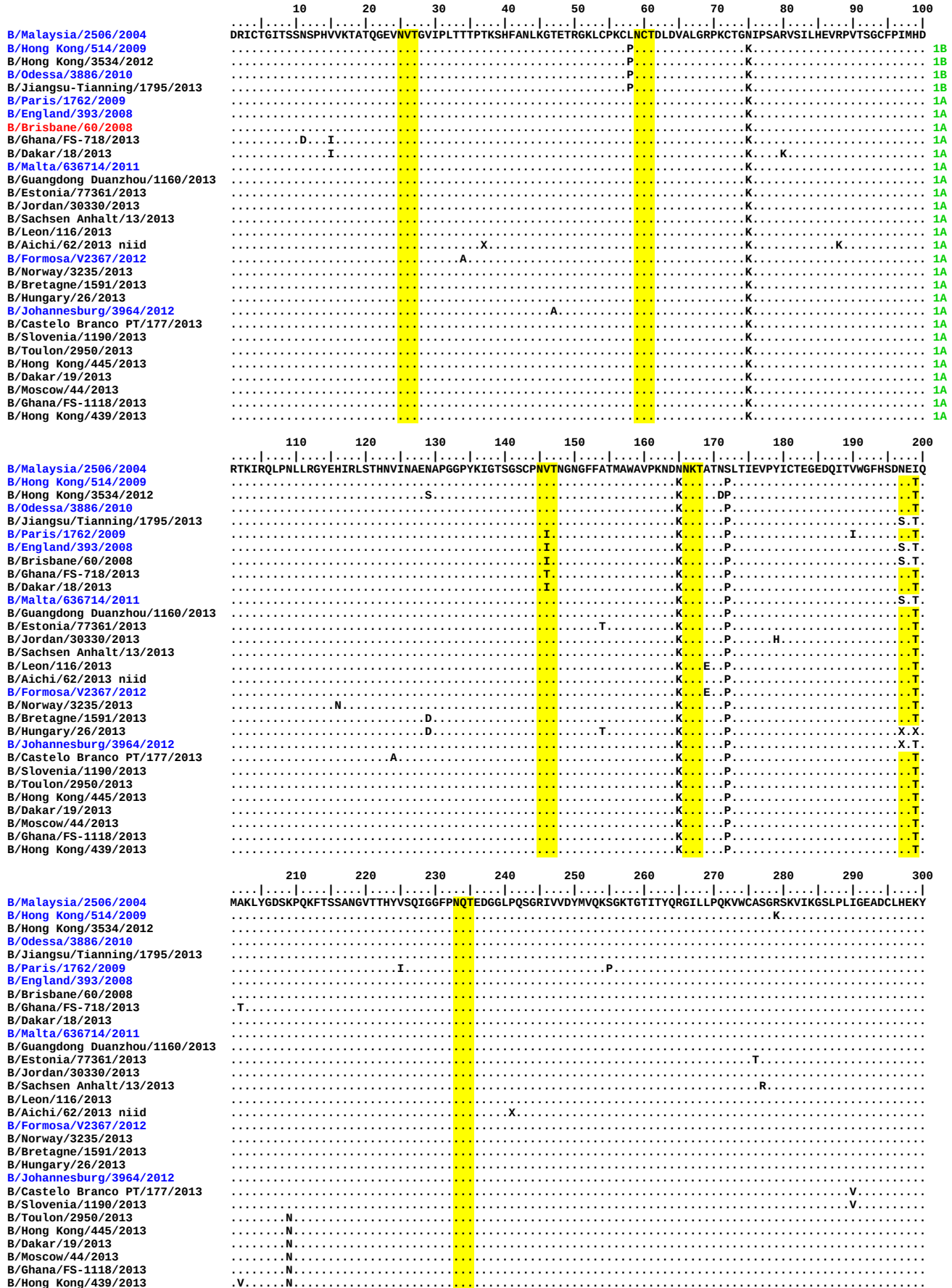


Figure 16. HA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Malaysia/2506/2004	GGLN	SKPY	TGEHAKAIG	NCPIWKTPLK	LANG	TKYRPPAKLLK	ERGFFGAIAGF	LEGGWEGMIAGWHGYT	SHGAHGVAAADL	KSTQEAINKITKNLN
B/Hong Kong/514/2009										
B/Hong Kong/3534/2012										
B/Odessa/3886/2010										
B/Jiangsu/Tianning/1795/2013										
B/Paris/1762/2009										
B/England/393/2008										
B/Brisbane/60/2008										
B/Ghana/FS-718/2013										
B/Dakar/18/2013										
B/Malta/636714/2011										
B/Guangdong Duanzhou/1160/2013										
B/Estonia/77361/2013										
B/Jordan/30330/2013										
B/Sachsen Anhalt/13/2013										
B/Leon/116/2013										
B/Aichi/62/2013 niid										
B/Formosa/V2367/2012										
B/Norway/3235/2013										
B/Bretagne/1591/2013										
B/Hungary/26/2013										
B/Johannesburg/3964/2012										
B/Castelo Branco PT/177/2013										
B/Slovenia/1190/2013										
B/Toulon/2950/2013										
B/Hong Kong/445/2013										
B/Dakar/19/2013										
B/Moscow/44/2013										
B/Ghana/FS-1118/2013										
B/Hong Kong/439/2013										

	410	420	430	440	450	460	470	480	490	500
B/Malaysia/2506/2004	SLS	ELVKNLQRLSGAMDELHNEILELDEKVDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLLKMLGPSAVEIGNGCFETKHKCNQTC	LDRIA							
B/Hong Kong/514/2009										
B/Hong Kong/3534/2012										
B/Odessa/3886/2010										
B/Jiangsu/Tianning/1795/2013										
B/Paris/1762/2009			D							
B/England/393/2008										
B/Brisbane/60/2008										
B/Ghana/FS-718/2013										
B/Dakar/18/2013										
B/Malta/636714/2011										
B/Guangdong Duanzhou/1160/2013			S							
B/Estonia/77361/2013										
B/Jordan/30330/2013										
B/Sachsen Anhalt/13/2013										
B/Leon/116/2013										
B/Aichi/62/2013 niid										
B/Formosa/V2367/2012										
B/Norway/3235/2013										
B/Bretagne/1591/2013										
B/Hungary/26/2013										
B/Johannesburg/3964/2012										
B/Castelo Branco PT/177/2013										
B/Slovenia/1190/2013										
B/Toulon/2950/2013		R								
B/Hong Kong/445/2013										
B/Dakar/19/2013										
B/Moscow/44/2013										
B/Ghana/FS-1118/2013										
B/Hong Kong/439/2013										

	510	520	530	540	550	560	570
B/Malaysia/2506/2004	AGTFDAGEFS	LPTFDSLNI	TAASLND	DGLDNH	ITILLYSTA	AASSLAV	TLMLIAIFVVMVSRDNVSCICL
B/Hong Kong/514/2009							
B/Hong Kong/3534/2012							
B/Odessa/3886/2010							
B/Jiangsu/Tianning/1795/2013							
B/Paris/1762/2009							
B/England/393/2008							
B/Brisbane/60/2008							
B/Ghana/FS-718/2013							
B/Dakar/18/2013							
B/Malta/636714/2011							
B/Guangdong Duanzhou/1160/2013							
B/Estonia/77361/2013							
B/Jordan/30330/2013							
B/Sachsen Anhalt/13/2013							
B/Leon/116/2013							
B/Aichi/62/2013 niid							
B/Formosa/V2367/2012							
B/Norway/3235/2013							
B/Bretagne/1591/2013							
B/Hungary/26/2013							
B/Johannesburg/3964/2012							
B/Castelo Branco PT/177/2013							
B/Slovenia/1190/2013							
B/Toulon/2950/2013							
B/Hong Kong/445/2013							
B/Dakar/19/2013							
B/Moscow/44/2013							
B/Ghana/FS-1118/2013							
B/Hong Kong/439/2013							

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 17. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

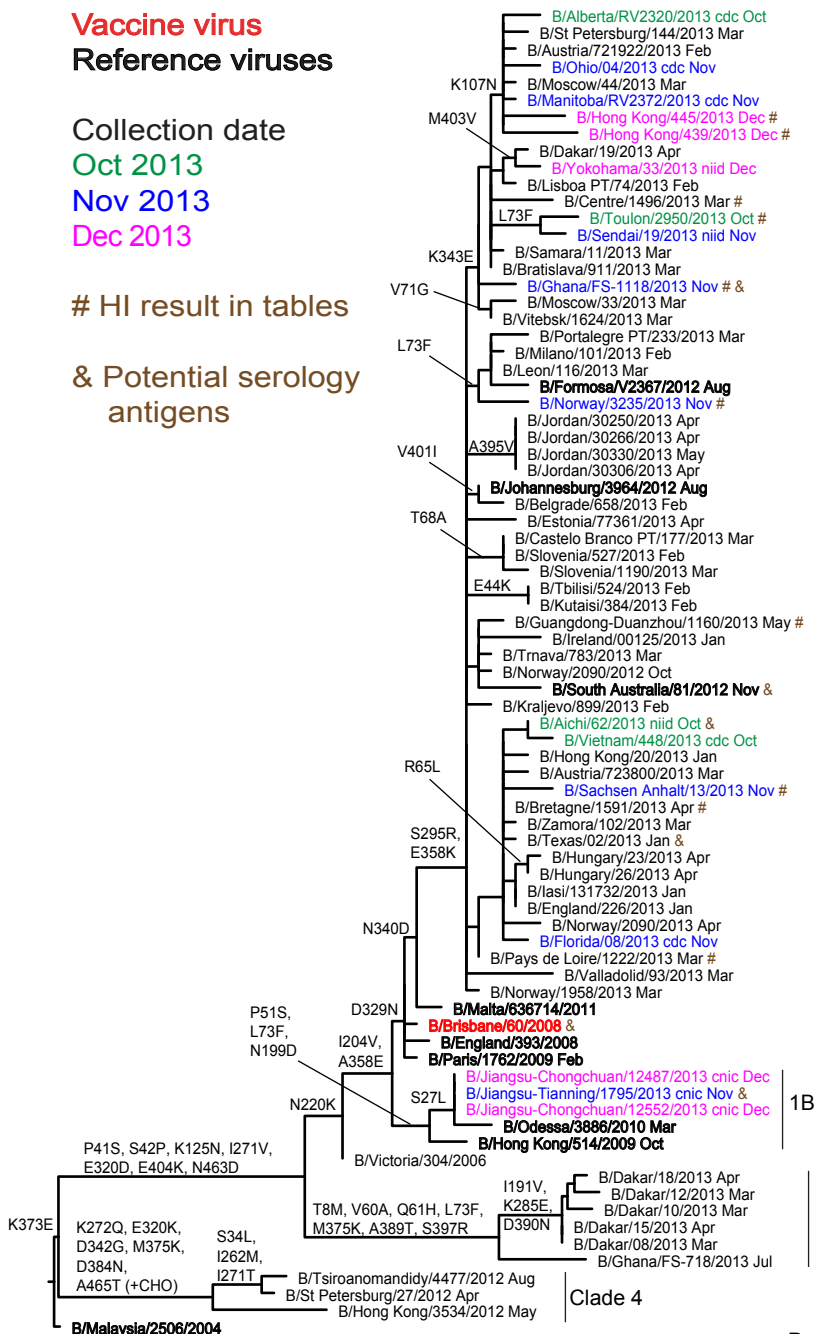


Figure 18. NA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

[illegible]

	110	120	130	140	150	160	170	180	190	200
B/Malaysia/2506/2004	HRRGETKGNSAPLIIREPFIAICGPKCKHFA	LTHYAAQPGGY	N	GTRDRNKLRLHSVKLGKIPTVENSIFHMAAWSGSACHDGKEWITYIGVDGPDNN	A					
B/Hong Kong/514/2009	.	N.	.	.	.	.	.	.	.	D.
B/Hong Kong/3534/2012	.	.	.	.	.	.	.	.	.	.
B/Odessa/3886/2010	.	N.	.	.	.	.	.	.	.	D.
B/Jiangsu-Tianning/1795/2013	.	N.	.	.	.	.	.	.	.	D.
B/Paris/1762/2009	.	N.	.	.	.	.	.	.	.	D.
B/England/393/2008	.	N.	.	.	.	.	.	.	.	.
B/Brisbane/60/2008	.	N.	.	.	.	.	.	.	.	.
B/Ghana/FS-718/2013	.	N.	.	.	.	.	.	.	.	.
B/Dakar/18/2013	.	N.	.	.	.	.	.	.	V.	.
B/Malta/636714/2011	.	N.	.	.	.	.	.	.	.	.
B/Guangdong-Duanzhou/1160/2013	.	N.	.	.	.	.	.	.	.	.
B/Estonia/77361/2013	.	N.	.	.	.	.	.	.	.	.
B/Jordan/30330/2013	.	N.	.	.	.	.	.	.	.	.
B/Sachsen Anhalt/13/2013	.	S.	.	.	.	.	.	.	.	.
B/Aichi/62/2013 niid	.	N.	.	.	.	.	.	.	.	.
B/Leon/116/2013	.	N.	.	.	.	.	.	.	.	.
B/Formosa/V2367/2012	.	N.	.	.	.	.	.	.	.	.
B/Norway/3235/2013	.	N.	.	.	.	.	.	.	.	.
B/Bretagne/1591/2013	.	N.	.	.	.	.	.	.	.	.
B/Hungary/26/2013	.	N.	.	.	.	.	.	.	.	.
B/Johannesburg/3964/2012	.	N.	.	.	.	.	.	.	.	.
B/Slovenia/1190/2013	.	N.	.	.	.	.	.	.	.	.
B/Castelo Branco PT/177/2013	.	N.	.	.	.	.	.	.	.	.
B/Toulon/2950/2013	.	N.	.	.	.	.	.	.	.	.
B/Hong Kong/445/2013	.	N.	.	.	.	.	.	.	.	.
B/Dakar/19/2013	.	N.	.	.	.	.	.	.	.	.
B/Moscow/44/2013	.	N.	.	.	.	.	.	.	.	.
B/Ghana/FS-1118/2013	.	N.	.	.	.	.	.	.	.	.
B/Hong Kong/439/2013	.	N.	.	.	.	.	.	.	.	.

	210	220	230	240	250	260	270	280	290	300
B/Malaysia/2506/2004	LLKIKYGEAYTD	YHSYANNILRT	QESACNCIGG	NCYLMITDGS	ASGVSECRFL	KIREGRIKEIF	PTGRIKHTEEC	CGFASN	KITIECACR	DNSTYAKR
B/Hong Kong/514/2009	..V.	..K.					..V.			
B/Hong Kong/3534/2012							..Q.			
B/Odessa/3886/2010	..V.	..K.					..V.			
B/Jiangsu-Tianning/1795/2013	..V.	..K.					..V.			
B/Paris/1762/2009	..V.	..K.					..V.			
B/England/393/2008	..V.	..K.					..V.			
B/Brisbane/60/2008	..V.	..K.					..V.			
B/Ghana/FS-718/2013							..V.	..N.		
B/Dakar/18/2013							..V.		..E.	
B/Malta/636714/2011	..V.	..K.					..V.			
B/Guangdong-Duanzhou/1160/2013	..V.	..K.					..V.			..R.
B/Estonia/77361/2013	..V.	..K.					..V.			..R.
B/Jordan/30330/2013	..V.	..K.					..V.			..R.
B/Sachsen Anhalt/13/2013	..V.	..K.					..V.			..R.
B/Aichi/62/2013 niid	..V.	..K.					..V.			..R.
B/Leon/116/2013	..V.	..K.					..V.			..R.
B/Formosa/V2367/2012	..V.	..K.					..V.			..R.
B/Norway/3235/2013	..V.	..K.					..V.			..R.
B/Bretagne/1591/2013	..V.	..K.					..V.			..R.
B/Hungary/26/2013	..V.	..K.					..V.			..R.
B/Johannesburg/3964/2012	..V.	..K.					..V.			..R.
B/Slovenia/1190/2013	..V.	..K.					..V.			..R.
B/Castelo Branco PT/177/2013	..V.	..K.					..V.			..R.
B/Toulon/2950/2013	..V.	..K.					..V.			..R.
B/Hong Kong/445/2013	..V.	..K.					..V.			..R.
B/Dakar/19/2013	..V.	..K.					..V.			..R.
B/Moscow/44/2013	..V.	..K.					..V.			..R.
B/Ghana/FS-1118/2013	..V.	..K.					..V.			..R.
B/Hong Kong/439/2013	..V.	..K.					..V.			..R.

Table 13. Summary of influenza B (Yamagata-lineage) samples analysed, collected after 2013-08-31

MONTH	B Yamagata lineage				
Country	Number received	Number propagated ¹	≤2 fold ²	4 fold ²	≥8 fold ²
2013					
SEPTEMBER					
OCTOBER					
Denmark	1	0			
France	1	1		1	
Iran	4	4		2	2
Portugal	1	1		1	
United Kingdom	1	1		1	
NOVEMBER					
France	3	3		1	2
Ghana	1	0			
Ireland	1	1		1	
Israel	1	1			1
Norway	2	2		1	1
Turkey	1	1			1
DECEMBER					
Algeria	1	in process			
Belgium	1	0			
Bosnia and Herzegovina	1	1			1
Denmark	1	0			
France	2	2			1
Hong Kong	3	3		1	2
Iceland	1	1		1	
Ireland	1	1		1	
Norway	4	4			4
Sweden	1	1			1
Turkey	1	1			1
2014					
JANUARY					
Italy	1	1			1
Turkey	2	in process			1
	37	30	0	11	19
18 Countries			0.0%	36.7%	63.3%

1. Propagated to sufficient titre to perform HI assay

2. Reduction in HI titre with ferret antiserum raised against B/Massachusetts/02/2012 (egg-grown vaccine virus)

Table 14-1. Antigenic analyses of influenza B viruses (Yamagata lineage) (2013-09-04, 2013-09-24, 2013-10-30)

				Haemagglutination Inhibition Titre									
Viruses	Genetic Group	Collection date	Passage History	Post-infection ferret sera									
				B/FI ³	B/FI ¹	B/Bris ²	B/Wis ²	B/Stock ²	B/Estonia ²	B/Novo ²	B/HK ²	B/Mass ²	B/Mass ²
				4/06 SH479	4/06 F1/10	3/07 F21/12	1/10 F24/12	12/11 F12/12	55669/11 F26/11	1/12 F31/12	3577/12 F33/12	2/12 Egg F2/13	2/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	320	640	160	40	320	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	160	40	320	1280	320
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	1280	320	320	320	640	20	40	40	320	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	2560	320	160	160	640	40	80	80	640	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	320	160	80	160	640	80	640	320	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	1280	320	160	320	320	80	160	160	320	160
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	320	160	160	320	640	160	640	320	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	5120	1280	640	320	640	160	40	640	640	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	640	640	80	640	640	640
TEST VIRUSES													
B/Cameroon/13V-015/2013	3	2012-12-27	MDCK3	2560	160	80	160	160	160	80	80	40	80
B/Hawaii/01/2013	2	2013-01-05	C1/MDCK1	2560	160	160	80	160	320	80	640	160	320
B/Cameroon/13V-450/2013		2013-01-25	MDCK1/MDCK1	2560	160	160	160	320	80	160	160	320	160
B/Cameroon/13V-453/2013		2013-01-28	MDCK1/MDCK1	2560	160	160	160	160	640	80	640	320	320
B/Kazakhstan/3246/2013	2	2013-02-14	MDCK2	2560	320	320	160	320	640	80	640	640	640
B/Cameroon/13V-930/2013		2013-02-18	MDCK1/MDCK1	1280	80	80	160	160	80	160	160	160	160
B/Guyane/138/2013	2	2013-02-25	MDCK1/MDCK1	2560	320	320	160	320	640	80	640	320	640
B/Cameroon/13V-1103/2013		2013-02-26	MDCK1/MDCK1	2560	160	160	160	320	640	160	640	320	640
B/Lithuania/6663/2013	2	2013-03-02	SIAT4	1280	160	160	80	160	320	40	320	320	320
B/Caen/1430/2013	2	2013-03-06	MDCK1/MDCK1	2560	160	320	160	320	640	80	640	320	640
B/Cameroon/13V-1548 GVFI/2013		2013-03-10	MDCK1/MDCK1	2560	320	320	160	320	640	80	640	640	640
B/Cameroon/13V-1549 GVFI/2013	2	2013-03-11	MDCK1/MDCK1	2560	320	320	160	320	640	80	640	320	640
B/Cameroon/13V-1405/2013		2013-03-13	MDCK2/MDCK1	2560	160	160	160	320	640	80	640	320	640
B/Cameroon/13V-1687/2013		2013-03-21	MDCK1/MDCK1	2560	160	160	160	160	640	80	640	320	640
B/Haute Normandie/1459/2013	2	2013-03-25	MDCK1/MDCK1	2560	160	320	160	320	640	80	640	320	320
B/Paris/1451/2013	2	2013-03-26	MDCK1/MDCK1	2560	160	160	160	160	640	80	640	320	320
B/Pays de Loire/1494/2013	2	2013-03-27	MDCK1/MDCK1	2560	160	320	160	320	640	80	640	320	640
B/Nord Pas de Calais/1539/2013	3	2013-04-04	MDCK1/MDCK1	2560	160	160	160	320	80	160	160	320	160
B/Pays de Loire/1534/2013	2	2013-04-05	MDCK1/MDCK1	5120	160	320	160	320	640	80	1280	640	640
B/Paris/1574/2013	2	2013-04-12	MDCK1/MDCK1	5120	160	320	160	320	640	80	640	320	640
B/Haute Normandie/1584/2013	2	2013-04-15	MDCK1/MDCK1	2560	160	320	160	320	640	80	640	320	640
B/Cameroon/13V-2053/2013	2	2013-04-15	MDCK2	640	80	80	40	160	320	40	640	160	320
B/Cameroon/13V-2198/2013	2	2013-04-22	MDCK1/MDCK1	2560	160	160	160	160	640	80	640	320	640
B/Fukui/24/2013	2	2013-05-07	MDCK1/MDCK1/MDCK1	1280	160	80	40	160	320	20	320	80	80
B/Sapporo/21/2013	2	2013-05-07	MDCK1/MDCK1/MDCK1	2560	160	160	160	160	320	40	640	160	160
B/Cameroon/13V-2891/2013	2	2013-05-27	MDCK1/MDCK1	2560	320	320	160	320	640	80	640	320	640
B/Cameroon/13V-3209/2013	2	2013-06-12	MDCK1/MDCK1	2560	320	320	160	320	640	80	640	320	640

1. <= <40; 2. <= <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 14-2. Antigenic analyses of influenza B viruses (Yamagata lineage) (2013-09-06, 2013-09-10, 2013-09-17, 2013-10-16)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination Inhibition Titre									
				Post-infection ferret sera									
				B/FI ³ 4/06 SH479	B/FI ¹ 4/06 F1/10	B/Bris ² 3/07 F21/12	B/Wis ² 1/10 F24/12	B/Stock ² 12/11 F34/11	B/Estonia ² 55669/11 F26/11	B/Novo ² 1/12 F31/12	B/HK ² 3577/12 F33/12	B/Mass ² 2/12 Egg F2/13	B/Mass ² 2/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	1280	320	640	320	40	320	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	320	40	640	1280	320
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	640	320	160	320	320	10	40	40	320	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	320	160	160	320	10	40	40	320	40
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	160	80	320	640	40	640	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK2	640	160	80	160	320	80	160	160	160	160
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	160	160	80	320	1280	80	640	160	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	160	320	160	40	320	640	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	320	160	320	640	80	640	640	640
TEST VIRUSES													
B/Shanghai-Putuo/120/2013	2	2013-01-16	C1/E4/E1	1280	320	320	160	320	80	20	160	640	80
B/Kazakhstan/2994/2013	2	2013-02-03	MDCK2	2560	320	320	160	640	640	80	640	320	320
B/Kazakhstan/2995/2013		2013-02-03	MDCK2	2560	320	320	160	320	640	80	640	320	320
B/Moldova/130/2013	2	2013-02-11	MDCK1/MDCK1	2560	160	160	160	320	640	80	640	320	640
B/Paris/1305/2013	2	2013-02-18	MDCK1/MDCK1	1280	160	160	80	320	320	40	640	160	160
B/Picardie/1175/2013	2	2013-02-19	MDCK1/MDCK1	2560	320	320	160	640	640	80	640	320	320
B/Paris/1168/2013	2	2013-02-21	MDCK1/MDCK1	2560	320	320	160	640	640	80	1280	320	160
B/Lithuania/6530/2013	3	2013-02-25	MDCK3	1280	80	40	40	320	20	40	40	<	20
B/Moldova/279/2013	2	2013-02-26	MDCK3	640	80	80	40	160	320	20	320	160	320
B/Lithuania/6453/2013		2013-02-28	MDCK2	2560	320	160	80	320	640	80	1280	160	320
B/Lithuania/6942/2013	3	2013-03-04	MDCK2	5120	320	320	320	1280	160	160	160	160	80
B/Lithuania/6935/2013	2	2013-03-04	MDCK3	1280	160	160	80	320	640	40	640	40	160
B/Lithuania/6931/2013	2	2013-03-05	MDCK2	2560	320	160	160	320	320	80	640	160	160
B/Lithuania/7845A/2013		2013-03-06	MDCK2	1280	160	160	160	320	320	80	640	160	160
B/Lithuania/7848/2013	3	2013-03-12	MDCK2	1280	160	80	160	320	20	80	80	80	40
B/Moldova/366/2013	2	2013-03-13	MDCK2	1280	80	160	80	320	640	40	640	160	320
B/Lithuania/7850/2013	2	2013-03-13	MDCK2	5120	320	320	160	640	640	80	640	320	320
B/Moldova/395/2013		2013-03-21	MDCK2	1280	160	160	80	320	640	40	640	320	320
B/Moldova/431/2013	2	2013-03-27	MDCK2	320	40	80	40	320	640	20	320	160	640
B/Moldova/450/2013		2013-04-01	MDCK3	1280	80	80	40	160	320	40	320	160	320
B/Moldova/458/2013	2	2013-04-07	MDCK2	320	20	80	40	160	640	40	640	160	640
B/Moldova/459/2013		2013-04-08	MDCK2	160	20	80	80	320	40	80	80	160	160
B/Moldova/462/2013	2	2013-04-10	MDCK2	160	20	80	40	160	640	40	320	160	320
B/Guangxi-Xixiangtang/1311/2013	3	2013-06-17	C2/C1/MDCK1	1280	320	320	320	640	80	160	160	320	80
B/Kazakhstan/2998/2013		unknown	MDCK2	2560	320	320	160	320	640	80	640	320	320

1. <= <40; 2. <= <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 14-3. Antigenic analyses of influenza B viruses (Yamagata lineage) (2013-11-15)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination Inhibition Titre									
				Post infection ferret sera									
				B/FI ³	B/FI ¹	B/Bris ²	B/Wis ²	B/Stock ²	B/Estonia ²	B/Novo ²	B/HK ²	B/Mass ²	B/Mass ²
				4/06 SH479	4/06 F1/10	3/07 F21/12	1/10 F24/12	12/11 F12/12	55669/11 F32/12	1/12 F31/12	3577/12 F33/12	2/12 Egg F2/13	2/12 T/C F15/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	320	1280	320	640	80	80	640	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	80	40	640	1280	320
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	640	320	320	320	640	20	40	40	320	80
B/Stockholm/12/2011	3	2007-08-07	E4/E1	2560	320	320	320	640	20	80	80	320	80
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	160	160	320	320	160	1280	320	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	320	160	320	640	320	320	640	320	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK3	1280	160	80	80	160	160	40	640	160	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	640	320	640	80	40	640	1280	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	320	160	80	640	640	640
TEST VIRUSES													
B/La Reunion/2503/2013		2013-06-17	MDCK2/MDCK1	2560	320	160	160	320	640	80	640	320	320
B/Lyon-chu/43.328/2013	3	2013-10-23	MDCK2/MDCK1	2560	320	160	160	320	160	160	160	320	320
B/England/639/2013	3	2013-10-24	SIAT1/MDCK1	1280	160	160	160	320	40	80	80	320	160
B/Lisboa_PT/235/2013	2	2013-10-29	SIAT2/MDCK1	2560	160	160	160	320	640	160	640	320	640
B/Norway/3044/2013	3	2013-11-03	MDCK1/MDCK1	640	80	40	80	160	40	80	80	320	80

1. <= <40; 2. <= <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Vaccine

Table 14-4. Antigenic analyses of influenza B viruses (Yamagata lineage) (2014-01-21)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination Inhibition Titre									
				Post infection ferret sera									
				B/FI ³	B/FI ¹	B/Bris ²	B/Wis ²	B/Stock ²	B/Estonia ²	B/Novo ²	B/HK ²	B/Mass ²	B/Mass ²
				4/06 SH479	4/06 F1/10	3/07 F21/12	1/10 F10/13	12/11 F12/12	55669/11 F26/11	1/12 F31/12	3577/12 F33/12	B/Mass ² Egg F2/13	2/12 T/C F3/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	640	320	640	160	40	320	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	160	320	160	40	320	640	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	640	160	80	80	160	<	20	20	160	20
B/Stockholm/12/2011	3	2007-08-07	E4/E1	1280	160	160	80	160	10	40	40	160	40
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	80	40	80	320	80	640	80	320
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	1280	160	80	80	160	80	160	160	80	320
B/Hong Kong/3577/2012	2	2012-06-13	MDCK3	2560	160	160	80	80	640	40	640	320	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	80	320	80	20	160	640	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	320	320	160	160	320	80	320	320	320
TEST VIRUSES													
B/Guatemala/260/2013	2	2013-07-11	Cx/MDCK1	1280	160	80	40	80	320	40	320	160	160
B/Arizona/08/2013	3	2013-09-27	C2/MDCK1	2560	160	160	160	320	160	320	320	160	320
B/Franche Comte/2166/2013	3	2013-11-04	MDCK1/MDCK1	1280	160	80	160	160	160	320	160	160	640
B/Ireland/13M95631/2013	3	2013-11-21	Cx/MDCK1	1280	160	80	80	160	80	160	160	160	320
B/Paris/2327/2013	2	2013-11-25	MDCK1/MDCK1	640	80	40	20	20	160	20	160	80	160
B/Paris/2375/2013	3	2013-11-26	MDCK1/MDCK1	640	80	40	40	80	20	40	40	80	40
B/Ireland/13M98449/2013	3	2013-12-04	Cx/MDCK1	640	80	40	80	80	40	40	40	160	80
B/Hong Kong/424/2013	3	2013-12-10	MDCK1/MDCK1	1280	160	80	160	160	40	80	80	80	80
B/Hong Kong/432/2013	3	2013-12-11	MDCK1/MDCK1	640	80	40	80	80	20	40	40	80	80
B/Hong Kong/436/2013	3	2013-12-12	MDCK1/MDCK1	1280	160	80	160	160	80	160	160	160	160
B/Paris/2535/2013	3	2013-12-28	MDCK1/MDCK1	320	80	40	40	80	20	40	40	80	40

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 14-5. Antigenic analyses of influenza B viruses (Yamagata lineage) (2014-01-29, 2014-02-04)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination Inhibition Titre									
				Post infection ferret sera									
				B/Fi ³	B/Fi ¹	B/Bris ²	B/Wis ²	B/Stock ²	B/Estonia ²	B/Novo ²	B/HK ²	B/Mass ²	B/Mass ²
				4/06 SH479	4/06 F1/10	3/07 F24/07	1/10 F24/12	12/11 F12/12	55669/11 F26/11	1/12 F31/12	3577/12 F33/12	2/12 Egg F2/13	2/12 T/C F3/13
				1	1	2	3	3	2	3	2	2	2
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	1280	320	640	160	80	320	1280	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	640	640	320	640	160	80	320	640	160
B/Wisconsin/1/2010	3	2007-08-07	E3/E2	640	160	160	160	320	<	40	40	320	40
B/Stockholm/12/2011	3	2007-08-07	E4/E1	2560	160	80	80	160	10	40	40	160	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	80	80	80	640	80	640	160	640
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	160	160	160	320	320	160	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	80	80	80	80	640	80	640	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	1280	320	640	160	40	320	1280	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	320	320	160	320	320	80	320	640	320
TEST VIRUSES													
B/Sarajevo/162/2013	2	2013-02-27	MDCK2	2560	160	80	80	160	640	80	640	160	320
B/Tehran/63141/2013	3	2013-10-27	E2/E1	320	40	40	20	40	<	20	20	80	40
B/Tehran/63364/2013	3	2013-10-28	MDCK2/MDCK1	2560	160	160	160	320	320	320	320	320	640
B/Tehran/63364/2013	3	2013-10-28	E2/E2	160	40	20	20	40	<	10	<	40	20
B/Tehran/63141/2013	3	2013-10-30	MDCK2/MDCK1	5120	320	320	640	640	320	640	640	320	1280
B/Diyarbakir/1200/2013		2013-11-07	SIAT1/MDCK1	1280	80	40	20	160	40	160	160	160	160
B/Israel/2/2013		2013-11-13	C2/MDCK1	640	80	40	20	40	320	20	640	80	320
B/Norway/3200/2013	2	2013-11-22	MDCK1/MDCK1	640	80	80	40	80	320	20	320	160	160
B/Stockholm/14/2013	2	2013-12-02	MDCK2/MDCK4	1280	80	40	40	40	640	80	320	160	320
B/Norway/3229/2013	3	2013-12-04	MDCK2/MDCK1	5120	160	160	160	320	320	320	640	160	1280
B/Lyon/3021/2013	3	2013-12-16	MDCK2/MDCK1	640	80	40	80	80	20	80	40	160	40
B/Sarajevo/513/2013	3	2013-12-18	MDCK2	640	160	80	80	160	40	80	80	160	80
B/Norway/3323/2013	2	2013-12-19	MDCK1/MDCK1	2560	160	80	80	80	320	40	320	160	160
B/Norway/3324/2013		2013-12-19	MDCK1/MDCK2	1280	80	80	20	80	320	20	640	80	320
B/Iceland/66/2013	2	2013-12-23	MDCK1/MDCK1	1280	160	160	80	160	320	40	320	320	320
B/Norway/3309/2013	2	2013-12-24	MDCK1	1280	80	80	40	80	160	40	320	160	160
B/Isparta/4225/2013		2013-12-27	SIAT1/MDCK1	1280	80	40	20	80	320	80	640	80	640
B/Trabzon/29/2014		2014-01-03	SIAT1/MDCK1	1280	80	80	40	80	320	80	640	80	640
B/Firenze/1/2014	3	2014-01-07	MDCK2	640	80	80	80	160	10	80	80	160	80
												Vaccine	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Figure 19. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Sep - Oct 2013

Nov 2013

Dec 2013

Jan 2014

HA2 numbering

HI result in tables

& Potential serology
antigens

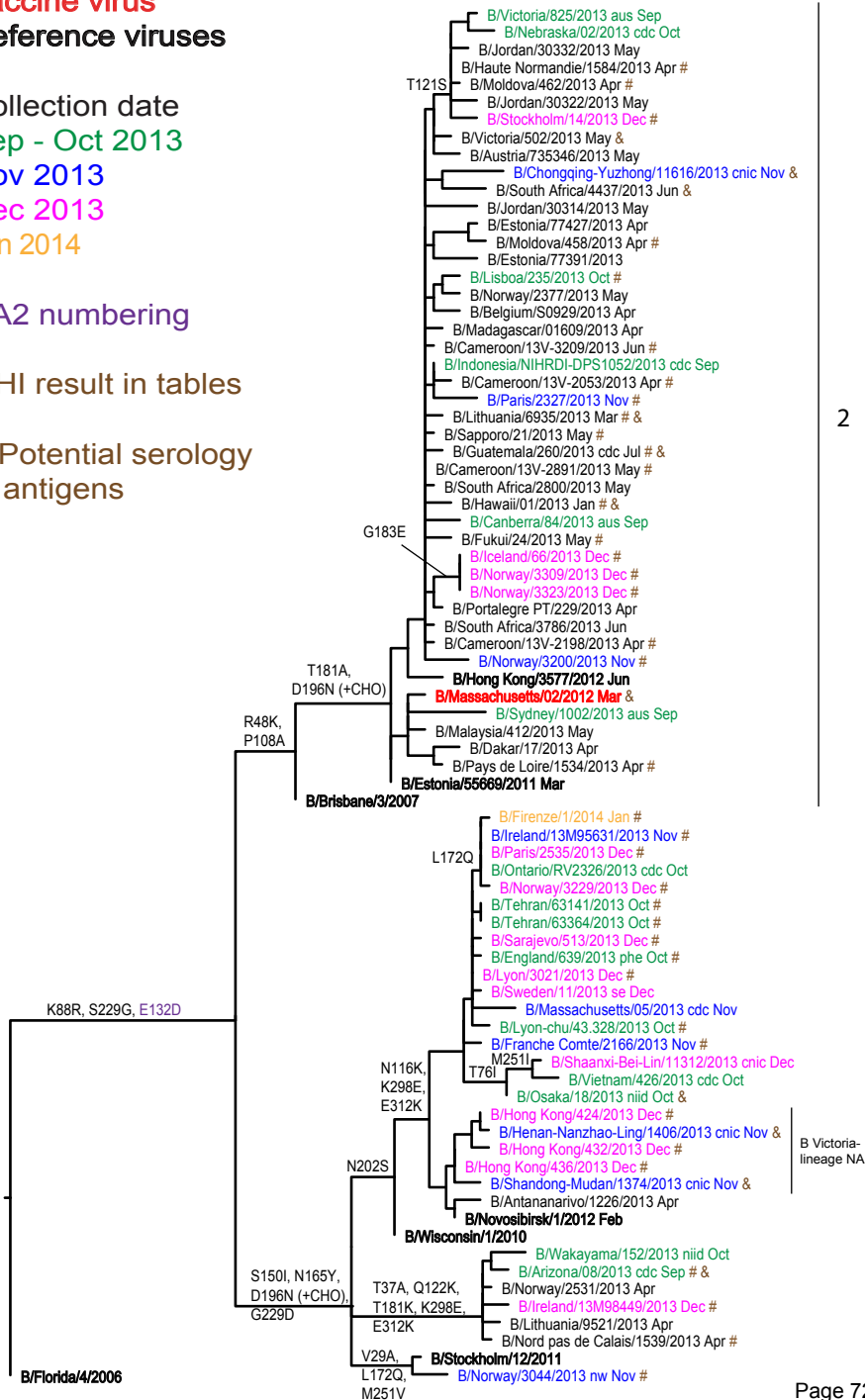


Figure 20. HA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Florida/4/2006	DRC	TG	TSSNSPHVVKTA	TQGEV	NVTGV	IPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDL	DVALGRPMCVGTTPSAKASILHEVKPVTSGCFPIIMHD	.	.	.
B/Norway/3044/2013 nw	.	.	.	.	A.	.	.	.	R.	.
B/Stockholm/12/2011	.	.	.	.	A.	.	.	.	R.	.
B/Ireland/13M98449/2013	.	.	.	.	A.	.	.	.	R.	.
B/Norway/2531/2013	.	.	.	.	A.	.	.	.	R.	.
B/Arizona/08/2013 cdc	.	.	.	.	A.	.	.	.	R.	.
B/Wisconsin/1/2010	.	.	.	.	.	.	.	.	R.	.
B/Novosibirsk/1/2012	.	.	.	.	.	.	.	.	R.	.
B/Hong Kong/436/2013	.	.	.	.	.	.	.	.	R.	.
B/Osaka/18/2013 niid	.	.	.	.	.	.	.	I.	R.	.
B/Franche Comte/2166/2013	.	.	.	.	.	.	.	.	R.I.	.
B/Lyon/3021/2013	.	.	.	.	.	.	.	.	R.	.
B/Sarajevo/513/2013	.	.	.	.	.	.	.	.	R.	.
B/Tehran/63364/2013	.	.	.	.	.	.	.	.	R.	.
B/Paris/2535/2013	.	.	.	.	.	.	.	.	R.	.
B/Firenze/1/2014	.	.	.	.	.	.	.	.	R.	.
B/Brisbane/3/2007	.	.	.	.	.	K.	.	.	R.	.
B/Estonia/55669/2011	.	.	.	.	.	K.	.	.	R.	.
B/Massachusetts/02/2012	.	.	.	.	.	K.	.	.	R.	.
B/Hong Kong/3577/2012	.	.	.	.	.	K.	.	.	R.	.
B/Norway/3323/2013	.	.	.	.	.	K.	.	.	R.	.
B/Iceland/66/2013	.	.	.	.	.	K.	.	.	R.	.
B/Guatemala/260/2013 cdc	.	.	.	.	.	K.	.	.	R.	.
B/Paris/2327/2013	.	.	.	.	.	K.	.	.	R.	.
B/Cameroon/13V-3209/2013	.	.	.	.	.	K.	.	.	R.	.
B/Lisboa/235/2013	.	.	.	.	.	K.	.	.	R.	.
B/South Africa/4437/2013	.	.	.	.	.	K.	.	.	R.	.
B/Austria/735346/2013	.	.	.	.	.	K.	.	.	R.	.
B/Moldova/462/2013	.	.	.	.	.	K.	.	.	R.	M.
B/Jordan/30332/2013	.	.	.	.	.	K.	.	.	R.	.

	110	120	130	140	150	160	170	180	190	200
B/Florida/4/2006	RTKIRQLPNLLR	GYENIRLSTQ	NIVDAEKAPGG	PYRLGTS	SGSCP	NATSKSGFFATMA	WVPKDNHKNAT	NPLTVEV	VPYICTEGEDQ	ITVWGFHSDDKTQM
B/Norway/3044/2013 nw		T				I		Y	Q	N
B/Stockholm/12/2011						I		Y	Q	
B/Ireland/13M98449/2013		K				I		Y		K
B/Norway/2531/2013		K				I		Y		K
B/Arizona/08/2013 cdc		K				I		Y		K
B/Wisconsin/1/2010						I		Y		
B/Novosibirsk/1/2012		K				I		Y		
B/Hong Kong/436/2013		K				I		Y		
B/Osaka/18/2013 niid		K				I		Y		
B/Franche Comte/2166/2013		K				I		Y		
B/Lyon/3021/2013		K				I		Y		
B/Sarajevo/513/2013		K				I		Y		
B/Tehran/63364/2013		K				I		Y		
B/Paris/2535/2013		K				I		Y	Q	
B/Firenze/1/2014		K				I		Y	Q	
B/Brisbane/3/2007	A									
B/Estonia/55669/2011	A								A	
B/Massachusetts/02/2012	A								A	
B/Hong Kong/3577/2012	A								A	
B/Norway/3323/2013	A								A	E
B/Iceland/66/2013	A								A	E
B/Guatemala/260/2013 cdc	A								A	
B/Paris/2327/2013	A								A	
B/Cameroon/13V-3209/2013	A								A	
B/Lisboa/235/2013	A								I	A
B/South Africa/4437/2013	A		N						A	
B/Austria/735346/2013	A								A	
B/Moldova/462/2013	A		S						A	
B/Jordan/30332/2013	A		S						A	

	210	220	230	240	250	260	270	280	290	300
B/Florida/4/2006	KNLYGDSNPQKFTSSANGVTTHYVSQIGSPFDQTEDGGLPQSGRIVVDYMMQKPgKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEKYG	D	V	I	.	.	.	.	.	.
B/Norway/3044/2013 nw	.	.	D	.	V	.	.	.	.	.
B/Stockholm/12/2011	.	.	D	.	V	.	.	.	.	.
B/Ireland/13M98449/2013	S.	.	D	.	.	.	E.	.	.	.
B/Norway/2531/2013	.	.	D	.	E.	.	.	.	.	.
B/Arizona/08/2013 cdc	.	.	D	.	E.	.	.	.	.	.
B/Wisconsin/1/2010	S.	.	D	.	.	.	.	.	.	.
B/Novosibirsk/1/2012	S.	.	D	.	.	.	E.	.	.	.
B/Hong Kong/436/2013	S.	.	D	.	.	.	E.	.	.	.
B/Osaka/18/2013 niid	S.	.	D	.	.	.	E.	.	.	.
B/Franche Comte/2166/2013	S.	.	D	.	.	.	E.	.	.	.
B/Lyon/3021/2013	S.	.	D	.	.	.	E.	.	.	.
B/Sarajevo/513/2013	S.	.	D	.	.	.	E.	.	.	.
B/Tehran/63364/2013	S.	.	D	.	.	.	E.	.	.	.
B/Paris/2535/2013	S.	.	D	.	.	.	E.	.	.	.
B/Firenze/1/2014	S.	.	D	.	.	.	E.	.	.	.
B/Brisbane/3/2007	G.	.	G.	.	.	.	.	.	.	.
B/Estonia/55669/2011	G.	.	G.	.	.	.	.	.	.	.
B/Massachusetts/02/2012	G.	.	G.	.	.	.	.	.	.	.
B/Hong Kong/3577/2012	G.	.	G.	.	.	.	.	.	.	.
B/Norway/3323/2013	G.	.	G.	.	.	.	.	.	.	.
B/Iceland/66/2013	G.	.	G.	.	.	.	.	.	.	.
B/Guatemala/260/2013 cdc	G.	.	G.	.	.	.	.	.	.	.
B/Paris/2327/2013	G.	.	G.	.	.	.	.	.	.	.
B/Cameroon/13V-3209/2013	G.	.	G.	.	.	.	.	.	.	.
B/Lisboa/235/2013	G.	.	G.	.	.	.	.	.	.	.
B/South Africa/4437/2013	G.	.	G.	.	.	.	.	.	.	.
B/Austria/735346/2013	G.	.	G.	.	.	.	.	.	.	.
B/Moldova/462/2013	G.	.	G.	.	.	.	.	.	.	.
B/Jordan/30332/2013	G.	.	G.	.	.	.	.	.	.	.

	310	320	330	340	350	360	370	380	390	400	
B/Florida/4/2006	GLN	KSK	PYYT	GEHAKA	IGNCPI	IWVK	TP	KL	NGT	KYRPPAKLLKE	RGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVA
B/Norway/3044/2013 nw											ADLSTQEAINKITKNLS
B/Stockholm/12/2011											
B/Ireland/13M98449/2013											
B/Norway/2531/2013											
B/Arizona/08/2013 cdc											
B/Wisconsin/1/2010											
B/Novosibirsk/1/2012											
B/Hong Kong/436/2013											
B/Osaka/18/2013 niid											
B/Franche Comte/2166/2013											
B/Lyon/3021/2013											
B/Sarajevo/513/2013											
B/Tehran/63364/2013											
B/Paris/2535/2013											
B/Firenze/1/2014											
B/Brisbane/3/2007											
B/Estonia/55669/2011											
B/Massachusetts/02/2012											
B/Hong Kong/3577/2012											
B/Norway/3323/2013											
B/Iceland/66/2013											
B/Guatemala/260/2013 cdc											
B/Paris/2327/2013											
B/Cameroon/13V-3209/2013											
B/Lisboa/235/2013											
B/South Africa/4437/2013											
B/Austria/735346/2013											
B/Moldova/462/2013											
B/Jordan/30332/2013											

	410	420	430	440	450	460	470	480	490	500	
B/Florida/4/2006	LSE	EVKNLQ	RLSG	AMDELH	NETLEL	DEKVD	DLRAD	TISQ	IELAV	LLSNEG	IINSEDEHLLALERKLLKMLGPS
B/Norway/3044/2013 nw											SAVEIGNGCFETKHKC
B/Stockholm/12/2011											QTCLDRIAA
B/Ireland/13M98449/2013											
B/Norway/2531/2013											
B/Arizona/08/2013 cdc											
B/Wisconsin/1/2010											
B/Novosibirsk/1/2012											
B/Hong Kong/436/2013											
B/Osaka/18/2013 niid											
B/Franche Comte/2166/2013											
B/Lyon/3021/2013											
B/Sarajevo/513/2013											
B/Tehran/63364/2013											
B/Paris/2535/2013											
B/Firenze/1/2014											
B/Brisbane/3/2007											
B/Estonia/55669/2011											
B/Massachusetts/02/2012											
B/Hong Kong/3577/2012											
B/Norway/3323/2013											
B/Iceland/66/2013											
B/Guatemala/260/2013 cdc											
B/Paris/2327/2013											
B/Cameroon/13V-3209/2013											
B/Lisboa/235/2013											
B/South Africa/4437/2013											
B/Austria/735346/2013											
B/Moldova/462/2013											
B/Jordan/30332/2013											

	510	520	530	540	550	560
B/Florida/4/2006	GTF	NAGEFS	LPTF	DSL	NI	TAASLND
B/Norway/3044/2013 nw						DGLDNH
B/Stockholm/12/2011						ITLLYYSTAASSLAVTLMLAIFIVYMVSRDNV
B/Ireland/13M98449/2013						SCICL
B/Norway/2531/2013						
B/Arizona/08/2013 cdc						
B/Wisconsin/1/2010						
B/Novosibirsk/1/2012						
B/Hong Kong/436/2013						
B/Osaka/18/2013 niid						
B/Franche Comte/2166/2013						
B/Lyon/3021/2013						
B/Sarajevo/513/2013						
B/Tehran/63364/2013						
B/Paris/2535/2013						
B/Firenze/1/2014						
B/Brisbane/3/2007						
B/Estonia/55669/2011						
B/Massachusetts/02/2012						
B/Hong Kong/3577/2012						
B/Norway/3323/2013						
B/Iceland/66/2013						
B/Guatemala/260/2013 cdc						
B/Paris/2327/2013						
B/Cameroon/13V-3209/2013						
B/Lisboa/235/2013						
B/South Africa/4437/2013						
B/Austria/735346/2013						
B/Moldova/462/2013						
B/Jordan/30332/2013						

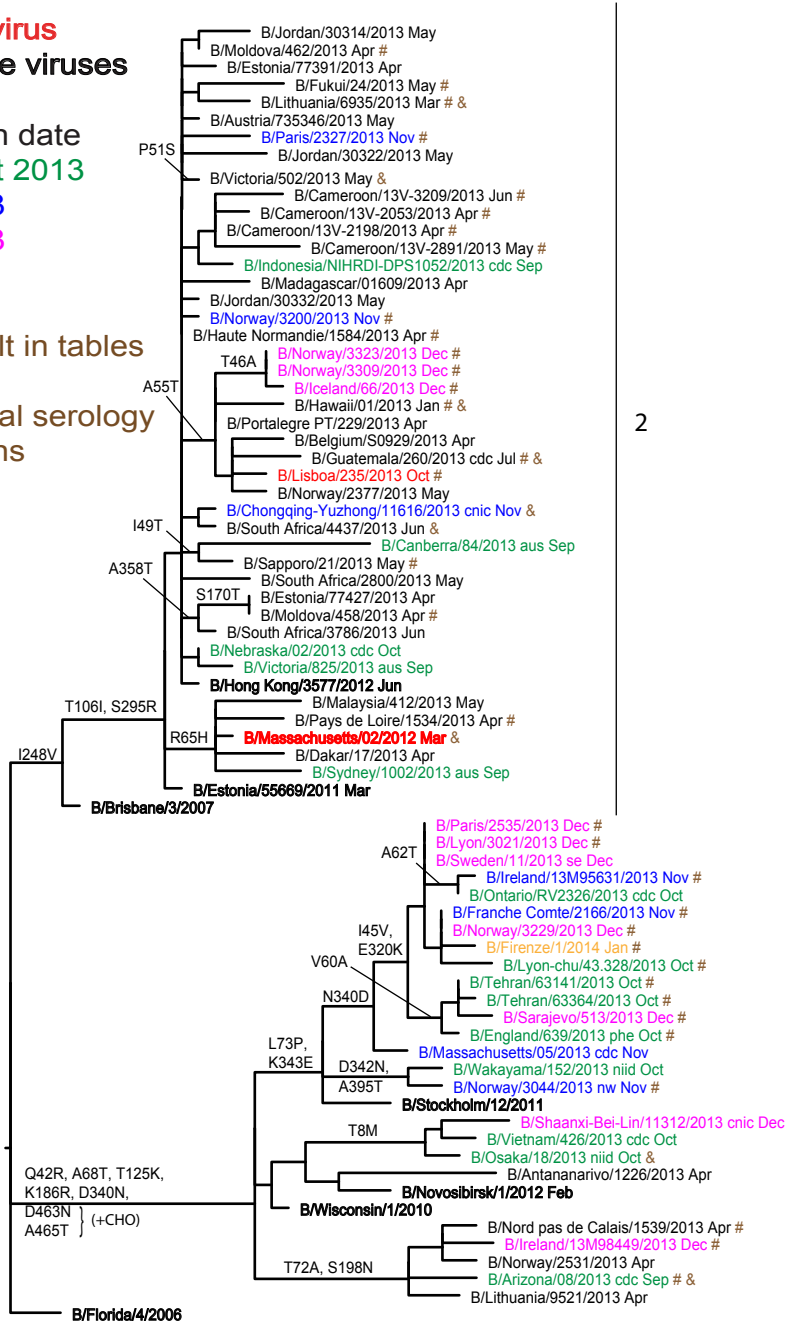
Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 21. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus
Reference viruses

Collection date
Sep - Oct 2013
Nov 2013
Dec 2013
Jan 2014

HI result in tables
& Potential serology
antigens



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400	
B/Florida/4/2006	P	F	V	K	L	N	V	E	T	D	T
B/Norway/3044/2013 nw						N	.	N	E	.	
B/Stockholm/12/2011						N	.	E	.		
B/Ireland/13M98449/2013				I	.	N	.				T
B/Norway/2531/2013						N	.				
B/Arizona/08/2013 cdc						N	.				
B/Wisconsin/1/2010						N	.				
B/Novosibirsk/1/2012						N	.				
B/Osaka/18/2013 niid						N	.				
B/Franche Comte/2166/2013								N	.		
B/Lyon/3021/2013				K	.			E	.		
B/Sarajevo/513/2013				K	.			E	.		
B/Tehran/63364/2013				K	.			E	.		
B/Paris/2535/2013				K	.			E	.		
B/Firenze/1/2014				K	.			E	.		
B/Brisbane/3/2007											
B/Estonia/55669/2011											
B/Massachusetts/02/2012											
B/Hong Kong/3577/2012											
B/Norway/3323/2013											
B/Iceland/66/2013											
B/Guatemala/260/2013 cdc											
B/Paris/2327/2013											
B/Cameroon/13V-3209/2013											
B/Lisboa/235/2013											
B/South Africa/4437/2013											
B/Austria/735346/2013											
B/Moldova/462/2013											
B/Jordan/30332/2013											

	410	420	430	440	450	460
B/Florida/4/2006	V	S	M	E	E	P
B/Norway/3044/2013 nw						N
B/Stockholm/12/2011						N
B/Ireland/13M98449/2013						N
B/Norway/2531/2013						N
B/Arizona/08/2013 cdc						N
B/Wisconsin/1/2010						N
B/Novosibirsk/1/2012						N
B/Osaka/18/2013 niid						N
B/Franche Comte/2166/2013						N
B/Lyon/3021/2013						N
B/Sarajevo/513/2013						N
B/Tehran/63364/2013						N
B/Paris/2535/2013						N
B/Firenze/1/2014						N
B/Brisbane/3/2007						
B/Estonia/55669/2011						
B/Massachusetts/02/2012						
B/Hong Kong/3577/2012						
B/Norway/3323/2013						
B/Iceland/66/2013						
B/Guatemala/260/2013 cdc						
B/Paris/2327/2013						
B/Cameroon/13V-3209/2013			K	.		
B/Lisboa/235/2013					P	.
B/South Africa/4437/2013						
B/Austria/735346/2013						
B/Moldova/462/2013						
B/Jordan/30332/2013						

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 23. Phylogenetic comparison of influenza A(H1N1)pdm09 M genes

Vaccine virus

Reference viruses

Collection date

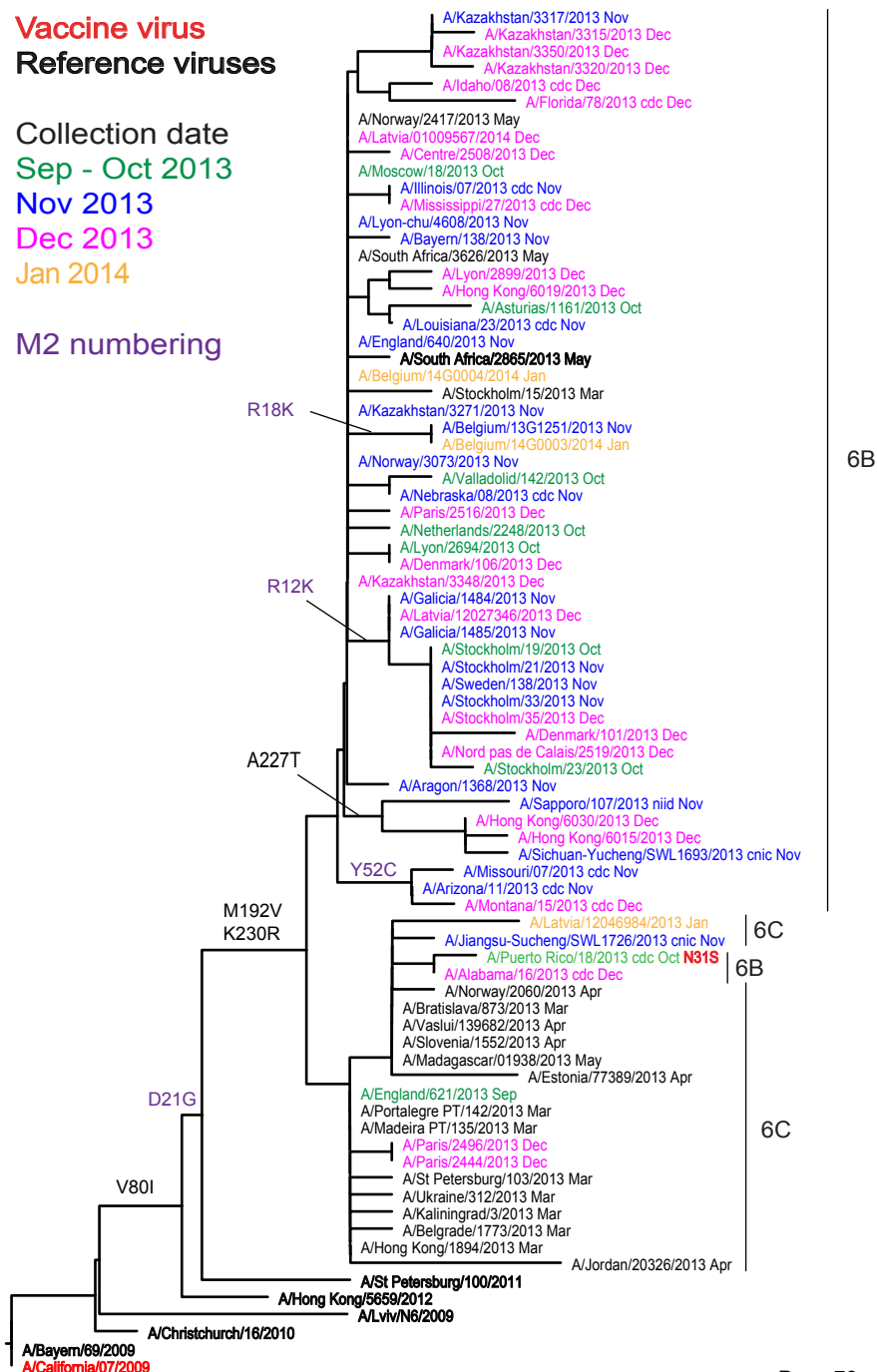
Sep - Oct 2013

Nov 2013

Dec 2013

Jan 2014

M2 numbering



Vaccine virus
Reference viruses

Sep - Oct 2013

Nov 2013

Dec 2013

Jan 2014

M2 numbering

For collection

dates from

2013-09-01:

@ HA-3C.2

unmarked HA-3C.3

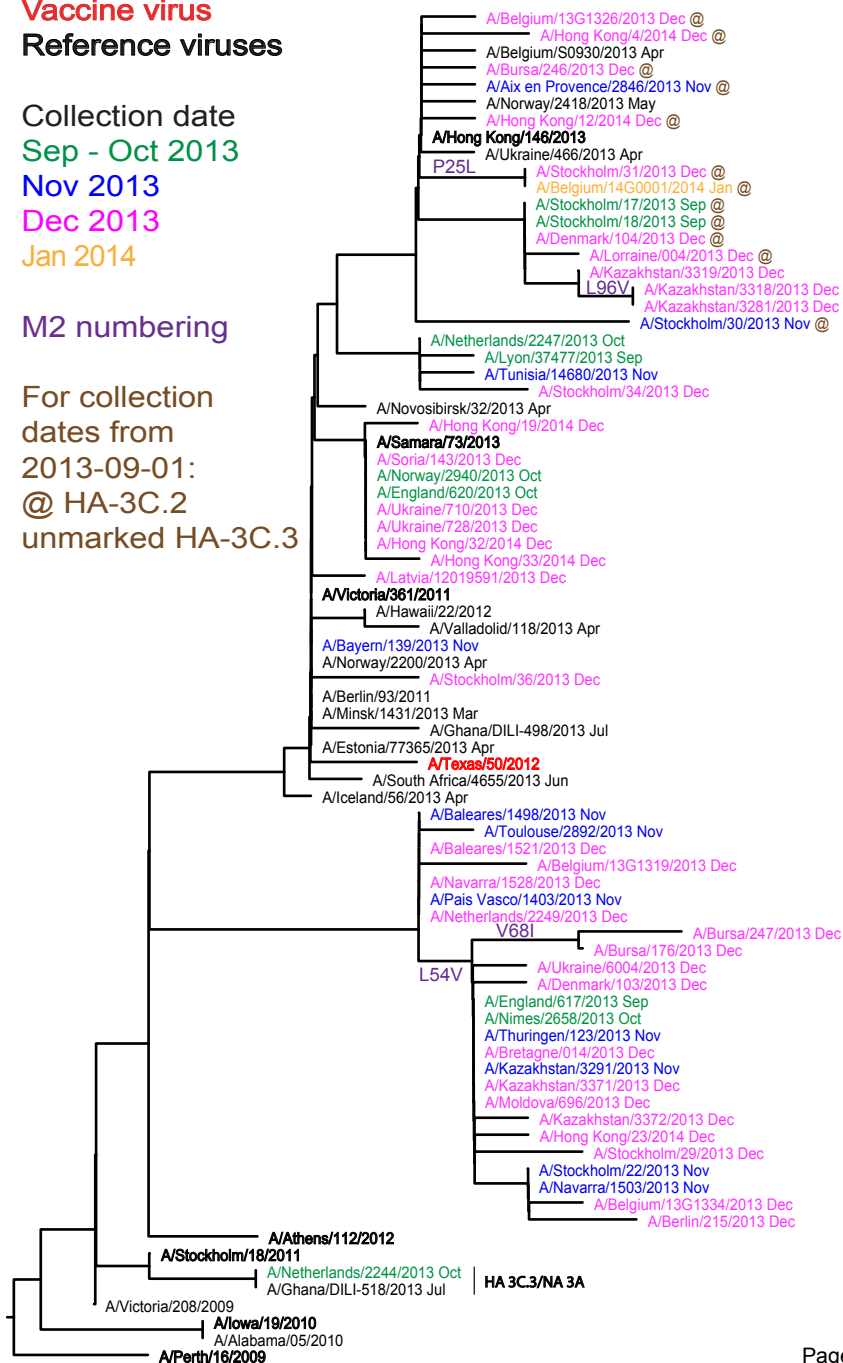


Table 15. Antiviral (NI) testing of isolates collected after 2012-12-31

Month of Collection	Virus Type/subtype						
	A(H1N1)pdm09		A(H3N2)		Flu B		
	Total	RI/HRI ¹	Total	RI/HRI ²	Total	RI/HRI	
2013							
January	195	2	96		109		402
February	121		55		143		319
March	92		71		158		321
April	23		23		85		131
May	24		3		17		44
June	9		10		5		24
July	1		8		1		10
August	0		5		0		5
Total	465	2	271	0	518	0	1256
September	0		8		0		8
October	7		14		5		26
November	10		16		8		34
December	15		79	1	14		109
2014							
January	1		13		1		15
Total	33	0	130	1	28	0	192
Grand Total	498	2	401	1	546	0	1448

All viruses were subjected to phenotypic testing against oseltamivir and zanamivir

¹ A(H1N1)pdm09 viruses carried H275Y substitutions causing HRI

² The A(H3N2) virus carried E119V substitutions causing RI